DIFFERENTIAL GEOMETRY AND LIE GROUPS

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Fifth Semester

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Chapter 1

PRELIMINARIES AND MANIFOLDS

July 20th.

The primary reference for this course will be Loring Tu's *An Introduction to Manifolds*. Prerequisites include Analysis of Several Variables, Linear Algebra, and Topology. The mid-term will account for 35% of the final grade, quizzes and student lectures will account for 15%, and the final exam will account for 50%.

Simply put, a manifold is any object that is locally Euclidean.

1.1 Calculus on $\mathbb{R}^n$

A point p or element in $\mathbb{R}^n$ is associated with an n-tuple of coordinates $x^1, x^2, \ldots, x^n$. Let U be an open neighbourhood of p in $\mathbb{R}^n$. For a non-negative integer k, we say that a function $f : U \to \mathbb{R}$ is C^k*-continuous* at $p \in U$ if all partial derivatives of f of order $\leq k$ exist and are continuous at p. f is said to be C^∞ or *smooth* if f is C^k for all $k \geq 0$. f is C^k (or C^∞)on U if it is C^k (or C^∞) at every point in U.

If f happens to be a vector-valued function, i.e. $f : U \to \mathbb{R}^m$, we say $f = (f^1, f^2, \ldots, f^m)$ is C^k (or C^∞) at p if each component function $f^i : U \to \mathbb{R}$ is C^k (or C^∞) at p. f is C^k (or C^∞) on U if it is C^k (or C^∞) at every point in U.

Definition 1.1. Let R be a commutative ring with unity. We term A to be an R*-algebra* if A is a commutative ring with unity and an R-module, and for $r \in R$ and $a, b \in A$, we have $r(ab) = (ra)b = a(rb)$.

In this course, we will be primarily be concerned with the case when R is a field, and A is an R-vector space.

For a subset $U \subseteq \mathbb{R}^n$, we denote $C^k(U)$ (or $C^\infty(U)$) as the set of all C^k (or C^∞) functions from U to $\mathbb{R}$. Note that this is a commutative ring with unity. It is also a vector space over $\mathbb{R}$. Together, $C^k(U)$ (or $C^\infty(U)$) is a (commutative) algebra over $\mathbb{R}$. Moreover, we have the descending chain of subalgebras

$$C^0(U) \supseteq C^1(U) \supseteq C^2(U) \supseteq \cdots C^k(U) \supseteq C^{k+1}(U) \supseteq \cdots \tag{1.1}$$

with

$$\bigcap_{k \geq 0} C^k(U) = C^\infty(U). \tag{1.2}$$

$f : U \to \mathbb{R}$ is termed *real analytic* at $p \in U$ if there exists a neighbourhood $V \subseteq U$ of p such that

$$f(x) = f(p) + \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(p)(x^i - p^i) + \cdots + \frac{1}{k!} \sum_{1 \leq i_1, \ldots, i_k \leq n} \frac{\partial^k f}{\partial x^{i_1} \cdots \partial x^{i_k}}(p)(x^{i_1} - p^{i_1}) \cdots (x^{i_k} - p^{i_k}) + \cdots \tag{1.3}$$

for all $x \in V$. f is said to be real analytic on U if it is real analytic at every point in U. We denote the set of all real analytic functions on U as $C^\omega(U)$. Note that $C^\omega(U) \subseteq C^\infty(U)$.

Theorem 1.2 (*Taylor's theorem*)**.** *Let $p \in U \subseteq \mathbb{R}^n$, and $f : U \to \mathbb{R}$ be a C^∞ function. Further assume that for all $x \in U$, the line segment from p to x is contained in U (U in this case is said to be* star-shaped *with respect to p). Then there exist $g_1, \ldots, g_n \in C^\infty(U)$ such that, for all $x \in U$,*

$$f(x) = f(p) + \sum_{i=1}^{n} g_i(x)(x^i - p^i), \tag{1.4}$$

with

$$g_i(p) = \frac{\partial f}{\partial x^i}(p), \quad 1 \leq i \leq n. \tag{1.5}$$

Proof. Let $x \in U$. Then the line segment $\{p + t(x-p) \mid t \in [0,1]\}$ is contained in U. Composing f with the map $t \mapsto p + t(x-p)$, we obtain

$$\frac{df}{dt}(p + t(x-p)) = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i}(p + t(x-p))(x^i - p^i). \tag{1.6}$$

Integrating from 0 to 1, we get

$$f(x) - f(p) = \int_0^1 \frac{df}{dt}(p + t(x-p))dt = \sum_{i=1}^{n}(x^i - p^i)\int_0^1 \frac{\partial f}{\partial x^i}(p + t(x-p))dt = \sum_{i=1}^{n}(x^i - p^i)g_i(x). \tag{1.7}$$

Clearly, because f is C^∞, each g_i is also C^∞. Further, we have $g_i(p) = \int_0^1 \frac{\partial f}{\partial x^i}(p)dt = \frac{\partial f}{\partial x^i}(p)$. ∎

Let $p = (p^1, \ldots, p^n), v \in \mathbb{R}^n$, and $f : U \to \mathbb{R}$ be C^∞ at p. The *directional derivative* of$f$ at p in the direction of v is defined as

$$\lim_{t\to 0} \frac{f(p+tv) - f(p)}{t} = \frac{d}{dt} f(p+tv)\Big|_{t=0} = \sum_{i=1}^{n} \frac{df}{dx^i}(p)v^i. \tag{1.8}$$

From here, we obtain the operator

$$D_v = \sum_{i=1}^{n} v^i \frac{\partial}{\partial x^i}\Big|_p \tag{1.9}$$

Now consider all pairs of the form (f, U) where $p \in U \subseteq \mathbb{R}^n$ is an open neighbourhood of p and $f : U \to \mathbb{R}$ is C^∞. Define an equivalence relation $\sim$ on this set by declaring $(f, U) \sim (g, V)$ if and only if there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$. The equivalence class of (f, U) is denoted by $[(f, U)]$, and the set of all such equivalence classes is denoted by $C_p^\infty(\mathbb{R}^n)$. Such an equivalence class $[(f, U)]$ is called the *germ* of f at p. We can give $C_p^\infty(\mathbb{R}^n)$ the structure of a commutative ring with unity by defining

$$[(f,U)] + [(g,V)] = [(f+g, U\cap V)], \quad [(f,U)]\cdot[(g,V)] = [(fg, U\cap V)], \quad 1 = [(1, \mathbb{R}^n)]. \tag{1.10}$$

It can also be given the structure of a vector space over $\mathbb{R}$ by defining

$$\lambda[(f,U)] = [(\lambda f, U)], \quad \lambda \in \mathbb{R}. \tag{1.11}$$

Thus, $C_p^\infty(\mathbb{R}^n)$ is an $\mathbb{R}$-algebra. Now, given a germ $\alpha = [(f, U)] \in C_p^\infty(\mathbb{R}^n)$ and a vector $v \in \mathbb{R}^n$, we define

$$D_v\alpha = \sum_{i=1}^{n} v^i \frac{\partial f}{\partial x^i}\Big|_p = v \cdot \nabla f(p). \tag{1.12}$$

Note that the choice of f is immaterial, since if $[(f, U)] = [(g, V)]$, then there exists an open neighbourhood $W \subseteq U \cap V$ of p such that $f|_W = g|_W$, and hence $\nabla f(p) = \nabla g(p)$. One can infer that $D_{\lambda v}(f) = \lambda D_v(f) =$

$D_v(\lambda f)$ and $D_v(f+g) = D_v(f) + D_v(g)$ for all $\lambda \in \mathbb{R}$ and $f, g \in C_p^\infty(\mathbb{R}^n)$. Thus, D_v is an $\mathbb{R}$-linear map from $C_p^\infty(\mathbb{R}^n)$ to $\mathbb{R}$ (hereforth, we shall use f to also denote the germ $[(f,U)]$ of f at p). Also,

$$D_v(fg) = \sum_{i=1}^n v^i \frac{\partial(fg)}{\partial x^i}(p) = \sum_{i=1}^n v^i \left(f(p)\frac{\partial g}{\partial x^i}(p) + g(p)\frac{\partial f}{\partial x^i}(p) \right) = f(p)D_v(g) + g(p)D_v(f). \tag{1.13}$$

Such a property makes D_v what we call a derivation.

Definition 1.3. Any $\mathbb{R}$-linear map $T : C_p^\infty(\mathbb{R}^n) \to \mathbb{R}$ satisfying the Leibniz rule $T(fg) = f(p)T(g) + g(p)T(f)$ for all $f, g \in C_p^\infty(\mathbb{R}^n)$ is called a *derivation* at p.

One may show that if T is a derivation at p, then $T(1) = 0$.

The set of all derivations at p is denoted by $\mathfrak{D}_p(\mathbb{R}^n)$. It is easy to see that $\mathfrak{D}_p(\mathbb{R}^n)$ is a vector space over $\mathbb{R}$. Now consider the map

$$\theta : \mathbb{R}^n \to \mathfrak{D}_p(\mathbb{R}^n), \quad v \mapsto D_v. \tag{1.14}$$

This θ is an $\mathbb{R}$-linear map (verify). We claim that θ is an isomorphism of real vector spaces. To this end, let $v \in \mathbb{R}^n$ such that $D_v = 0 \in \mathfrak{D}_p(\mathbb{R}^n)$. Then $D_v f = 0$ for all $f \in C_p^\infty(\mathbb{R}^n)$. In particular, for the coordinate functions x^i, we have $D_v x^i = v_i = 0$ for all $1 \le i \le n$. Thus, $v = 0$, and hence θ is injective. To show surjectivity, we appeal to Taylor's theorem; let $D \in \mathfrak{D}_p(\mathbb{R}^n)$. We claim that $D = D_v$ where $v = (D(x^1), \dots, D(x^n))$. Indeed, for any $f \in C_p^\infty(\mathbb{R}^n)$, we have

$$\begin{aligned} D(f) &= D(f(p)) + D\left(\sum_{i=1}^n (x^i - p^i) g_i(x) \right) = 0 + \sum_{i=1}^n \left((x^i - p^i)(p) Dg_i(x) + D(x^i - p^i) g_i(p) \right) \\ &= \sum_{i=1}^n (Dx^i) g_i(p) = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p) Dx^i = D_v(f). \end{aligned} \tag{1.15}$$

Hence, θ is surjective, and we have the desired isomorphism $\mathfrak{D}_p(\mathbb{R}^n) \cong \mathbb{R}^n$.

Note that a germ $f \in C_p^\infty(\mathbb{R}^n)$ has a valid value at p, but no other point. Hence, we have the evaluation map

$$\mathrm{ev}_p : C_p^\infty(\mathbb{R}^n) \to \mathbb{R}, \quad [(f,U)] \mapsto f(p). \tag{1.16}$$

The kernel here is $\{[(f,U)] \in C_p^\infty(\mathbb{R}^n) \mid f(p) = 0\}$, which is a ideal in $C_p^\infty(\mathbb{R}^n)$. For any other element $[(f,U)] \in C_p^\infty(\mathbb{R}^n)$ with $f(p) \neq 0$, a valid inverse is given by $[(f^{-1}, U \setminus \{f = 0\})]$. This makes $C_p^\infty(\mathbb{R}^n)$ a local ring.

1.2 Multilinear Algebra

July 22nd.

Hereforth, V shall denote a finite-dimensional real vector space, and V^V shall denote its *dual space* $\mathrm{Hom}_{\mathbb{R}}(V, \mathbb{R})$. If $e_1, \dots, e_n$ are a basis for V, then the *dual basis* of V^V is denoted by $\alpha^1, \dots, \alpha^n \in V^V$, and is defined by $\alpha^i(e_j) = \delta_{i,j}$ for all $1 \le i, j \le n$. One may verify that the dual basis $\{\alpha^1, \dots, \alpha^n\}$, induced by the basis $\{e_1, \dots, e_n\}$, is indeed a basis for V^V. S_k shall denote the permutation group on k symbols, satisfying $|S_k| = k!$. For any $\sigma \in S_k$, the sign of σ is denoted by $\mathrm{sgn}(\sigma)$, and is defined as $+1$ if σ is an even permutation, and -1 if σ is an odd permutation.

With notation out of the way, let $k \ge 1$ be an integer. We shall consider the k-tuple $V^k = V \times \cdots \times V$ (k times).

Definition 1.4. A mapping $f : V^k \to \mathbb{R}$ is said to be *k-linear* if f is linear in each of its k arguments.

A special case is when $k = 2$, in which case f is called a *bilinear form*. Such an $f : V \times V \to \mathbb{R}$ satisfies

$$f(\lambda v + \mu w, u) = \lambda f(v,u) + \mu f(w,u), \quad f(u, \lambda v + \mu w) = \lambda f(u,v) + \mu f(u,w). \tag{1.17}$$

A common example of a bilinear form is any valid inner product $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ on V. Another example is the determinant $\det : \mathbb{R}^n \to \mathbb{R}$, where the vectors are interpreted as the columns of a matrix. We can also write

$$L_k(V) = \{f : V^k \to \mathbb{R} \mid f \text{ is } k\text{-linear}\}. \tag{1.18}$$

It is easy to see that $L_k(V)$ is an $\mathbb{R}$-vector space.

Definition 1.5. A k-linear map $f : V^k \to \mathbb{R}$ is called a *symmetric k-linear form* if for all $\sigma \in S_k$,

$$f(v_1, \ldots, v_k) = f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}). \tag{1.19}$$

That is, f is invariant under any permutation of its arguments. Moreover, f is called an *alternating k-linear form* if for all $\sigma \in S_k$,

$$f(v_1, \ldots, v_k) = \operatorname{sgn}(\sigma) f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}). \tag{1.20}$$

One can see that the standard inner product on $\mathbb{R}^n$ is a symmetric bilinear form, and the determinant is an alternating n-linear form. Related to these definitions, we can immediately define the following subspaces of $L_k(V)$:

$$S_k(V) = \{f \in L_k(V) \mid f \text{ is symmetric}\} \subseteq L_k(V), \quad A_k(V) = \{f \in L_k(V) \mid f \text{ is alternating}\} \subseteq L_k(V). \tag{1.21}$$

Currently, our main interest of study is $A_k(V)$, the space of alternating k-linear forms on V. Elements of $A_k(V)$ are termed *alternating k-tensors*. Similarly, those of $S_k(V)$ and $L_k(V)$ are called *symmetric k-tensors* and *k-tensors*, respectively. For $k = 0$, we define $L_0(V) = S_0(V) = A_0(V) = \mathbb{R}$. For $k = 1$, we have $L_1(V) = S_1(V) = A_1(V) = V^V$. We can define the (left) action of S_k on $L_k(V)$ by

$$(\sigma \cdot f)(v_1, \ldots, v_k) = f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}). \tag{1.22}$$

Thus, f is symmetric if $\sigma \cdot f = f$ for all $\sigma \in S_k$, and f is alternating if $\sigma \cdot f = \operatorname{sgn}(\sigma) f$ for all $\sigma \in S_k$.

Given an $f \in L_k(V)$, we can define the *symmetrization* $Sf \in S_k(V)$, of f, as

$$Sf = \sum_{\sigma \in S_k} \sigma \cdot f. \tag{1.23}$$

One may, indeed, check that Sf is symmetric. Similarly, we can define the *alternation* $Af \in A_k(V)$, of f, as

$$Af = \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma) \sigma \cdot f. \tag{1.24}$$

1.2.1 The Tensor and Wedge Products

July 24th.

For any $f \in L_k(V)$ and $g \in L_l(V)$, we can define the *tensor product* $f \otimes g \in L_{k+l}(V)$ as

$$(f \otimes g)(v_1, \ldots, v_{k+l}) = f(v_1, \ldots, v_k) g(v_{k+1}, \ldots, v_{k+l}). \tag{1.25}$$

Verify that $f \otimes g$ is indeed $(k+l)$-linear. We are interested in the tensor product of alternating tensors, so let $f \in A_k(V)$ and $g \in A_l(V)$. Then, we can define the *wedge product* $f \wedge g \in A_{k+l}(V)$ as

$$f \wedge g = \frac{1}{k!l!} A(f \otimes g) = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \sigma \cdot (f \otimes g). \tag{1.26}$$

The associativity of the tensor product holds easily: for $f \in L_k(V)$, $g \in L_l(V)$, and $h \in L_m(V)$,

$$(f \otimes g) \otimes h = f \otimes (g \otimes h). \tag{1.27}$$

Proposition 1.6. *For $f \in A_k(V)$ and $g \in A_l(V)$, we have*

$$f \wedge g = (-1)^{kl} g \wedge f. \tag{1.28}$$

Proof. We shall show that $A(f \otimes g) = (-1)^{kl} A(g \otimes f)$. To this end, fix the permutation $\tau \in S_{k+l}$ defined by

$$\tau = \begin{pmatrix} 1 & 2 & \cdots & l & l+1 & l+2 & \cdots & l+k \\ k+1 & k+2 & \cdots & k+l & 1 & 2 & \cdots & k \end{pmatrix}. \tag{1.29}$$

We have

$$\begin{aligned} A(f \otimes g)(v_1, \ldots, v_{k+l}) &= \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \ldots, v_{\sigma(k+l)}) \\ &= \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) f(v_{\sigma\tau(l+1)}, \ldots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \ldots, v_{\sigma\tau(l)}) \\ &= \operatorname{sgn}(\tau) \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma\tau) f(v_{\sigma\tau(l+1)}, \ldots, v_{\sigma\tau(l+k)}) g(v_{\sigma\tau(1)}, \ldots, v_{\sigma\tau(l)}) \\ &= \operatorname{sgn}(\tau) A(g \otimes f)(v_1, \ldots, v_{k+l}). \end{aligned} \tag{1.30}$$

With $\operatorname{sgn}(\tau) = (-1)^{kl}$, the proof is complete. ∎

Corollary 1.7. *For $f \in A_k(V)$, we have $f \wedge f = 0$ if k is odd.*

Proof. Easy since $f \wedge f = (-1)^{k^2} f \wedge f = -f \wedge f$ if k is odd, so $f \wedge f = 0$. ∎

Lemma 1.8. *For $f \in L_k(V)$ and $g \in L_l(V)$, we have*

$$A(Af \otimes g) = k! A(f \otimes g), \quad A(f \otimes Ag) = l! A(f \otimes g). \tag{1.31}$$

Proof. We shall prove the first equality; the second follows similarly. We have

$$\begin{aligned} A(Af \otimes g) &= \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \sigma \cdot \left(\sum_{\tau \in S_k} \operatorname{sgn}(\tau)(\tau \cdot f) \otimes g \right) = \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \sigma \cdot \left(\sum_{\tau \in S_k} \operatorname{sgn}(\tau) \tau \cdot (f \otimes g) \right) \\ &= \sum_{\tau \in S_k} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma\tau) \sigma\tau \cdot (f \otimes g) = k! \sum_{\alpha \in S_{k+l}} \operatorname{sgn}(\alpha) \alpha \cdot (f \otimes g) = k! A(f \otimes g). \end{aligned} \tag{1.32}$$

∎

Proposition 1.9. *The wedge product is associative, i.e. for $f \in A_k(V)$, $g \in A_l(V)$, and $h \in A_m(V)$, we have*

$$(f \wedge g) \wedge h = f \wedge (g \wedge h). \tag{1.33}$$

Proof. We have

$$\begin{aligned} (f \wedge g) \wedge h &= \frac{1}{(k+l)!m!} A((f \wedge g) \otimes h) = \frac{1}{(k+l)!m!k!l!} A(A(f \otimes g) \otimes h) = \frac{(k+l)!}{(k+l)!l!k!m!} A((f \otimes g) \otimes h) \\ &= \frac{1}{k!l!m!} A(f \otimes g \otimes h) = f \wedge (g \wedge h). \end{aligned} \tag{1.34}$$

∎

Corollary 1.10. *$f \wedge g \wedge h = \frac{1}{k!l!m!} A(f \otimes g \otimes h)$ for $f \in A_k(V)$, $g \in A_l(V)$, and $h \in A_m(V)$. Inductively,*

$$f_1 \wedge \cdots \wedge f_k = \frac{1}{k_1! \cdots k_k!} A(f_1 \otimes \cdots \otimes f_k), \tag{1.35}$$

where $f_i \in A_{k_i}(V)$ for $1 \leq i \leq k$.

Proposition 1.11. *Let $\alpha^1, \ldots, \alpha^n \in V^V = A_1(V) = L_1(V)$, and $v_1, \ldots, v_k \in V$. Then*

$$(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \ldots, v_k) = \det(\alpha^i(v_j))_{1 \leq i,j \leq k}. \tag{1.36}$$

Proof. We have

$$\begin{aligned}(\alpha^1 \wedge \cdots \wedge \alpha^k)(v_1, \ldots, v_k) &= \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)(\alpha^1 \otimes \cdots \otimes \alpha^k)(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) \\ &= \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)\alpha^1(v_{\sigma(1)})\alpha^2(v\sigma(2)) \cdots \alpha^k(v_{\sigma(k)}) = \det(\alpha^i(v_j))_{1 \leq i,j \leq k}. \end{aligned} \tag{1.37}$$

■

July 27th.

Let V be an $\mathbb{R}$-vector space with basis $e_1, \ldots, e_n$, and V^V be its dual space with dual basis $\alpha^1, \ldots, \alpha^n$. Moreover, let I be a multi-index of length k, i.e. $I = (i_1, \ldots, i_k)$ with $1 \leq i_1, \ldots, i_k \leq n$. We define $e_I := (e_{i_1}, \ldots, e_{i_k}) \in V^k$ and $\alpha^I := \alpha^{i_1} \wedge \cdots \wedge \alpha^{i_k} \in A_k(V)$. Then, a k-linear form $f \in L_k(V)$ is completely determined by its values on the e_I, and an alternating k-linear form $f \in A_k(V)$ is completely determined by its values on the e_I with $i_1 < i_2 < \cdots < i_k$.

To this end, we first claim that $\alpha^I(e_J) = \delta^I_J$ for all multi-indices I, J of length k. Indeed, if $J = (j_1, \ldots, j_k)$ with $j_1 < j_2 < \cdots < j_k$, then

$$\alpha^I(e_J) = \det(\alpha^{i_m}(e_{j_n}))_{1 \leq m,n \leq k}; \tag{1.38}$$

. this is 1 for $I = J$. Suppose $I \neq J$. Let $1 \leq l \leq k$ be the first index where $i_l \neq j_l$. Then the lth row of $\alpha^{i_m}(e_{j_n})_{1 \leq m,n \leq k}$ is an all zeroes row, and hence $\det(\alpha^{i_m}(e_{j_n}))_{1 \leq m,n \leq k} = 0$. Thus, $\alpha^I(e_J) = \delta^I_J$. We now claim that α^I, for strictly increasing multi-indices I of length k, form a basis for $A_k(V)$. For linear independence, we simply have (the sums in this context are over strictly increasing multi-indices I of length k):

$$\sum_I c_I \alpha^I = 0 \implies \sum_I c_I \alpha^I(e_J) = 0 \implies c_J = 0, \quad \forall J. \tag{1.39}$$

To show that they span $A_k(V)$, let $f \in A_k(V)$. Then consider the k-linear form

$$g = \sum_I f(e_I)\alpha^I. \tag{1.40}$$

We show that $f = g$; it is enough to prove that $f(e_J) = g(e_J)$ for all strictly increasing multi-indices J of length k. We have

$$g(e_J) = \sum_I f(e_I)\alpha^I(e_J) = f(e_J)\alpha^J(e_J) + \sum_{I \neq J} f(e_I)\alpha^I(e_J) = f(e_J). \tag{1.41}$$

Corollary 1.12. *$A_k(V) = 0$ if $k > n$. Further, $\dim A_k(V) = \binom{n}{k}$ if $1 \leq k \leq n$.*

1.3 Building Blocks for Manifolds

Recall that we had defined the class of germs $C^\infty_p(\mathbb{R}^n)$ of C^∞ functions at a point $p \in \mathbb{R}^n$, which agree on some neighbourhood of p. One can show that $C^\infty_p(\mathbb{R}^n)$ and $C^\infty_p(U)$ are isomorphic as $\mathbb{R}$-algebras, where $U \subseteq \mathbb{R}^n$ is an open neighbourhood of p. Note that we term $T_p(U) = T_p(\mathbb{R}^n) = \mathbb{R}^n$ as the *tangent space* of U (or $\mathbb{R}$) at p.

Definition 1.13. Let $U \subseteq \mathbb{R}^n$ be an open subset. A *vector field* on U is a function $X : U \to \amalg_{p\in U} T_p(\mathbb{R}^n)$ such that $X(p) \in T_p(\mathbb{R}^n)$ for all $p \in U$. In particular,

$$X(p) = \sum_{i=1}^{n} a^i(p) \frac{\partial}{\partial x^i}\Big|_p, \quad a^i(p) \in \mathbb{R}. \tag{1.42}$$

We can rewrite this as

$$X = \sum_{i=1}^{n} a^i \frac{\partial}{\partial x^i}, \quad a^i : U \to \mathbb{R}. \tag{1.43}$$

X is called a *smooth vector field* if each a^i is a smooth function on U. The set of all smooth vector fields on U is denoted by $\mathfrak{X}(U)$.

This is a real vector space; note that for $X, Y \in \mathfrak{X}(U)$, we have $X + Y \in \mathfrak{X}(U)$, and for $\lambda \in \mathbb{R}$, we have $\lambda X \in \mathfrak{X}(U)$. Further, $\mathfrak{X}(U)$ is a $C^\infty(U)$-module since for $f \in C^\infty(U)$ and $X \in \mathfrak{X}(U)$, we have $fX \in \mathfrak{X}(U)$, where

$$(fX)(p) = f(p)X(p), \quad p \in U. \tag{1.44}$$

Notice that $X \in \mathfrak{X}(U)$ describes a map $C^\infty(U) \to C^\infty(U)$, where $f \mapsto Xf$, being defined as $(Xf)(p) = X_p f$. This is a derivation on $C^\infty(U)$, since for $f, g \in C^\infty(U)$, we have

$$(X(fg))(p) = X_p(fg) = f(p)X_p g + g(p)X_p f = (fXg)(p) + (gXf)(p). \tag{1.45}$$

We refine our definition of a derivation as follows.

Definition 1.14. Let K be a field, and A be a K-algebra. A *derivation* $D : A \to A$ is an $\mathbb{R}$-linear map satisfying the Leibniz rule $D(ab) = aD(b) + bD(a)$ for all $a, b \in A$.

Definition 1.15. A *differential 1-form* on U is a smooth function $\omega : U \to \amalg_{p\in U} T_p(\mathbb{R}^n)^V$ such that $\omega(p) \in T_p(\mathbb{R}^n)^V =: T_p^*(\mathbb{R}^n)$ for all $p \in U$.

For example, for $f \in C^\infty(U)$, we can define the differential 1-form df on U by $df(p) = df_p \in \mathrm{Hom}_{\mathbb{R}}(T_p\mathbb{R}^n, \mathbb{R})$, where $df_p(X) = X_p f$ for all $X \in T_p\mathbb{R}^n$. In particular, dx^i is a differential 1-form on U for all $1 \le i \le n$. In fact, $\{dx^1, \dots, dx^n\}$ is the dual basis of $T_p^*\mathbb{R}^n$ corresponding to the basis $\{\frac{\partial}{\partial x^1}\Big|_p, \dots, \frac{\partial}{\partial x^n}\Big|_p\}$ of $T_p\mathbb{R}^n$. Thus, a differential 1-form ω on U can be expressed as

$$\omega(p) = \sum_{i=1}^{n} a_i(p) dx^i\Big|_p, \quad a_i : U \to \mathbb{R}. \tag{1.46}$$

$T_p^*(\mathbb{R}^n)$, here, is termed the *cotangent space* of $\mathbb{R}^n$ at p. Thus, our current setup is as follows: $V = T_p\mathbb{R}^n$, and $V^V = T_p^*\mathbb{R}^n$. The basis for $T_p\mathbb{R}^n$ considered is $\{\frac{\partial}{\partial x^1}\Big|_p, \dots, \frac{\partial}{\partial x^n}\Big|_p\}$, and the corresponding dual basis for $T_p^*\mathbb{R}^n$ is $\{dx^1\Big|_p, \dots, dx^n\Big|_p\}$. And finally, we have

$$A_k(T_p\mathbb{R}^n) = \text{the space of alternating } k\text{-linear forms on } T_p\mathbb{R}^n \tag{1.47}$$

which has the basis $\{dx^{i_1}\Big|_p \wedge \cdots \wedge dx^{i_k}\Big|_p \mid 1 \le i_1 < \cdots < i_k \le n\}$.

Definition 1.16. A *differential k-form* on U is a smooth function $\omega : U \to \amalg_{p\in U} A_k(T_p\mathbb{R}^n)$ such that $\omega(p) \in A_k(T_p\mathbb{R}^n)$ for all $p \in U$.

Any such ω may be written as $\omega = \sum_I a_I dx_I$. It termed a smooth differential k-form if each a_I is a smooth function on U. The set of all smooth differential 1-forms on U is denoted by $\Omega^1(U)$, and the set

of all smooth differential k-forms on U is denoted by $\Omega^k(U)$. Both these sets are $\mathbb{R}$-vector spaces, and $\Omega^k(U)$ is a free $C^\infty(U)$-module of rank $\binom{n}{k}$. The isomorphism may be given as

$$\Omega^1(U) \cong C^\infty(U)^{\oplus\binom{n}{k}}, \quad \omega = \sum_I a_I dx_I \mapsto (a_1, a_2, \ldots, a_{\binom{n}{k}}). \tag{1.48}$$

We can endow $\Omega^k(U)$ with the wedge product $\wedge : \Omega^k(U) \times \Omega^l(U) \to \Omega^{k+l}(U)$, defined as

$$(\omega \wedge \eta)(p) = \omega(p) \wedge \eta(p), \quad p \in U, \tag{1.49}$$

or

$$\omega \wedge \eta = \sum_{I,J} a_I b_J dx_I \wedge dx_J \tag{1.50}$$

where I and J are strictly increasing multi-indices of length k and l, respectively. Note that if $I \cap J \neq \emptyset$, then $dx_I \wedge dx_J = 0$. Thus, the sum may be taken over all strictly increasing multi-indices I and J of length k and l, respectively, such that $I \cap J = \emptyset$. Also,

$$A_*(V) = \bigoplus_{k=0}^{n} A_k(V) \tag{1.51}$$

is defined as the *exterior algebra* of V. The wedge product on here is associative and anti-commutative. We define $\Omega^0(U) = C^\infty(U)$. The map $d : \Omega^k(U) \to \Omega^{k+1}(U)$ is defined as the *exterior derivative*; for $k = 0$, we have

$$d : C^\infty(U) \to \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i. \tag{1.52}$$

For a k-form $\omega = \sum_I a_I dx_I \in \Omega^k(U)$, we have

$$d\omega = \sum_I da_I \wedge dx_I = \sum_I \sum_{j=1}^{n} \frac{\partial a_I}{\partial x^j} dx^j \wedge dx_I. \tag{1.53}$$

Example 1.17. Let $U \subseteq \mathbb{R}^3$. Let us call the coordinates $x^1 = x$, $x^2 = y$, and $x^3 = z$. The only non-zero forms are $\Omega^0(U)$, $\Omega^1(U)$, $\Omega^2(U)$, and $\Omega^3(U)$. Note that $\Omega^0(U) = C^\infty(U)$, and there's nothing to say about this. For $\Omega^1(U)$, a form looks like $f\,dx + g\,dy + h\,dz$ for $f, g, h \in C^\infty(U)$. For $\Omega^2(U)$, a form looks like $R\,dx \wedge dy + Q\,dx \wedge dz + P\,dy \wedge dz$ for $f, g, h \in C^\infty(U)$.

July 29th.

The map $\Omega^1(U) \times \mathfrak{X}(U) \to C^\infty(U)$, defined as $(\omega, X) \mapsto \omega(X)$, is $\mathbb{R}$-bilinear. In particular, for $\omega = \sum_{i=1}^n a_i dx^i$ and $X = \sum_{j=1}^n b^j \frac{\partial}{\partial x^j}$, we have

$$\omega(X) = \sum_{i=1}^{n} a_i dx^i \left(\sum_{j=1}^{n} b^j \frac{\partial}{\partial x^j} \right) = \sum_{i=1}^{n} a_i b^i \in C^\infty(U). \tag{1.54}$$

Extending this, we have the map $\Omega^k(U) \times \mathfrak{X}(U)^k \to C^\infty(U)$, defined as $(\omega, X_1, \ldots, X_k) \mapsto \omega(X_1, \ldots, X_k)$, which is $\mathbb{R}$-multilinear. In particular, for $\omega = \sum_I b_I dx^I$ and $X_i = \sum_{i=1}^n a_i^j \frac{\partial}{\partial x^j}$, we have

$$\omega(X_1, \ldots, X_k) = \sum_I b_I a_{i_1}^1 \cdots a_{i_k}^k \in C^\infty(U) \tag{1.55}$$

using the fact that $dx^I(\frac{\partial}{\partial x^{i_1}}, \ldots, \frac{\partial}{\partial x^{i_k}}) = 1$ if the multi-index matches, and 0 otherwise.

1.3.1 Exterior Differentiation

Here, d_k shall denote the map $d_k : \Omega^k(U) \to \Omega^{k+1}(U)$, defined as the *exterior derivative*. The subscript k is often omitted, and we simply write $d : \Omega^k(U) \to \Omega^{k+1}(U)$. For $k = 0$, we have

$$d : C^\infty(U) \to \Omega^1(U), \quad f \mapsto df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i. \tag{1.56}$$

Analogously, for a general k-form $\omega = \sum_I a_I dx^I \in \Omega^k(U)$, we have

$$d : \Omega^k(U) \to \Omega^{k+1}(U), \quad \omega \mapsto d\omega = \sum_I da_I \wedge dx^I = \sum_I \sum_{j=1}^n \frac{\partial a_I}{\partial x^j} dx^j \wedge dx^I. \tag{1.57}$$

All the d maps are $\mathbb{R}$-linear, but not $C^\infty(U)$-linear. Also, for two forms $\omega \in \Omega^k(U)$ and $\eta \in \Omega^l(U)$, we have

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta. \tag{1.58}$$

This is known as *antiderivation*. Moreover, $d_{k+1} \circ d_k = d^2 = 0$. Both these properties may be verified by the reader. Now suppose we have another map $D : \Omega^k(U) \to \Omega^{k+1}(U)$ which is $\mathbb{R}$-linear, satisfies antiderivation, and $D^2 = 0$, and $Df = df$ for all $f \in C^\infty(U)$. Then $D = d$; the exterior derivative is unique. Thus, a chain of maps is formed:

$$0 \xrightarrow{d} C^\infty(U) \xrightarrow{d} \Omega^1(U) \xrightarrow{d} \Omega^2(U) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(U) \xrightarrow{d} 0 \quad \text{for open } U \subseteq \mathbb{R}^n. \tag{1.59}$$

An $\omega \in \Omega^k(U)$ is called an *exact k-form* if $\omega = d\eta$ for some $\eta \in \Omega^{k-1}(U)$. It is called a *closed k-form* if $d\omega = 0$. Note that every exact form is closed, but the converse is not true in general. We will refine the following later: quotienting the space of closed k-forms by the space of exact k-forms gives us the *de Rham cohomology* of U, denoted by $H^k_{\mathrm{dR}}(U)$.

1.4 Manifolds

July 31st.

Definition 1.18. A topological space M is termed a *topological manifold* of dimension n if

1. M is Hausdorff,
2. M is secound countable (that is, has a countable basis), and
3. for all $p \in M$, there exists an open neighbourhood $U \subseteq M$ and a homemorphism $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$, where $\varphi(U)$ is open. This makes a manifold *locally euclidean*.

The *invariance of dimension* property states that if we have two open sets $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ such that U is homeomorphic to V, then $m = n$. Thus, the dimension of a topological manifold is well-defined.

Definition 1.19. A pair (U, φ) is called a *coordinate chart* on M if $U \subseteq M$ is open and $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism.

A trivial example is $\mathbb{R}^n$ with one (coordinate) chart $(\mathbb{R}^n, \mathrm{id}_{\mathbb{R}^n})$. In fact, if $U \subseteq \mathbb{R}^n$ is open, then (U, id_U) is a coordinate chart. Note that subspaces inherit the properties of Hausdorffness and second countability. We state some properties without proof; if X is locally Euclidean but not T_2, then X being a topological manifold implies that the number of connected (or path connected) components is at most countable.

Definition 1.20. Two charts (U, φ) and (V, ψ) on a topological manifold M are said to be C^∞-*compatible* if either

1. $U \cap V = \emptyset$, or
2. $U \cap V \neq \emptyset$, and the transition maps $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ and $\varphi \circ \psi^{-1} : \psi(U \cap V) \to \varphi(U \cap V)$ are C^∞-maps between open subsets of $\mathbb{R}^n$.

Similarly, C^k-compatibility may be defined for $k \geq 0$. A C^ω-manifold is termed to be a real analytic manifold. In case of $k = 2m$, then $\mathbb{R}^{2m}$ is identified with $\mathbb{C}^m$, and the transition maps are required to be holomorphic.

Definition 1.21. M is a C^∞-manifold or a *smooth manifold* if M is a topological manifold and there exists a family of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ such that

1. $\{U_\alpha\}_{\alpha\in\Lambda}$ is an open cover of M, and
2. $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) are C^∞-compatible for all $\alpha, \beta \in \Lambda$.

Similarly, C^k-manifolds, C^ω-manifolds, and complex manifolds may be defined. The family of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ is called an *atlas* of M.

The most basic non-trivial example is perhaps the circle $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$, which is a C^∞-manifold of dimension 1. We can think of $\mathbb{R}^2$ as $\mathbb{C}$. Let $U = S^1 \setminus \{1\} = \{e^{it} \mid 0 < t < 2\pi\}$ and $V = S^1 \setminus \{-1\} = \{e^{it} \mid -\pi < t < \pi\}$. Then (U, φ) and (V, ψ) are charts, where $\varphi : U \to (0, 2\pi) \subseteq \mathbb{R}$ is defined as $e^{it} \mapsto t$ and $\psi : V \to (-\pi, \pi) \subseteq \mathbb{R}$ is defined as $e^{it} \mapsto t$. Here, $U \cap V = \{e^{it} \mid 0 < t < \pi\} \cup \{e^{-t} \mid \pi < t < 2\pi, -\pi < t < 0\}$. The transition maps are $\psi \circ \varphi^{-1} : (0, \pi) \to (0, \pi)$ defined as $t \mapsto t$ and $\psi \circ \varphi^{-1} : (\pi, 2\pi) \to (-\pi, 0)$ defined as $t \mapsto t - 2\pi$. Both these maps are smooth, and hence the charts are compatible. Thus, S^1 is a smooth manifold of dimension 1.

August 3rd.

Proposition 1.22. *Let $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ be an atlas on M. Suppose (V, ψ) and (W, σ) are two charts on M which are C^∞-compatible with all charts in $\mathcal{U}$. Then (V, ψ) and (W, σ) are C^∞-compatible.*

Proof. Let $p \in V \cap W$. Then there exists $\alpha \in \Lambda$ such that $p \in U_\alpha$. Since (V, ψ) and $(U_\alpha, \varphi_\alpha)$ are compatible, we have that $\psi \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap V) \to \psi(U_\alpha \cap V)$ is smooth. Similarly, since (W, σ) and $(U_\alpha, \varphi_\alpha)$ are compatible, we have that $\sigma \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap W) \to \sigma(U_\alpha \cap W)$ is smooth. Now, we have

$$(\sigma \circ \psi^{-1})|_{\psi(U_\alpha\cap V\cap W)} = (\sigma \circ \varphi_\alpha^{-1})|_{\varphi_\alpha(U_\alpha\cap V\cap W)} \circ (\varphi_\alpha \circ \psi^{-1})|_{\psi(U_\alpha\cap V\cap W)}. \tag{1.60}$$

The right-hand side is a composition of smooth maps, and hence is smooth. Thus, $\sigma \circ \psi^{-1}$ is smooth on an open neighbourhood of p. Since p was arbitrary, we have that $\sigma \circ \psi^{-1}$ is smooth on $V \cap W$. Similarly, we can show that $\psi \circ \sigma^{-1}$ is smooth on $V \cap W$. Thus, (V, ψ) and (W, σ) are compatible. ∎

This leads to the following definition.

Definition 1.23. A *maximal atlas* $\mathcal{M}$ on a smooth manifold M is an atlas which is maximal with respect to inclusion; that is, if $\mathcal{M} \subseteq \mathcal{U}$ for some atlas $\mathcal{U}$, then $\mathcal{M} = \mathcal{U}$. Equivalently, $\mathcal{M}$ is a maximal atlas if it contains all charts which are compatible with all charts in $\mathcal{M}$.

Any atlas $\mathcal{U}$ on M is contained in a unique maximal atlas $\mathcal{M}$. For a proof of this, let $\mathcal{M} = \{(U, \varphi) \mid$ compatible with all charts in $\mathcal{U}\}$. Then $\mathcal{M}$ is an atlas. If $\mathcal{M} \subseteq \mathcal{M}'$, then any chart in $\mathcal{M}'$ must be compatible with all charts in $\mathcal{M}$; it must be compatible with all charts in $\mathcal{U}$. By definition of $\mathcal{M}$, such a chart must be in $\mathcal{M}$, so $\mathcal{M} = \mathcal{M}'$. For uniqueness, if $\mathcal{U} \subseteq \mathcal{M}, \mathcal{M}'$, then $\mathcal{M}, \mathcal{M}' \subseteq \mathcal{M} \cup \mathcal{M}'$ tells us $\mathcal{M} = \mathcal{M}'$ by maximality. The unique maximal atlas $\mathcal{M} \supseteq \mathcal{U}$ is also called a differentiable structure on M.

Example 1.24.

1. Let $U \subseteq \mathbb{R}^n$, and $F : U \to \mathbb{R}^m$ be a C^∞-map. The graph of F is defined as $\Gamma(F) = \{(x, F(x)) \mid x \in U\} \subseteq \mathbb{R}^{n+m}$. At atlas on $\Gamma(F)$ is given by $\{(U, \varphi)\}$, where $\varphi : U \to \Gamma(F)$ is defined as $x \mapsto (x, F(x))$. The transition map is $\varphi^{-1} : \Gamma(F) \to U$ defined as $(x, F(x)) \mapsto x$. This is smooth, and hence $\Gamma(F)$ is a smooth manifold of dimension n.
2. Let M and N be smooth manifolds of dimension m and n, respectively. Then $M \times N$ is a smooth manifold of dimension $m + n$; if $\mathcal{U}$ and $\mathcal{V}$ are atlases on M and N, respectively, then $\mathcal{U} \times \mathcal{V} = \{(U \times V, \varphi \times \psi) \mid (U, \varphi) \in \mathcal{U}, (V, \psi) \in \mathcal{V}\}$ is an atlas on $M \times N$. The transition maps are $(\varphi_2 \times \psi_2) \circ (\varphi_1 \times \psi_1)^{-1} = (\varphi_2 \circ \varphi_1^{-1}) \times (\psi_2 \circ \psi_1^{-1})$, which is smooth since both $\varphi_2 \circ \varphi_1^{-1}$ and $\psi_2 \circ \psi_1^{-1}$ are smooth.

1.4.1 Smooth Functions and Smooth Maps

We will often omit 'smooth' but still mean smooth in the context of manifolds.

Definition 1.25. Let M be a manifold of dimension n. A function $f : M \to \mathbb{R}$ is smooth at $p \in M$ if there exists a chart (U, φ) of M such that $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth at $\varphi(p)$. A function $f : M \to \mathbb{R}$ is smooth if it is smooth at all $p \in M$.

The definition of the smooth function $f : M \to \mathbb{R}$ is independent of the choice of chart (U, φ), since if (V, ψ) is another chart containing p, then $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ is smooth, and hence $f \circ \psi^{-1} = (f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1})$ is smooth at $\psi(p)$.

Proposition 1.26. *Let M be a manifold of dimension n, and $f : M \to \mathbb{R}$ be a function. Then the following are equivalent:*

1. *f is a smooth function on M.*
2. *M has an atlas $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ such that $f \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha) \to \mathbb{R}$ is smooth for all $\alpha \in \Lambda$.*
3. *For all charts (U, φ) on M, the map $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth.*

Proof. Note that 2. implies 1. by definition, and 3. implies 2. trivially. We show that 1. implies 3. Let (U, φ) be a chart on M, and $p \in U$; it is enough to prove that $f \circ \varphi^{-1}$ is smooth at $\varphi(p)$. Since f is smooth, there exists a chart (V, ψ) of M such that $p \in V$ and $f \circ \psi^{-1} : \psi(V) \to \mathbb{R}$ is smooth at $\psi(p)$. Now, we have

$$f \circ \varphi^{-1} = (f \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1}). \tag{1.61}$$

The right-hand side is a composition of smooth maps, and hence is smooth at $\varphi(p)$. Thus, $f \circ \varphi^{-1}$ is smooth at $\varphi(p)$. ∎

A function $f : M \to \mathbb{R}$ is smooth on a chart (U, φ) if and only if the pullback $\varphi^* f := f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth. Also note that the definition of a smooth function f does not assume that f is continuous; however, one may show that the definition itself implies that f is continuous. This is because if f is smooth at $p \in M$, then there exists a chart (U, φ) of M such that $f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}$ is smooth at $\varphi(p)$, and hence continuous at $\varphi(p)$. Since φ is a homeomorphism, we have that f is continuous at p. Since p was arbitrary, we have that f is continuous on M.

Definition 1.27. Let M and N be smooth manifolds of dimension m and n, respectively. A continuous map $F : N \to M$ is smooth at $p \in N$ if there exist charts (U, φ) on N, with $p \in U$, and (V, ψ) on M, with $F(p) \in V$, such that $\psi \circ F \circ \varphi^{-1} : \varphi(U) \to \psi(V)$ is smooth at $\varphi(p)$. A continuous map $F : N \to M$ is smooth if it is smooth at all $p \in N$.

The definition of a map f being smooth at p is independent of the choice of charts (U, φ) and (V, ψ), since if (U', φ') and (V', ψ') are other charts containing p and $F(p)$, respectively, then $\psi' \circ \psi^{-1} : \psi(V \cap V') \to \psi'(V \cap V')$ and $\varphi' \circ \varphi^{-1} : \varphi(U \cap U') \to \varphi'(U \cap U')$ are smooth, and hence

$$\psi' \circ F \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1}) \tag{1.62}$$

Proposition 1.28. *Let $F : N \to M$ be a continuous map between smooth manifolds. Then the following are equivalent:*

1. *F is smooth.*
2. *There exist atlases $\mathcal{U}$ of N and $\mathcal{V}$ of M such that for all $(U, \varphi) \in \mathcal{U}$ and $(V, \psi) \in \mathcal{V}$, the map $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \to \psi(V)$ is smooth.*
3. *For all charts (U, φ) on N and (V, ψ) on M, the map $\psi \circ F \circ \varphi^{-1} : \varphi(U \cap F^{-1}(V)) \to \psi(V)$ is smooth.*

Proof. Again, note that 2. implies 1. by definition, and 3. implies 2. trivially. A similar argument as in the previous proposition shows that 1. implies 3. ■

Proposition 1.29. *Let $F : N \to M$ and $G : M \to P$ be smooth maps between smooth manifolds. Then the composition $G \circ F : N \to P$ is smooth.*

Proof. Let (U, φ), (V, ψ), and (W, σ) be charts on N, M, and P, respectively, such that $p \in U$, $F(p) \in V$, and $G(F(p)) \in W$. Then we have

$$\sigma \circ (G \circ F) \circ \varphi^{-1} = (\sigma \circ G \circ \psi^{-1}) \circ (\psi \circ F \circ \varphi^{-1}). \tag{1.63}$$

The right-hand side is a composition of smooth maps, and hence is smooth at $\varphi(p)$. Thus, $G \circ F$ is smooth at p. ■

Finally, a smooth map $F : N \to M$ is termed a *diffeomorphism* if F is a bijection, and F and F^{-1} are smooth. Some properties of diffeomorphisms may be inferred.

Proposition 1.30. *The following hold true.*

1. *Let (U, φ) be a chart on a smooth manifold M of dimension n. Then $\varphi : U \to \varphi(U)$ is a diffeomorphism.*
2. *Suppose $U \subseteq M$ is open, and that $F : U \to F(U)$ is a diffeomorphism. Then (U, F) is a chart in the maximal atlas of M.*

Proof.

1. Note that U is a manifold since it is open in M, and $\varphi(U)$ is a manifold since it is open in $\mathbb{R}^n$. The map $\varphi : U \to \varphi(U)$ is a homeomorphism by definition of a chart, so it is bijective. Note that $\{(U, \varphi)\}$ is an atlas on U, and $\{(\varphi(U), \mathrm{id}_{\varphi(U)})\}$ is an atlas on $\varphi(U)$. Then φ is smooth since $\mathrm{id}_{\varphi(U)} \circ \varphi \circ \varphi^{-1} = \mathrm{id}_{\varphi(U)}$ is smooth, and φ^{-1} is smooth since $\varphi \circ \varphi^{-1} \circ \mathrm{id}_{\varphi(U)} = \mathrm{id}_{\varphi(U)}$ is smooth. Thus, φ is a diffeomorphism.

2. Since U is open in M, it is a smooth manifold. Because $F : U \to F(U)$ is a diffeomorphism, it is in particular a homeomorphism. Hence F is bijective, and its image $F(U)$ is an open subset of $\mathbb{R}^n$ (as required for a chart). Let (V, ψ) be any chart on M with $U \cap V \neq \varnothing$. Then the transition maps

$$\psi \circ F^{-1} : F(U \cap V) \to \psi(V) \quad \text{and} \quad F \circ \psi^{-1} : \psi(U \cap V) \to F(U \cap V) \tag{1.64}$$

are smooth, since they are compositions of smooth maps. Therefore (U, F) is compatible with every chart in the maximal atlas of M, and so (U, F) is a chart in that maximal atlas.

■

Proposition 1.31. *Let N be a smooth manifold of dimension n, and $F : N \to \mathbb{R}^m$ be a continuous map. Then the following are equivalent.*

1. *F is smooth.*
2. *N has an atlas $\mathcal{U}$ such that for all $(U, \varphi) \in \mathcal{U}$, the map $F \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^m$ is smooth.*
3. *For all charts (U, φ) on N, the map $F \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^m$ is smooth.*

Proof. 3. implies 2. is obvious. For 2. implies 1., note that there exist atlases $\mathcal{U}$ and $\mathcal{V}$ of N and M respectively such that $(U, \varphi) \in \mathcal{U}$ and $(V, \psi) \in \mathcal{V}$ satisfy the smoothness condition. Since $\mathbb{R}^m$ has a global chart $(\mathbb{R}^m, \mathrm{id}_{\mathbb{R}^m})$, we can take $\mathcal{V} = \{(\mathbb{R}^m, \mathrm{id}_{\mathbb{R}^m})\}$. Then for any $(U, \varphi) \in \mathcal{U}$, we have $\mathrm{id}_{\mathbb{R}^m} \circ F \circ \varphi^{-1} = F \circ \varphi^{-1}$ is smooth, which implies that F is smooth. For 1. implies 3., F is smooth and φ is a diffeomorphism; thus, φ^{-1} is smooth, and $F \circ \varphi^{-1}$ is smooth. ■

Proposition 1.32. *Let $F : N \to M$ be a continuous map between smooth manifolds. Then the following are equivalent.*

1. *F is smooth.*

2. *M has an atlas $\mathcal{V}$ such that for all $(V, \psi) \in \mathcal{V}$, the map $\psi \circ F : F^{-1}(V) \to \psi(V)$ is smooth.*
3. *For any chart (V, ψ) of M, the map $\psi \circ F : F^{-1}(V) \to \psi(V)$ is smooth.*

The proof follows a similar structure to the one above.

Proposition 1.33. *Let $F : N \to \mathbb{R}^m$ be a continuous map. Then the following are equivalent.*

1. *F is smooth.*
2. *$F^1, \dots, F^m : N \to \mathbb{R}$ are smooth, where $F = (F^1, \dots, F^m)$.*

Examples

August 5th.

We look at some examples of smooth manifolds and smooth maps.

Example 1.34. For $S^n \subseteq \mathbb{R}^{n+1}$, the *$n$-sphere*, define $U_i^+ = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_i > 0\}$ and $U_i^- = \{(x_1, \dots, x_{n+1}) \in S^n \mid x_i < 0\}$ for $i = 1, \dots, n+1$. Then $\{(U_i^+, \varphi_i^+), (U_i^-, \varphi_i^-)\}_{i=1}^{n+1}$ is an atlas on S^n, where $\varphi_i^+ : U_i^+ \to \mathbb{R}^n$ is defined as $(x_1, \dots, x_{n+1}) \mapsto (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{n+1})$, and $\varphi_i^- : U_i^- \to \mathbb{R}^n$ is defined similarly. The transition maps are smooth, and hence S^n is a smooth manifold of dimension n.

Example 1.35. Recall $\mathbb{R}P^n$, the *real projective space* of dimension n, defined as the quotient $\mathbb{R}^{n+1} \setminus \{0\} / \sim$, where $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$ if there exists $\lambda \in \mathbb{R} \setminus \{0\}$ such that $(x_0, \dots, x_n) = (\lambda y_0, \dots, \lambda y_n)$. Let $(a_1 : a_2 : \dots : a_{n+1})$ denote the equivalence class of $(a_1, \dots, a_{n+1}) \in \mathbb{R}^{n+1} \setminus \{0\}$. This is a smooth manifold of dimension n; an atlas is given by $\{(U_i, \varphi_i)\}_{i=1}^{n+1}$, where $U_i = \{(x_1 : \dots : x_{n+1}) \in \mathbb{R}P^n \mid x_i \neq 0\}$ and $\varphi_i : U_i \to \mathbb{R}^n$ is defined as

$$\varphi_i(x_1 : \dots : x_{n+1}) = \left(\frac{x_1}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_{n+1}}{x_i} \right). \tag{1.65}$$

Verify that the transition maps $\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$ are smooth for all $i, j = 1, \dots, n+1$. We can also provide a smooth map $F : S^n \to \mathbb{R}P^n$ defined as $(x_1, \dots, x_{n+1}) \mapsto (x_1 : \dots : x_{n+1})$. This is a smooth map, and it is a 2-to-1 covering map.

Example 1.36. Similar to the above, we can define $\mathbb{P}_k^n$ to be the projective space of dimension n over the field k; that is, $\mathbb{P}_k^n = k^{n+1} \setminus \{0\} / \sim$, where $(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1})$ if there exists $\lambda \in k \setminus \{0\}$ such that $(x_1, \dots, x_{n+1}) = (\lambda y_1, \dots, \lambda y_{n+1})$. In particular, if $k = \mathbb{C}$, then $\mathbb{P}_{\mathbb{C}}^n$ consists of complex 1-dimensional subspaces of $\mathbb{C}^{n+1}$. This is a smooth manifold of dimension $2n$. $\mathbb{R}P^n$ may be, thus, interpreted as the space of real 1-dimensional subspaces of $\mathbb{R}^{n+1}$. Building upon this, the *Grassmannian space* $\mathrm{Gr}(n, k)$ is defined as the space of k-dimensional subspaces of $\mathbb{R}^n$. This is a smooth manifold of dimension $k(n-k)$. The Grassmannian space $\mathrm{Gr}(n+1, 1)$ is precisely $\mathbb{R}P^n$, and $\mathrm{Gr}(n, 0)$ is a single point. $\mathrm{Gr}(n, n-1)$ is the space of hyperplanes in $\mathbb{R}^n$. Again, the Grassmannian construction may be generalized to any field.

1.4.2 Quotient Manifolds

August 10th.

Recall that for an equivalence relation $\sim$ on a set S, the quotient set $S/\sim$ is defined as the set of equivalence classes $[s] = \{t \in S \mid t \sim s\} \in S/\sim$. If S happens to be a topological space, then the projection map $\pi : S \to S/\sim$ defined as $s \mapsto [s]$ is continuous if we endow $S/\sim$ with the quotient topology; that is, a subset $U \subseteq S/\sim$ is open if and only if $\pi^{-1}(U) \subseteq S$ is open. This is the finest topology on $S/\sim$ which makes π continuous. Also recall the universal property; that is, if $f : S \to X$ is a set map between topological spaces, and $\pi : S \to S/\sim$ is the projection map, then there exists a unique map $\tilde{f} : S/\sim \to X$

such that $f = \tilde{f} \circ \pi$ if and only if f is constant on equivalence classes. In this case, $\tilde{f}$ is continuous if and only if f is continuous.

The equivalence relation $\sim$ on S is called open if the projection map $\pi : S \to S/\sim$ is open; that is, if $U \subseteq S$ is open, then $\pi(U) \subseteq S/\sim$ is open. Usually, $S/\sim$ is not T_2 is general. The following is a sufficient condition for $S/\sim$ to be T_2, or Hausdorff.

Proposition 1.37. *Let $\sim$ be an open equivalence relation on a topological space S. Then $S/\sim$ is T_2 if and only if $\{(x, y) \in S \times S \mid x \sim y\}$ is closed in $S \times S$.*

Proof. Suppose the set $R = \{(x, y) \in S \times S \mid x \sim y\}$ is closed in $S \times S$. That is, for all $(x, y) \in S \times S$ such that $x \not\sim y$, there exist open neighbourhoods U of x and V of y such that $U \times V \subseteq S \times S \setminus R$. Since $\sim$ is open, we have that $\pi(U)$ and $\pi(V)$ are open neighbourhoods of $[x]$ and $[y]$, respectively. We claim that $\pi(U) \cap \pi(V) = \emptyset$. Suppose not; then there exists $[z] \in \pi(U) \cap \pi(V)$. Then there exist $u \in U$ and $v \in V$ such that $[z] = [u] = [v]$, which implies that $u \sim v$. But this contradicts the fact that $(u, v) \in U \times V \subseteq S \times S \setminus R$. Thus, $\pi(U) \cap \pi(V) = \emptyset$, and hence $S/\sim$ is Hausdorff.

On the other hand, suppose $S/\sim$ is Hausdorff. Let $(x, y) \in S \times S$ such that $x \not\sim y$. Then there exist open neighbourhoods U of $[x]$ and V of $[y]$ in $S/\sim$ such that $U \cap V = \emptyset$. Since $\pi : S \to S/\sim$ is continuous, we have that $\pi^{-1}(U)$ and $\pi^{-1}(V)$ are open neighbourhoods of x and y, respectively. We claim that $\pi^{-1}(U) \times \pi^{-1}(V) \subseteq S \times S \setminus R$. Suppose not; then there exists $(u, v) \in \pi^{-1}(U) \times \pi^{-1}(V)$ such that $u \sim v$. Then we have $[u] = [v]$, which implies that $[u] \in U \cap V$, contradicting the fact that $U \cap V = \emptyset$. Thus, $\pi^{-1}(U) \times \pi^{-1}(V) \subseteq S \times S \setminus R$, and hence R is closed in $S \times S$. ∎

Example 1.38. Let $\mathbb{R}^{n+1} \setminus \{(0, \ldots, 0)\}$ be the punctured space, and define an equivalence relation $\sim$ on it as $(x_1, \ldots, x_{n+1}) \sim (y_1, \ldots, y_{n+1})$ if there exists $\lambda \in \mathbb{R} \setminus \{0\}$ such that $(x_1, \ldots, x_{n+1}) = (\lambda y_1, \ldots, \lambda y_{n+1})$. Then $\mathbb{R}^{n+1} \setminus \{(0, \ldots, 0)\}/\sim$ is precisely $\mathbb{R}P^n$. The projection map $\pi : \mathbb{R}^{n+1} \setminus \{(0, \ldots, 0)\} \to \mathbb{R}P^n$ is open; let $U \subseteq \mathbb{R}^n \setminus \{(0, \ldots, 0)\}$ be open. Then $\pi^{-1}(\pi(U)) = \bigcup_{\lambda \in \mathbb{R} \setminus \{0\}} \lambda U$ is open, and hence $\pi(U)$ is open in $\mathbb{R}P^n$. Also, the set $R = \{(x, y) \in (\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\}) \mid x \sim y\}$ is closed in $(\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\})$, since if $y_i = \lambda x_i$ for some $\lambda \in \mathbb{R} \setminus \{0\}$, then $x_i y_j - x_j y_i = 0$ for all $i, j = 1, \ldots, n+1$. Thus, R is the zero set of the continuous functions $f_{ij} : (\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\}) \to \mathbb{R}$ defined as $(x, y) \mapsto x_i y_j - x_j y_i$ for all $i, j = 1, \ldots, n+1$. Hence, R is closed in $(\mathbb{R}^{n+1} \setminus \{0\}) \times (\mathbb{R}^{n+1} \setminus \{0\})$. By the previous proposition, we have that $\mathbb{R}P^n$ is Hausdorff.

Proposition 1.39. *Let $\sim$ be an open equivalence relation on a topological space S. Then S is second countable if and only if $S/\sim$ is second countable.*

Proof. Suppose S is second countable, and let $\mathcal{B}$ be a countable basis for S. Then $\mathcal{B}' = \{\pi(B) \mid B \in \mathcal{B}\}$ is a countable collection of open sets in $S/\sim$. We claim that $\mathcal{B}'$ is a basis for $S/\sim$. Let $U \subseteq S/\sim$ be open, and let $[x] \in U$. Then $\pi^{-1}(U) \subseteq S$ is open, and hence there exists $B \in \mathcal{B}$ such that $x \in B \subseteq \pi^{-1}(U)$. Then we have $[x] \in \pi(B) \subseteq U$, and hence $\mathcal{B}'$ is a basis for $S/\sim$. Thus, $S/\sim$ is second countable.

On the other hand, suppose $S/\sim$ is second countable, and let $\mathcal{B}'$ be a countable basis for $S/\sim$. Then $\mathcal{B} = \{\pi^{-1}(B') \mid B' \in \mathcal{B}'\}$ is a countable collection of open sets in S. We claim that $\mathcal{B}$ is a basis for S. Let $U \subseteq S$ be open, and let $x \in U$. Then $\pi(U) \subseteq S/\sim$ is open, and hence there exists $B' \in \mathcal{B}'$ such that $[x] \in B' \subseteq \pi(U)$. Then we have $x \in \pi^{-1}(B') \subseteq U$, and hence $\mathcal{B}$ is a basis for S. Thus, S is second countable. ∎

Example 1.40. Consider the disc $D^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_1^2 + \ldots + x_n^2 \leq 1\}$, and the hemisphere $H^n = \{(x_1, \ldots, x_{n+1}) \in S^n \mid x_{n+1} \geq 0\}$. On H^n, define an equivalence relation as $(x_1, \ldots, x_n, x_{n+1}) \sim (y_1, \ldots, y_n, y_{n+1})$ if either

1. $x_{n+1} = y_{n+1} = 0$ and $(y_1, \ldots, y_n, 0) = (-x_1, \ldots, -x_n, 0)$, or
2. $(x_1, \ldots, x_n, x_{n+1}) = (y_1, \ldots, y_n, y_{n+1})$.

On D^n, define an equivalence relation simply as $x \sim y$ if $x = y$ or $x = -y$. Then, one can show that $\mathbb{R}P^n \cong S^n/\sim \cong D^n/\sim \cong H^n/\sim$ as smooth manifolds, where $\sim$ on S^n is the antipodal equivalence relation.

1.5 Differential Structure

Let M be a smooth manifold of dimension n. For a point $p \in M$, let (U, φ) be a chart of M such that $p \in U$. The functions $r^1, \dots, r^n : \mathbb{R}^n \to \mathbb{R}$ defined as $r^i(x_1, \dots, x_n) = x_i$ are smooth, and hence, we can interpret (U, φ) as $(U, x^1, \dots, x^n)$, a chart with smooth coordinate functions $x^i = r^i \circ \varphi : U \to \mathbb{R}$ for $i = 1, \dots, n$. Then, given a map $f \in C^\infty(U)$, we define the partial derivative of f with respect to the coordinate function x^i at p as

$$\frac{\partial f}{\partial x^i}(p) := \frac{\partial (f \circ \varphi^{-1})}{\partial r^i}(\varphi(p)). \tag{1.66}$$

In fact, given any p in this particular U, we have

$$(\varphi^{-1})^* \left(\frac{\partial f}{\partial x^i} \right) = \frac{\partial f}{\partial x^i} \circ \varphi^{-1} = \frac{\partial (f \circ \varphi^{-1})}{\partial r^i} \in C^\infty(\varphi(U)). \tag{1.67}$$

This implies that $\frac{\partial f}{\partial x^i} \in C^\infty(U)$, and hence the partial derivative of f with respect to x^i is smooth on U. Moreover, one can show that $\frac{\partial x^i}{\partial x^j}(p) = \frac{\partial r^i}{\partial r^j}(\varphi(p)) = \delta^i_j$.

Let $F : N \to M$ be a smooth map between smooth manifolds of dimension n and m, respectively. Let (U, φ) be a chart on N with $p \in U$, and (V, ψ) be a chart on M with $F(p) \in V$. Then we can interpret (U, φ) as $(U, x^1, \dots, x^n)$, and (V, ψ) as $(V, y^1, \dots, y^m)$, where $x^i = r^i \circ \varphi : U \to \mathbb{R}$ and $y^j = r^j \circ \psi : V \to \mathbb{R}$ are smooth coordinate functions. Then, we have $F^i = y^i \circ F = r^i \circ \psi \circ F \in C^\infty(U)$ for $i = 1, \dots, m$. The *Jacobian matrix* of F at p, with respect to the charts (U, φ) and (V, ψ), is defined as the $m \times n$ matrix

$$J_F(p) = \left(\frac{\partial F^i}{\partial x^j}(p) \right)_{1 \le i \le m, 1 \le j \le n}. \tag{1.68}$$

Each entry here can be written as

$$\frac{\partial F^i}{\partial x^j}(p) = \frac{\partial (\psi \circ F \circ \varphi^{-1})^i}{\partial r^j}(\varphi(p)) = \frac{\partial (r^i \psi F \varphi^{-1})}{\partial r^j}(\varphi(p)) = \frac{\partial (y^i F \varphi^{-1})}{\partial r^j}(\varphi(p)) = \frac{\partial (F^i \circ \varphi^{-1})}{\partial r^j}(\varphi(p)). \tag{1.69}$$

Recall the inverse function theorem, which states that if $F : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is smooth, and the Jacobian matrix $J_F(p)$ is invertible at $p \in U$, then there exists an open neighbourhood V of $F(p)$ such that $F : U \to V$ is a diffeomorphism. This motivates the following definition. Thus, if $m = n$ and $J_F(p)$ is invertible, then F is a local diffeomorphism at p; $\psi circ F \circ \varphi^{-1}$ is a local diffeomorphism at $\varphi(p)$ if and only if F is a local diffeomorphism at p.

Let M be a smooth manifold of dimension n, and let $p \in M$. Suppose $F^1, F^2, \dots, F^n$ are smooth functions on some open neighbourhood $U \subseteq M$ of p. Then consider $F = (F^1, \dots, F^n) : U \to \mathbb{R}^n$, which is a smooth map. Then $\left(\frac{\partial F^i}{\partial x^j}(p) \right)_{n \times n}$ is invertible if and only if F is a local diffeomorphism at p if and only if $F : W \to F(W) \subseteq \mathbb{R}^n$ is a diffeomorphism for some open neighbourhood $W \subseteq U$ of p. Then, on some open set around p, we can use $F^1, \dots, F^n$ as local coordinates. This motivates the following definition.

Definition 1.41. Let M be a smooth manifold, with $p \in M$. Recall $C^\infty_p(M) \equiv C^\infty_p$, the local ring of germs of smooth functions at p, which is also an $\mathbb{R}$-algebra. Also define T_pM to be the $\mathbb{R}$-vector space of derivations at p; that is, $T_pM = \{D : C^\infty_p \to \mathbb{R} \mid D \text{ is linear and satisfies the Leibniz rule}\}$. This T_pM is called the *tangent space* of M at p.

August 12th.

We leave it as an exercise to check that $\frac{\partial}{\partial x^i}|_p$ is a derivation from C^∞_p to $\mathbb{R}$, and hence $\frac{\partial}{\partial x^i}|_p \in T_pM$ for all valid i. Given a smooth map $F : N \to M$ between smooth manifolds, we can define the map $F_{*,p} : T_pN \to T_{F(p)}M$ as $D \mapsto F_{*,p}(D)$, where $F_{*,p}(D)(f) = D(f \circ F)$ for all $f \in C^\infty_{F(p)}$. This is the pushforward of f at p. Moreover, if $G : M \to P$ is another smooth map, then $(G \circ F)_{*,p} = G_{*,F(p)} \circ F_{*,p}$. The map $F_{*,p}$ is linear, and hence a linear transformation between the tangent spaces T_pN and $T_{F(p)}M$.

If $F : N \to M$ were a diffeomorphism, then there exists a diffeomorphism $G : M \to N$ such that $G \circ F = \mathrm{id}_N$ and $F \circ G = \mathrm{id}_M$. Now look at the chain

$$T_pN \xrightarrow{F_{*,p}} T_{F(p)}M \xrightarrow{G_{*,F(p)}} T_{G(F(p))}N = T_pN. \tag{1.70}$$

Here, we have $(G \circ F)_{*,p} = G_{*,F(p)} \circ F_{*,p} = (\mathrm{id}_N)_{*,p} = \mathrm{id}_{T_pN}$, and similarly $F_{*,p} \circ G_{*,F(p)} = \mathrm{id}_{T_{F(p)}M}$. Thus, $F_{*,p}$ is a linear isomorphism between the tangent spaces T_pN and $T_{F(p)}M$.

Let $p \in M$, a smooth manifold, and let (U, φ) be a chart containing p. Since $C^\infty_p(M) \equiv C^\infty_p(U)$, the diffeomorphism $\varphi : U \to \varphi(U)$ induces an isomorphism $\varphi_{*,p} : T_pM \to T_{\varphi(p)}\mathbb{R}^n$. We know that the tangent space on the right has the basis $\left\{\frac{\partial}{\partial r^1}|_{\varphi(p)}, \ldots, \frac{\partial}{\partial r^n}|_{\varphi(p)}\right\}$. Applying this map to a basis element, we have

$$\varphi_{*,p}\left(\frac{\partial}{\partial x^i}\Big|_p\right)(f) = \frac{\partial}{\partial x^i}\Big|_p (f \circ \varphi) = \frac{\partial}{\partial r^i}\Big|_{\varphi(p)} (f \circ \varphi \circ \varphi^{-1}) = \frac{\partial}{\partial r^i}\Big|_{\varphi(p)} (f) \quad \text{for all } f \in C^\infty_p(M). \tag{1.71}$$

Hence, $\varphi_{*,p}\left(\frac{\partial}{\partial x^i}|_p\right) = \frac{\partial}{\partial r^i}|_{\varphi(p)}$, and so $\left\{\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p\right\}$ is a basis for T_pM. Thus, we have shown that the tangent space T_pM has dimension n. This also shows that if $F : N \to M$ is a smooth map between smooth manifolds of dimension n and m that also happens to be a diffeomorphism, then $n = m$. Moreover, given a map $F : N \to M$ between smooth manifolds, and given a chart (U, φ) of $p \in N$ and a chart (V, ψ) of $F(p) \in M$, we have

$$F_{*,p} : T_pN \to T_{F(p)}M, \quad F_{*,p}\left(\frac{\partial}{\partial x^i}\Big|_p\right) = \sum_{j=1}^m b^i_j \frac{\partial}{\partial y^j}\Big|_{F(p)}, \quad b^i_j = \frac{\partial F^j}{\partial x^i}(p) \quad \text{for } i = 1, \ldots, n, j = 1, \ldots, m. \tag{1.72}$$

That is, the matrix of $F_{*,p}$ with respect to the bases $\left\{\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p\right\}$ and $\left\{\frac{\partial}{\partial y^1}|_{F(p)}, \ldots, \frac{\partial}{\partial y^m}|_{F(p)}\right\}$ is precisely the $m \times n$ real *Jacobian matrix* $(dF)_p := J_F(p) = \left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1 \le i \le m,\, 1 \le j \le n}$.

August 17th.

Suppose $0 \in (a, b) \subseteq \mathbb{R}$, and let $c : (a, b) \to M$ be a smooth map to a smooth manifold M of dimension n such that $c(0) = p \in M$. c is then called a *smooth curve* in M. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M with $p \in U$. Then $c^1(0) \in T_pM$ is defined as $c'(0) = c_{*,0}\left(\frac{d}{dt}\Big|_0\right)$, where $\frac{d}{dt}|_0 \in T_0\mathbb{R}$ is the standard basis element. Then $c'(0)$ is a tangent vector at p, and we have

$$\mathbb{R} \cong T_0((a, b)) \xrightarrow{c_{*,0}} T_pM \xrightarrow{\varphi_{*,p}} T_{\varphi(p)}\mathbb{R}^n \cong \mathbb{R}^n, \quad c'(0) \mapsto \varphi_{*,p}(c'(0)) = (\varphi \circ c)_{*,0}\left(\frac{d}{dt}\Big|_0\right) = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}\Big|_p \tag{1.73}$$

since $\{\frac{\partial}{\partial x^i}|_p\}_{i=1}^n$ is a basis for T_pM. Utilising $c'(0) = c_{*,0}\left(\frac{d}{dt}|_0\right)$, we have

$$a^i = \varphi_{*,p}(c'(0))(x^i) = c_{*,0}\left(\frac{d}{dt}\Big|_0\right)(x^i \circ \varphi^{-1}) = \frac{d}{dt}\Big|_0 (x^i \circ \varphi^{-1} \circ c) = \frac{d}{dt}\Big|_0 (x^i \circ c) = \frac{dc^i}{dt}(0). \tag{1.74}$$

Thus, a curve c can be thought of as a tangent vector at p, and the tangent vector is given by the derivative of the curve at 0. In particular, let us interpret $c : \mathbb{R} \to \mathbb{R}^n$ as a smooth curve in $\mathbb{R}^n$; we have $c(t) = (c_1(t), \ldots, c_n(t))$, and hence $c'(0) = (\frac{dc_1}{dt}(0), \ldots, \frac{dc_n}{dt}(0)) \in \mathbb{R}^n$. This is consistent with the above definition of $c'(0)$ as a tangent vector at p. From here, we wish to obtain a smooth map $c : (a, b) \to M$ such that $c(0) = p$ and $c'(0) = X_p = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}|_p$ for some $X_p \in T_pM$. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M with $p \in U$. In fact, choose a chart φ such that $\varphi(p) = 0 \in \mathbb{R}^n$ (translations are diffeomorphisms). Then we can define a smooth curve $c : (-\epsilon, \epsilon) \to M$ for some $\epsilon > 0$ such that $c(0) = p$ and $c'(0) = X_p$ as

$$c(t) = \varphi^{-1}(ta^1, \ldots, ta^n) \quad \text{for } t \in (-\epsilon, \epsilon). \tag{1.75}$$

Then, $c'(0) = c_{*,0}\left(\frac{d}{dt}|_0\right) = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}|_p = X_p$, as desired.

Coming back to the smooth map $F : N \to M$ between smooth manifolds, the rank of F at $p \in N$ is defined as the rank of the linear map $F_{*,p} : T_pN \to T_{F(p)}M$, which is also the rank of the matrix $(dF)_p = J_F(p) = \left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1 \le i \le m, 1 \le j \le n}$. The map $F : N \to M$ is called an *immersion* at $p \in N$ if $F_{*,p} : T_pN \to T_{F(p)}M$ is injective (that is, $\mathrm{rank}(F_{*,p}) = \dim(T_pN) = n$), and F is called an immersion if it is an immersion at all points $p \in N$. Similarly, F is called a *submersion* at $p \in N$ if $F_{*,p} : T_pN \to T_{F(p)}M$ is surjective (that is, $\mathrm{rank}(F_{*,p}) = \dim(T_{F(p)}M) = m$), and F is called a submersion if it is a submersion at all points $p \in N$. If $F : N \to M$ is a smooth map between smooth manifolds of dimension n and m, respectively, then $\mathrm{rank}(F_{*,p}) \le \min\{n, m\}$ for all $p \in N$. Thus, if $n < m$, then F cannot be a submersion, and if $n > m$, then F cannot be an immersion.

1.6 Regular Maps and Regular Submanifolds

Definition 1.42. Let $F : N \to M$ be a smooth map between smooth manifolds. $p \in N$ is called a *critical point* of F if $F_{*,p} : T_pN \to T_{F(p)}M$ is not surjective; that is, F is not a submersion at p. p is called a *regular point* of F if $F_{*,p} : T_pN \to T_{F(p)}M$ is surjective; that is, F is a submersion at p. Similarly, $q \in M$ is called a *critical value* of F if there exists a critical point $p \in N$ such that $F(p) = q$. $q \in M$ is called a *regular value* of F if it is not a critical value; that is, for all $p \in N$ such that $F(p) = q$, we have that p is a regular point of F. That is, either $F^{-1}(m) = \emptyset$, or $F_{*,p} : T_pN \to T_{F(p)}M$ is surjective for all $p \in F^{-1}(m)$.

From here, we can define something called a regular submanifold, which is useful in determining whether a certain set is a manifold or not. Let M be a manifold of dimension n, and $S \subseteq M$ be a subset with the subspace topology. Suppose that for all $p \in S$, there exists a chart $(U, \varphi) = (U, x^1, \ldots, x^n)$ of M with $p \in U$ and $\varphi(p) = 0 \in \varphi(U) \subseteq \mathbb{R}^n$ such that

$$U \cap S = \{(x^1, x^2, \ldots, x^n) \in U \mid x^{i_1} = x^{i_2} = \ldots = x^{i_{n-k}} = 0 \text{ for some } 1 \leq i_1 < i_2 < \ldots < i_{n-k} \leq n\} \tag{1.76}$$

for some fixed integer $0 \leq k \leq n$ (via switching of coordinates, we may assume that the last $n - k$ coordinates are zero). Such a chart (U, φ) of M is said to be adapted to S. Then, we say that S is a *regular submanifold* of M of dimension k if for all $p \in S$, there exists a chart (U, φ) of M with $p \in U$ such that (U, φ) is adapted to S. Therefore, if (U, φ) is adapted to S, then $\varphi(U \cap S)$ consists of only those points that look like $(r^1, \ldots, r^k, 0, \ldots, 0) \in \mathbb{R}^n$, and hence $\varphi(U \cap S)$ is an open subset of $\mathbb{R}^k$. If we define $\varphi_S : U \cap S \to \mathbb{R}^k$ as $\varphi_S(p) = (x^1(p), \ldots, x^k(p))$, then $\varphi_S(U \cap S) = \varphi(U \cap S) \cap \mathbb{R}^k$.

Theorem 1.43. *Consider the maximal atlas $\mathcal{U} = \{(U, \varphi)\}$ of M, and let $\mathcal{A} = \{(U \cap S, \varphi_S) \mid (U, \varphi) \in \mathcal{U}\}$, where (U, φ) is adapted to S and $\bigcup\{U \cap S \mid (U, \varphi) \in \mathcal{U}\} = S$. Then $\mathcal{A}$ is a smooth atlas on S, and hence S is a smooth manifold of dimension k.*

The *codimension* of a regular submanifold S of M is defined as $\operatorname{codim}_M S = \dim(M) - \dim(S) = n - k$.

Let $f : M \to \mathbb{R}$ be a smooth function on a smooth manifold M of dimension n. Also let $0 \in \mathbb{R}$ be a regular value of f with $f^{-1}(0) \neq \emptyset$. Then, $f^{-1}(0)$ is a regular submanifold of M of dimension $n - 1$, or codimension 1. Note that 0 here is a red herring; we can replace 0 with any regular value $c \in \mathbb{R}$ of f with $f^{-1}(c) \neq \emptyset$, and the same result holds.

For a proof of this, let $p \in f^{-1}(0) \neq \emptyset$ so that $f(p) = 0$. Since 0 is a regular value of f, we have that $f_{*,p} : T_pM \to T_0\mathbb{R}$ is surjective. T_pM has a basis $\left\{\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p\right\}$, and $T_0\mathbb{R}$ has a basis $\left\{\frac{d}{dt}|_0\right\}$. The $1 \times n$ Jacobian matrix of f at p is given by

$$(df)_p = \left(\frac{\partial f}{\partial x^1}(p), \ldots, \frac{\partial f}{\partial x^n}(p)\right). \tag{1.77}$$

Since $f_{*,p} : T_pM \to T_0\mathbb{R}$ is surjective, we have that $\operatorname{rank}(f_{*,p}) = 1$, and hence $(df)_p \neq 0$; at least coordinate function x^i has $\frac{\partial f}{\partial x^i}(p) \neq 0$. Without the loss of generality, we may assume that $\frac{\partial f}{\partial x^1}(p) \neq 0$. Then, by the inverse function theorem, there exists an open neighbourhood V of p such that $f : V \to f(V) \subseteq \mathbb{R}$ is a diffeomorphism. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M with $p \in U \subseteq V$. Then, we have

$$U \cap f^{-1}(0) = \{(x^1, \ldots, x^n) \in U \mid x^1 = 0\}, \tag{1.78}$$

which shows that (U, φ) is adapted to $f^{-1}(0)$. Since $p \in f^{-1}(0)$ was arbitrary, we have that $f^{-1}(0)$ is a regular submanifold of M of dimension $n - 1$, or codimension 1. In general, this result generalizes to the following theorem.

Theorem 1.44. *Let $F : N \to M$ be a smooth map between smooth manifolds of dimension n and m, respectivel, with $n \geq m$. Let $q \in M$ be a regular value of F with $F^{-1}(q) \neq \emptyset$. Then, $F^{-1}(q)$ is a regular submanifold of N of dimension $n - m$, or codimension m.*

Proof. Let $p \in F^{-1}(q)$; then $F_{*,p} : T_pN \to T_qM$ is surjective. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart in N such that $\varphi(p) = 0 \in \mathbb{R}^n$, and let $(V, \psi) = (V, y^1, \ldots, y^m)$ be a chart in M such that $\psi(q) = 0 \in \mathbb{R}^m$.

We may assume $F(U) \subseteq V$. The matrix of $F_{*,p}$ with respect to the bases $\left\{\frac{\partial}{\partial x^1}|_p, \ldots, \frac{\partial}{\partial x^n}|_p\right\}$ and $\left\{\frac{\partial}{\partial y^1}|_q, \ldots, \frac{\partial}{\partial y^m}|_q\right\}$ is given by $\left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1\leq i\leq m, 1\leq j\leq n}$; this has rank m since $F_{*,p}$ is surjective. Without the loss of generality, we may assume that the $m \times m$ submatrix $\left(\frac{\partial F^i}{\partial x^j}(p)\right)_{1\leq i,j\leq m}$ is invertible. Then, by the inverse function theorem, there exists an open neighbourhood W of p such that $F : W \to F(W)$ is a diffeomorphism. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on N with $p \in U \subseteq W$. Then, we have

$$U \cap F^{-1}(q) = \{(x^1, \ldots, x^n) \in U \mid x^1 = x^2 = \ldots = x^m = 0\}, \tag{1.79}$$

which shows that (U, φ) is adapted to $F^{-1}(q)$. Since $p \in F^{-1}(q)$ was arbitrary, we have that $F^{-1}(q)$ is a regular submanifold of N of dimension $n - m$, or codimension m. ■

August 19th.

Example 1.45. Consider the map $f : \mathbb{R}^3 \to \mathbb{R}$ defined as $f(x, y, z) = x^2 + y^2 + z^2$. Then, $S^2 = f^{-1}(1) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$. We have $f_{*,p} : T_p\mathbb{R}^3 \to T_1\mathbb{R} = \mathbb{R}$ given by $f_{*,p}\left(\frac{\partial}{\partial x^i}|_p\right) = \sum_{j=1}^3 \frac{\partial f}{\partial x^j}(p)\frac{\partial}{\partial y}|_1$, where y is the coordinate function on $\mathbb{R}$. Then, the Jacobian matrix of f at p is given by $(df)_p = \left(\frac{\partial f}{\partial x}(p), \frac{\partial f}{\partial y}(p), \frac{\partial f}{\partial z}(p)\right) = (2x, 2y, 2z)$. The origin is the only critical point of f and is excluded from S^2. Thus, $1 \in \mathbb{R}$ is a regular value of f, and hence $S^2 = f^{-1}(1)$ is a regular submanifold of $\mathbb{R}^3$ of dimension $3 - 1 = 2$.

Example 1.46. Consider the determinant map $\det : \mathbb{R}^{n^2} \to \mathbb{R}$, interpreting a point $p \in \mathbb{R}^{n^2}$ as an $n \times n$ matrix $A \in M_n(\mathbb{R})$. Look at $SL_n(\mathbb{R}) = \det^{-1}(1) = \{A \in M_n(\mathbb{R}) \mid \det(A) = 1\}$. We have $\det_{*,A} : T_A\mathbb{R}^{n^2} \to T_1\mathbb{R} = \mathbb{R}$, where the bases are $\{\frac{\partial}{\partial x_{ij}}\}_{1\leq i,j\leq n}$ and $\{\frac{d}{dt}|_1\}$, respectively. Note that $\frac{\partial \det}{\partial x_{ij}}(A)$ is the (i, j)-th cofactor of A; not all of these cofactors can be zero since otherwise we would have $\det(A) = 0$ but $A \in \det^{-1}(1)$. Thus, $1 \in \mathbb{R}$ is a regular value of $\det$, and hence $SL_n(\mathbb{R}) = \det^{-1}(1)$ is a regular submanifold of $\mathbb{R}^{n^2}$ of dimension $n^2 - 1$.

The following is a corollary of the inverse function theorem; it may be generalized to the case of smooth maps between smooth manifolds.

Corollary 1.47. *Let $p \in \mathbb{R}^n$ have an open neighbourhood $U \subseteq \mathbb{R}^n$, and let $f : U \to \mathbb{R}^m$ be a smooth map with $f = (f_1, \ldots, f_m)$. Suppose f has rank k in a neighbourhood of p in U; that is, $\left(\frac{\partial f_i}{\partial x_j}(q)\right)$ has rank k for all q in this possibly smaller neighbourhood of p. Then there exists a diffeomorphism $G : V \to G(V)$, with $G(V)$ open in $\mathbb{R}^n$ and $p \in V$ open in U, and a diffeomorphism $F : W \to F(W)$, with $F(W)$ open in $\mathbb{R}^m$ and $f(p) \in W$ open in $\mathbb{R}^m$, such that $f(V) \subseteq W$, and $(FfG^{-1})(x^1, \ldots, x^n) = (x^1, \ldots, x^k, 0, \ldots, 0)$ for all $(x^1, \ldots, x^n) \in G(V)$.*

Proof. Let $p \in \mathbb{R}^n$ have an open neighbourhood $U \subseteq \mathbb{R}^n$, and let $f : U \to \mathbb{R}^m$ be a smooth map with $f = (f_1, \ldots, f_m)$. Suppose f has rank k in a neighbourhood of p in U. Then, without the loss of generality, we may assume that the $k \times k$ submatrix $\left(\frac{\partial f_i}{\partial x_j}(q)\right)_{1\leq i,j\leq k}$ is invertible for all q in this possibly smaller neighbourhood of p. Define the smooth map $F : U \to \mathbb{R}^n$ as $F(x^1, \ldots, x^n) = (f_1(x^1, \ldots, x^n), \ldots, f_k(x^1, \ldots, x^n), x^{k+1}, \ldots, x^n)$. Then, the Jacobian matrix of F at $q \in U$ is given by

$$(dF)_q = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(q) & \cdots & \frac{\partial f_1}{\partial x_k}(q) & \frac{\partial f_1}{\partial x_{k+1}}(q) & \cdots & \frac{\partial f_1}{\partial x_n}(q) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_k}{\partial x_1}(q) & \cdots & \frac{\partial f_k}{\partial x_k}(q) & \frac{\partial f_k}{\partial x_{k+1}}(q) & \cdots & \frac{\partial f_k}{\partial x_n}(q) \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.80}$$

Thus, by the inverse function theorem, there exists an open neighbourhood V of p with $V \subseteq U$ such that $F : V \to F(V)$ is a diffeomorphism. Consider the map $F^{-1} : F(V) \to V$; a point $(x^1, \ldots, x^n) \in F(V)$ is

mapped to $(x^1,\dots,x^k,f^{k+1},\dots,f^m)$. The Jacobian matrix of F^{-1} at $(x^1,\dots,x^n)\in F(V)$ is given by

$$(dF^{-1})_{(x^1,\dots,x^n)}=\begin{pmatrix}1&0&0&0&0&0\\0&1&0&0&0&0\\0&0&1&0&0&0*&*&*&\frac{\partial f^{k+1}}{\partial x^{k+1}}&\dots&\frac{\partial f^{k+1}}{\partial x^n}*&*&*&\vdots&\ddots&\vdots*&*&*&\frac{\partial f^m}{\partial x^{k+1}}&\dots&\frac{\partial f^m}{\partial x^n}\end{pmatrix}. \tag{1.81}$$

We know this matrix to have rank k by hypothesis. Thus, the $(m-k)\times(n-k)$ submatrix $\left(\frac{\partial f^i}{\partial x^j}(q)\right)_{k+1\le i\le m,k+1\le j\le n}$ must have rank 0; that is, $\frac{\partial f^i}{\partial x^j}(q)=0$ for all $k+1\le i\le m$ and $k+1\le j\le n$. The functions $f^{k+1},\dots,f^m$ are then functions of only $x^1,\dots,x^k$. Define the smooth map $G:f(V)\to\mathbb{R}^m$ as $(y^1,\dots,y^m)=(y^1,\dots,y^k,y^{k+1}-f^{k+1},\dots,y^m-f^m)$. Then, the Jacobian matrix of G at $(y^1,\dots,y^m)\in f(V)$ is given by

$$(dG)_{(y^1,\dots,y^m)}=\begin{pmatrix}1&0&0&0&0&0\\0&1&0&0&0&0\\0&0&1&0&0&0\\-\frac{\partial f^{k+1}}{\partial y^1}&\dots&-\frac{\partial f^{k+1}}{\partial y^k}&1&0&0\\\vdots&\ddots&\vdots&0&1&0\\-\frac{\partial f^m}{\partial y^1}&\dots&-\frac{\partial f^m}{\partial y^k}&0&0&1\end{pmatrix}. \tag{1.82}$$

This matrix is of full rank, so G is a diffeomorphism from $f(V)$ to $G(f(V))$. Then, we have $(GF)(x^1,\dots,x^n)=(f_1(x^1,\dots,x^n),\dots,f_k(x^1,\dots,x^n),0,\dots,0)$ for all $(x^1,\dots,x^n)\in V$. Thus, we have found diffeomorphisms G and F such that $(GF)(x^1,\dots,x^n)=(x^1,\dots,x^k,0,\dots,0)$ for all $(x^1,\dots,x^n)\in G(V)$. ■

The generalization is as follows. Let $f:N\to M$ be a smooth map between smooth manifolds of dimension n and m, respectively. Let $p\in N$ be a point such that f has a rank k in a neighbourhood of p in N. Then there exists a chart (U,φ) centred at p in N and a chart (V,ψ) centred at $f(p)$ in M such that $f(U)\subseteq V$, and $\psi\circ f\circ\varphi^{-1}(x^1,\dots,x^n)=(x^1,\dots,x^k,0,\dots,0)$ for all $(x^1,\dots,x^n)\in\varphi(U)$.

August 24th.

Theorem 1.48. *Let $f:N\to M$ be a smooth map between smooth manifolds of dimension n and m, respectively. Let $c\in M$ be such that f has a rank k in a neighbourhood of $f^{-1}(c)\neq\emptyset$ in N. Then $f^{-1}(c)$ is a regular submanifold of N of dimension $n-k$.*

Example 1.49. Look at the map $f:GL_n(\mathbb{R})\to GL_n(\mathbb{R})$ define by $f(A)=A^tA$. Then, $O_n(\mathbb{R})=f^{-1}(I_n)=\{A\in GL_n(\mathbb{R})\mid A^tA=I_n\}$. For now, the dimension of $O_n(\mathbb{R})$ will remain unknown, but we can still show that it is a regular submanifold of $GL_n(\mathbb{R})$. Let $C\in GL_n(\mathbb{R})$. Then $f(AC)=C^tf(A)C$. Note that conjugation by C is a diffeomorphism from $GL_n(\mathbb{R})$ to itself, and hence f has the same rank at A and AC. We shall show that the ranks of $f_{*,A}$ and $f_{*,B}$ are the same for all $A,B\in GL_n(\mathbb{R})$. We can break down $B=AC$ for some $C\in GL_n(\mathbb{R})$. Then $f\circ r_C=l_{C^t}\circ r_c\circ f$, where $r_C:GL_n(\mathbb{R})\to GL_n(\mathbb{R})$ is right multiplication by C and $l_{C^t}:GL_n(\mathbb{R})\to GL_n(\mathbb{R})$ is left multiplication by C^t. Then, we have

$$f_{*,AC}(r_C)_{*,A}=(l_{C^t})_{*,A^tAC}(r_C)_{*,A^tA}f_{*,A}. \tag{1.83}$$

The rank of $f_{*,A}$ is the same as the rank of $f_{*,AC}$ since $(r_C)_{*,A}$ and $(l_{C^t})_{*,A^tAC}(r_C)_{*,A^tA}$ are both isomorphisms. Thus, f has the same rank at all points in $GL_n(\mathbb{R})$.

The previous theorem may be generalized to the following theorem.

Theorem 1.50. *Let $f:N\to M$ be a smooth map between smooth manifolds of dimension n and m, respectively. Also let $S\subseteq M$ be a regular submanifold of codimension k, or dimension $m-k$. Suppose $f_{*,p}(T_pN)+T_{f(p)}S=T_{f(p)}M$ for all $p\in f^{-1}(S)$. Then, $f^{-1}(S)$ is a regular submanifold of N of*

codimension k, or dimension $n-k$.

Theorem 1.51. *Let $f : N \to M$ be a smooth map between smooth manifolds of dimension n and m, respectively. Let $p \in N$.*

1. *Suppose* $\operatorname{rank}(f_{*,p}) = m$. *Then there exists a local chart (U, φ) centred at p in N and a local chart (V, ψ) centred at $f(p)$ in M such that $\psi f \varphi^{-1}(x^1, \ldots, x^n) = (x^1, \ldots, x^m)$ for all $(x^1, \ldots, x^n) \in \varphi(U)$.*

2. *Suppose* $\operatorname{rank}(f_{*,p}) = n$. *Then there exists a local chart (U, φ) centred at p in N and a local chart (V, ψ) centred at $f(p)$ in M such that $\psi f \varphi^{-1}(x^1, \ldots, x^n) = (x^1, \ldots, x^n, 0, \ldots, 0)$ for all $(x^1, \ldots, x^n) \in \varphi(U)$.*

Chapter 2

TANGENT SPACES, BUNDLES, AND VECTORS

2.1 Manifold Structure

August 31st.

Recall that given a smooth manifold M, the tangent space T_pM at a point $p \in M$ consists of all derivations from $C^\infty_p(M)$ to $\mathbb{R}$. T_pM exactly equals T_pU, where U is an open neighbourhood of p in M, since germs of functions need only match in a neighbourhood. If M is a manifold of dimension n, then any tangent space of a point in M is a vector space of dimension n, too. Now, the *tangent bundle* of the manifold is defined as the disjoint union of all tangent spaces:

$$TM = \bigsqcup_{p \in M} T_pM. \tag{2.1}$$

From here, let $\pi : TM \to M$ denote the projection map; that is, if $v \in T_pM \subseteq TM$ for some $p \in M$, then $\pi(v) = p$.

We can equip TM with a topology so that it becomes a smooth manifold is its own right, and so that π also happens to be a smooth map. For $p \in M$, let (U, φ) be a chart on M containing p. Note that for all $p \in U$, the tangent space T_pM has a basis $\{\frac{\partial}{\partial x^1}\big|_p, \dots, \frac{\partial}{\partial x^n}\big|_p\}$. Given any $v \in T_pM$, it can uniquely be written as

$$v = \sum_{i=1}^{n} c^i(v) \, \frac{\partial}{\partial x^i}\Big|_p. \tag{2.2}$$

Now, given $v \in TU$, we define a chart $(\tilde{U}, \tilde{\varphi})$ as $\tilde{U} = \pi^{-1}(U) = \bigsqcup_{p \in U} T_pM = TU$, and

$$\tilde{\varphi} : TU \to \varphi(U) \times \mathbb{R}^n, \quad v \mapsto (x^1(p), \dots, x^n(p), c^1(v), \dots, c^n(v)) = ((x^1 \circ \pi)(v), \dots, (x^n \circ \pi)(v), c^1(v), \dots, c^n(v)). \tag{2.3}$$

Note that $\tilde{\varphi}$ has a (set-theoretic) inverse since

$$\tilde{\varphi}^{-1} : \varphi(U) \times \mathbb{R}^n \to TU, \quad (\varphi(p), c^1, \dots, c^n) \mapsto v = \sum_{i=1}^{n} c^i \frac{\partial}{\partial x^i}\Big|_p. \tag{2.4}$$

On $\varphi(U) \times \mathbb{R}^n \subseteq \mathbb{R}^{2n}$, we put the subspace topology. Using the bijection $\tilde{\varphi}$, we put a topology on TU; that is, $A \subseteq TU$ is open if and only if $\tilde{\varphi}(A) \subseteq \varphi(U) \times \mathbb{R}^n$ is open.

With $\mathcal{U} = \{(U_\alpha, \varphi_\alpha)\}$ is a maximal atlas of M, let $\mathcal{B} = \{A \subseteq TU_\alpha \mid A \text{ open and } \alpha \text{ is any index}\}$. We show that $\mathcal{B}$ is a basis for a topology on TM. The union being M is clear since

$$TM = \bigcup_{\alpha} TU_\alpha \subseteq \bigcup_{A \in \mathcal{B}} A \subseteq TM. \tag{2.5}$$

To show the second property of a basis, let $A \subseteq TU$ be open and $B \subseteq TV$ be open. Then $TU \cap TV = \pi^{-1}(U) \cap \pi^{-1}(V) = \pi^{-1}(U \cap V) = T(U \cap V)$ and so $A \cap B \subseteq T(U \cap V)$. Since $A \cap T(U \cap V)$ and $B \cap T(U \cap V)$ are both open in $T(U \cap V)$, $A \cap B = A \cap B \cap T(U \cap V)$ is open in $T(U \cap V)$.

Lemma 2.1. *TM is a second countable Hausdorff space.*

Proof. Let $v, w \in TM$; so that $v \in T_pM$ and $w \in T_qM$, or $\pi(v) = p$ and $\pi(w) = q$. Two cases arise. If $p \neq q$, then since M is Hausdorff, there exists open neighbourhoods $p \in U$ and $q \in V$ such that $U \cap V = \emptyset$. This simply implies $v \in \pi^{-1}(U) = TU$ and $w \in \pi^{-1}(V) = TV$, and $TU \cap TV = \emptyset$. For the case when $p = q$, both $v, w \in T_pM$. Since this T_pM identifies with $\mathbb{R}^n$, n being the dimension of M, we can pick open neighbourhoods $U_1, U_2 \subseteq T_pM$ of v and w, respectively, such that $U_1 \cap U_2 = \emptyset$. If we now pick a coordinate chart W of $p \in M$, then $W \times U_1$ and $W \times U_2$ are two open sets in TM that separate v and w, with $W \times U_1 \cap W \times U_2 = \emptyset$.

Let $\{B_i\}_{i \geq 1}$ be a countable basis of M, and let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ be an atlas. For $p \in U_\alpha$, there exists $p \in B_{p,\alpha} \subseteq U_\alpha$. Varying p and α, we get $\{B_{p,\alpha}\}_{p \in M,\, \alpha \in \Lambda} \subseteq \{B_i\}_{i \geq 1}$. So, there exists a countable basis of M consisting of coordinate charts for M. Since TU_i identifies with $\varphi_i(U_i) \times \mathbb{R}^n$ for any i, it is second countable and has a countable basis $\mathcal{B}_i$. Therefore, $\mathcal{B} = \bigcup_{i=1}^{\infty} \mathcal{B}_i$ is a countable basis of TM. ∎

Finally, we show that $\{(TU_\alpha, \tilde{\varphi}_\alpha)\}$ gives an atlas consisting of smooth compatible charts on TM. Let $(U, \varphi) = (U, x^1 \ldots, x^n)$ and $(V, \psi) = (V, y^1, \ldots, y^n)$ be two charts in M such that $U \cap V \neq \emptyset$. Let $p \in U \cap V$ with $v \in T_pM$. $\{\frac{\partial}{\partial x^1}\big|_p, \ldots, \frac{\partial}{\partial x^n}\big|_p\}$ and $\{\frac{\partial}{\partial y^1}\big|_p, \ldots, \frac{\partial}{\partial y^n}\big|_p\}$ are two bases for T_pM, so $v = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}\big|_p = \sum_{i=1}^n b^i \frac{\partial}{\partial y^i}\big|_p$. Applying y^i on both sides gives

$$b^i = \sum_{j=1}^{n} \frac{\partial y^i}{\partial x^j}(p) a^j. \tag{2.6}$$

This is a smooth transformation, showing that switching from one set of coordinates to another is smooth. Moreover, the TU_α's cover TM. This is a valid atlas.

As a consequence of this, the projection map $\pi : TM \to M$ is a smooth map, since $\varphi(U) \times \mathbb{R}^n \cong TU \xrightarrow{\pi} U \cong \varphi(U)$ takes the first n coordinates of $\varphi(U) \times \mathbb{R}^n$, which is a smooth map.

Definition 2.2. A *vector field* X on a smooth manifold M is a map $X : M \to TM$ such that $X(p) \in T_pM$ for all $p \in M$. In other words, a vector field is a smooth assignment of a tangent vector to each point in the manifold. It is termed smooth if the map X is smooth, meaning that for any chart (U, φ) on M, the composition $\tilde{\varphi} \circ X|_U : U \to \varphi(U) \times \mathbb{R}^n$ is smooth.

Such a vector field X is a section of the tangent bundle TM, as it assigns to each point $p \in M$ a vector in the corresponding tangent space T_pM. The projection map $\pi : TM \to M$ turns out to be the left inverse of X, since $\pi(X(p)) = p$ for all $p \in M$.

Definition 2.3. Let M be a smooth manifold of dimension n. A *smooth vector bundle* on M of rank r is a triple (E, M, π), where E is a smooth manifold of dimension $n + r$, and $\pi : E \to M$ is a smooth surjective map such that for each $p \in M$, the fiber $\pi^{-1}(p)$ is an r-dimensional real vector space. Additionally, π satisfies the local triviality condition: for every $p \in M$, there exists an open neighborhood U of p in M and a diffeomorphism $\phi_U : E_U := \pi^{-1}(U) \to U \times \mathbb{R}^r$ such that

$$\pi|_{E_U} = \mathrm{pr}_1 \circ \phi_U, \tag{2.7}$$

where $\mathrm{pr}_1 : U \times \mathbb{R}^r \to U$ is the projection onto the first factor, and $\varphi_U|_{\pi^{-1}(p)} : \pi^{-1}(p) \to \{p\} \times \mathbb{R}^r$ is a linear isomorphism of real vector spaces for each $p \in U$. Here, E is termed the *total space* and M the *base space* of the vector bundle.

Definition 2.4. A *morphism* between two smooth vector bundles (E, M, π) and (F, N, φ) is a pair of smooth maps $(\tilde{f}, f)$, where $\tilde{f} : E \to F$ and $f : M \to N$, such that $\varphi \tilde{f} = f\pi$, and for each $p \in M$, the map $\tilde{f}|_{\pi^{-1}(p)} : \pi^{-1}(p) \to \varphi^{-1}(f(p))$ is a linear map of real vector spaces.

As a special case, if $M = N$, and $f = \mathrm{id}_M$, then $\tilde{f}$ is called a *bundle map* over M. If $\tilde{f}$ is a diffeomorphism, then $(\tilde{f}, \mathrm{id}_M)$ is an isomorphism of vector bundles, and the two bundles are said to be isomorphic. In particular, if (E, M, π) is isomorphic to $(M \times \mathbb{R}^r, M, \mathrm{pr}_1)$, then (E, M, π) is called a *trivial vector bundle* of rank r.

2.1.1 Sections and Frames

September 2nd.

As earlier with a vector field, we have the following.

Definition 2.5. A map $s : M \to E$ is called a *section* of the vector bundle (E, M, π) if $\pi(s(p)) = p$ for all $p \in M$. In other words, a section assigns to each point in the base space a vector in the corresponding fiber. A section is termed smooth if the map s is smooth, meaning that for any chart (U, φ) on M, the composition $\phi_U \circ s|_U : U \to U \times \mathbb{R}^r$ is smooth.

We use $\Gamma(E) = \Gamma(M, E)$ to denote the set of all (global) smooth sections of the vector bundle (E, M, π). For any open subset $U \subseteq M$, we denote by $\Gamma(U, E)$ the set of all smooth sections of (E, M, π) over U; that is, smooth maps $s : U \to E$ such that $\pi(s(p)) = p$ for all $p \in U$. Moreover, for $s, t \in \Gamma(U, E)$, we define the sum $s + t \in \Gamma(U, E)$ by $(s + t)(p) = s(p) + t(p)$ for all $p \in U$; this sum is indeed in $\Gamma(U, E)$ since $\pi((s + t)(p)) = \pi(s(p) + t(p)) = \pi(s(p)) = p$, and $s + t$ can also be shown to be smooth. For $f \in C^\infty(U)$, we define the product $fs \in \Gamma(U, E)$ by $(fs)(p) = f(p)s(p)$ for all $p \in U$; this product is also in $\Gamma(U, E)$ since $\pi((fs)(p)) = \pi(f(p)s(p)) = \pi(s(p)) = p$, and fs can also be shown to be smooth. Therefore, $\Gamma(U, E)$ is a module over the ring $C^\infty(U)$, and $\Gamma(E)$ is a module over the ring $C^\infty(M)$.

Definition 2.6. A *frame* of a vector bundle (E, M, π) over an open subset $U \subseteq M$ is a collection of r sections $s_1, \ldots, s_r \in \Gamma(U, E)$ such that for each $p \in U$, the set $\{s_1(p), \ldots, s_r(p)\}$ forms a basis of the fiber $E_p = \pi^{-1}(p)$. The frame is termed smooth if each section s_i is smooth.

Consider the tangent bundle (TM, M, π) of a smooth manifold M of dimension n. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M. Then the collection of sections $s_i : U \to TM$ defined by $s_i(p) = \frac{\partial}{\partial x^i}\big|_p$ for $i = 1, \ldots, n$ forms a smooth frame of the tangent bundle over U.

Lemma 2.7. *Let (E, M, π) be a vector bundle, and let $U \subseteq M$ be an open subset. Suppose $s_1, \ldots, s_r$ is a smooth frame of E over U, and $s = \sum_{i=1}^r c^i s_i$. Then s is a smooth section of E over U if and only if each coefficient function $c^i : U \to \mathbb{R}$ is smooth.*

Proof. The "if" direction is clear from the definition of smoothness. For the "only if" direction, let s be a smooth section of E over U such that $s = \sum_{i=1}^r c^i s_i$. Let $p \in U$. We show that each c^i is smooth at p. Since $s_1, \ldots, s_r$ is a smooth frame of E over U, there exists an open neighborhood $V \subseteq U$ of p and a diffeomorphism $\phi_V : E_V \to V \times \mathbb{R}^r$ such that $\pi|_{E_V} = \mathrm{pr}_1 \circ \phi_V$, and $\phi_V|_{\pi^{-1}(q)} : \pi^{-1}(q) \to \{q\} \times \mathbb{R}^r$ is a linear isomorphism for each $q \in V$. Let $\phi_V(s(q)) = (q, (c^1(q), \ldots, c^r(q)))$ for all $q \in V$. Since $\phi_V \circ s|_V : V \to V \times \mathbb{R}^r$ is smooth, it follows that each component function $c^i : V \to \mathbb{R}$ is smooth. Therefore, each coefficient function c^i is smooth at p, and since p was arbitrary, each c^i is smooth on all of U. ∎

September 4th.

Definition 2.8. Let $f : N \to M$ be a smooth map between smooth manifolds, such that f is also injective and an immersion. Then $f(N)$, equipped with the topology borrowed from N, is termed an *immersed submanifold*.

In such a case, $f(N)$ is diffeomorphic to N. Note, however, that $f(N)$ need not be a manifold in the subspace topology of M; the subspace topology on $f(N)$ is weaker than the topology on $f(N)$ borrowed from N.

Definition 2.9. A smooth map $f : N \to M$ between smooth manifolds is called an *embedding* if f is injective and an immersion, and the subspace topology from M and the topology inherited from N are the same on $f(N)$.

Theorem 2.10. *If $S \subseteq M$ is a regular submanifold of M, then the inclusion map $i : S \to M$ is an embedding. Conversely, if $f : N \to M$ is an embedding, then $f(N)$ is a regular submanifold of M.*

Proof. Note that $i : S \to M$ is a smooth map, taking $(x^1, x^2, \ldots, x^s) \mapsto (x^1, x^2, \ldots, x^s, 0, \ldots, 0)$, is injective and an immersion. Also, the topologies trivially coincide. Hence, i is an embedding. For the converse, suppose $f : N \to M$ is an embedding. Let $p \in N$, with $f(p) \in f(N)$. Since f is an immersion (at p), there exist charts $(U, \varphi) = (U, x^1, \ldots, x^n)$, centred at p, and $(V, \psi) = (V, y^1, \ldots, y^m)$, centred at $f(p)$, such that $f(U) \subseteq V$, and f acts as $(x^1, \ldots, x^n) \to (x^1, \ldots, x^n, 0, \ldots, 0)$. Now, since $f(U) \subseteq f(N) \cap V$, we will change V to be the smallest open set $W \subseteq V \subseteq M$ such that $f(U) = f(N) \cap W$; since U is open in N if and only if $f(U)$ is open in $f(N)$ if and only if $f(U) = V' \cap f(N)$ where V' is open in M, we can set $W = V \cap V' \subseteq V$, so that $f(U) = W \cap f(N)$. ■

Theorem 2.11. *Let $f : N^n \to M^m$ be a smooth map between smooth manifolds. Suppose $S^s \subseteq M$ is a regular submanifold of M, and $f(N) \subseteq S$; this induces a set-theoretic factoring map $\tilde{f} : N \to S$ such that $i \circ \tilde{f} = f$. Then $\tilde{f}$ is a smooth map between smooth manifolds.*

Proof. With $p \in N$, and $f(p) \in S \subseteq M$, there exists a chart $(V, \psi) = (V, y^1, \ldots, y^m)$, centred at $f(p)$ in M, such that $S \cap V = \{(y^1, \ldots, y^m) \in V \mid y^{s+1} = \cdots = y^m = 0\}$. There also exits a chart $(U, \varphi) = (U, x^1, \ldots, x^n)$, centred at $p \in N$, such that $f(U) \subseteq V$ and f is given by $(x^1, \ldots, x^n) \mapsto (f^1, f^2, \ldots, f^m)$, where each f^i is smooth on U. But this means $f^{s+1}(x^1, \ldots, x^n) = \cdots = f^m(x^1, \ldots, x^n) = 0$ for all $(x^1, \ldots, x^n) \subseteq U$. In S, we have a smooth chart given by $(V \cap S, \varphi_S)$ via which, the map $\tilde{f}$ on U is given as $(x^1, \ldots, x^n) \mapsto (f^1, \ldots, f^s)$; $\tilde{f}$ is smooth. ■

2.2 Bump Functions and Partitions of Unity

September 18th.

Definition 2.12. A *bump function* is a smooth non-negative function f on a smooth manifold M such that f has compact support, that is, the closure of the set $\{q \in M \mid f(q) \neq 0\}$ is compact in M. A bump function is said to be adapted to an open set $U \subseteq M$ if the support of f is contained in U.

If $p \in U \subseteq M$ is an open set, then there exists a bump function f adapted to U such that $f(x) = 1$ in a neighbourhood of p contained in U.

Definition 2.13. A collection of subsets $\{A_\alpha\}_{\alpha \in \Lambda}$ of a topological space S is said to be *locally finite* if for all $p \in S$, there exists a neighbourhood V of P such that $\{\alpha \in \Lambda \mid A_\alpha \cap V \neq \emptyset\}$ is finite. In other words, each point in S has a neighbourhood that intersects only finitely many of the sets A_α.

Definition 2.14. A (smooth) *partition of unity* is a collection of smooth functions $\{\rho_\alpha : M \to \mathbb{R}\}_{\alpha \in \Lambda}$ on a smooth manifold M such that the following conditions hold:

1. $\{\operatorname{supp} \rho_\alpha\}_{\alpha \in \Lambda}$ is a locally finite collection of subsets of M.
2. $\sum_{\alpha \in \Lambda} \rho_\alpha(p) = 1$ for all $p \in M$.

Note that the sum in the second condition is well-defined since for each $p \in M$, there exists a neighbourhood V of p such that $\{\alpha \in \Lambda \mid \operatorname{supp} \rho_\alpha \cap V \neq \emptyset\}$ is finite, and so only finitely many terms in the sum are non-zero.

Theorem 2.15. *Let M be a smooth manifold, and let $\{U_\alpha\}_{\alpha\in\Lambda}$ be an open cover of M. Then the following hold true.*

1. *There exists a countable smooth partition of unity $\{\varphi_i\}_{i=1}^\infty$ on M such that for each i, $\operatorname{supp}(\varphi_i)$ is compact and contained in some U_α.*
2. *There exists a smooth partition of unity $\{\rho_\alpha\}_{\alpha\in\Lambda}$ on M such that for each α, $\operatorname{supp}(\rho_\alpha)$ is contained in U_α.*

In the second statement, we say that the partition of unity is *subordinate* to the open cover $\{U_\alpha\}_{\alpha\in\Lambda}$. Before we prove the theorem, we require a few results from topology.

Firstly, since M is a smooth manifold, it is second countable and has a countable basis $\{B_i\}_{i\geq 1}$. Define $\mathcal{S} = \{B_i \mid \overline{B_i} \text{ is compact}\}$. Then $\mathcal{S}$ is a countable basis of M consisting of open sets with compact closure. We shall show that $\mathcal{S}$ also happens to be a basis of M; to this end, let $p \in U \subseteq M$ be a point in an open set U. Moreover, let V be an open neighbourhood of p such that $\overline{V}$ is compact and $\overline{V} \subseteq U$. Since $\mathcal{B}$ is a basis of M, there exists $B_i \in \mathcal{B}$ such that $p \in B_i \subseteq V$. But then $\overline{B_i} \subseteq \overline{V}$ is compact, and so $B_i \in \mathcal{S}$. Therefore, $\mathcal{S}$ is a basis of M consisting of open sets with compact closure. Thus, any smooth manifold has a countable basis of open sets with compact closure.

Secondly, given any smooth manifold M, there exists a sequence of open sets $\{V_i\}_{i\geq 1}$ such that $\overline{V_i}$ is compact and $\overline{V_i} \subseteq V_{i+1}$ for all i, and $M = \bigcup_{i=1}^\infty V_i$. Indeed, let $\{B_i\}_{i=1}^\infty$ be a countable basis of M consisting of open sets with compact closure. Define $V_1 = B_1$, so that $V_1 \subseteq \overline{V}_1$ is compact. Thus, it must be contained in a finite union of the open cover $\{U_\alpha\}_{\alpha\in\Lambda}$; that is, there exists a finite set $A_1 \subseteq \Lambda$ such that $\overline{V}_1 \subseteq \bigcup_{\alpha\in A_1} U_\alpha$; for simplicity, suppose $\overline{V}_1 \subseteq B_1 \cup B_2 \cup \cdots \cup B_{i_1}$ for some $i_1 > 1$ such that i_1 is the smallest integer with this property. Now, let $V_2 = B_1 \cup B_2 \cup \cdots \cup B_{i_1}$, an open set with compact closure. Thus, $V_1 \subseteq \overline{V}_1 \subseteq V_2$, and $\overline{V}_2$ is compact. Repeating the same argument, we have $\overline{V}_2 \subseteq B_1 \cup B_2 \cup \cdots \cup B_{i_1} \cup \cdots \cup B_{i_2}$ for some $i_2 > i_1$ such that i_2 is the smallest integer with this property. Let $V_3 = B_1 \cup B_2 \cup \cdots \cup B_{i_2}$, an open set with compact closure. Thus, $V_2 \subseteq \overline{V}_2 \subseteq V_3$, and $\overline{V}_3$ is compact. Continuing in this manner, we get a sequence of open sets $\{V_i\}_{i=1}^\infty$ such that $\overline{V}_i$ is compact and $\overline{V}_i \subseteq V_{i+1}$ for all i, and $M = \bigcup_{i=1}^\infty V_i$. If we define $V_{-1} = V_0 = \emptyset$, we obtain a sequence as

$$V_{-1} \subseteq \overline{V}_{-1} \subseteq V_0 \subseteq \overline{V}_0 \subseteq V_1 \subseteq \overline{V}_1 \subseteq V_2 \subseteq \overline{V}_2 \subseteq V_3 \subseteq \cdots \tag{2.8}$$

From here, we can infer that

$$\overline{V}_{i+1} \setminus V_i \subseteq V_{i+2} \setminus \overline{V}_{i-1} \tag{2.9}$$

where $\overline{V}_{i+1} \setminus V_i$ is compact and $\overline{V}_{i-1} \subseteq V_{i+2}$ is open. We shall use ths construction in the proof of the theorem.

Proof. Start with the construction of the sequence of open sets $\{V_i\}_{i=-1}^\infty$ as above, where $V_{-1} = V_0 = \emptyset$, and $\overline{V}_i \subseteq V_{i+1}$ for all i. Let $p \in U_\alpha$ for some $\alpha \in \Lambda$. Then there exists i such that $p \in U_\alpha \cap (V_{i+2} - \overline{V}_{i-1})$. Since $U_\alpha \cap (V_{i+2} - \overline{V}_{i-1})$ is open, there exists a smooth bump function $\rho_p : M \to \mathbb{R}$ such that $\operatorname{supp}(\rho_p) \subseteq U_\alpha \cap (V_{i+2} - \overline{V}_{i-1})$ and $\rho_p > 0$ in some neighbourhood W_p of p. The collection $\{W_p\}_{p\in\overline{V}_{i+1}\setminus V_i}$ is then an open cover of the compact set $\overline{V}_{i+1} \setminus V_i$; refine this cover to a finite subcover consisting of $W_1, \ldots, W_m$, and let $\rho_1, \ldots, \rho_m$ be the corresponding bump functions. Repeating this for every i, we obtain, for each $\overline{V}_{i+1} \setminus V_i$, a finite cover $W_1^i, W_2^i, \ldots, W_{m(i)}^i$ of $\overline{V}_{i+1} \setminus V_i$, and corresponding bump functions $\rho_1^i, \rho_2^i, \ldots, \rho_{m(i)}^i$ such that $\operatorname{supp}(\rho_j^i) \subseteq U_\alpha \cap (V_{i+2} - \overline{V}_{i-1})$ for some $\alpha \in \Lambda$, and $\rho_j^i \equiv 1$ in W_j^i for all $j = 1, \ldots, m(i)$. Now, let $\{\rho_k\}_{k=1}^\infty$ be the countable collection of all these bump functions, and $\{W_j^i\}$ be the corresponding open sets. Note that the following properties hold:

1. $\operatorname{supp}(\rho_j^i)$ is compact for all i, j.
2. $\operatorname{supp}(\rho_j^i) \subseteq U_{\alpha_{ij}} \cap (V_{i+2} - \overline{V}_{i-1})$ for some $\alpha_{ij} \in \Lambda$.
3. $\overline{V}_{i+1} \setminus V_i \subseteq \bigcup_{j=1}^{m(i)} W_j^i$ for all i.
4. $\rho_j^i > 0$ on W_j^i for all i, j.

Since $\operatorname{supp}(\rho_j^i) \subseteq V_{i+2} \setminus \overline{V}_{i-1}$, we must have $\operatorname{supp}(\rho_j^i) \cap \overline{V}_{i-1} = \emptyset$ and $\operatorname{supp}(\rho_j^i) \cap (M \setminus V_{i+2}) = \emptyset$. Therefore, for each $p \in M$, there exists i such that $p \in V_{i+2} \setminus \overline{V}_{i-1}$, and so p can only belong to the support of bump

functions ρ_j^k for $k = i-1, i, i+1$. This shows that the collection $\{\text{supp}(\rho_j^i)\}$ is locally finite. Finally, we define

$$\varphi_k = \frac{\rho_k}{\sum_{j=1}^{\infty} \rho_j} \tag{2.10}$$

where the denominator is smooth and positive, and only finitely many terms are non-zero in a neighbourhood of any point. Then $\{\varphi_k\}_{k=1}^{\infty}$ is a countable smooth partition of unity on M such that for each k, $\text{supp}(\varphi_k)$ is compact and contained in some U_α. This proves the first statement. For the second statement, we can define $\rho_\alpha = \sum_{k \in I_\alpha} \varphi_k$, where $I_\alpha = \{k \mid \text{supp}(\varphi_k) \subseteq U_\alpha\}$. Then $\{\rho_\alpha\}_{\alpha \in \Lambda}$ is a smooth partition of unity on M subordinate to the open cover $\{U_\alpha\}_{\alpha \in \Lambda}$. ■

September 21st.

Lemma 2.16. *Let M be a smooth manifold, and X be a vector field on M. Then X is smooth on M if and only if for any chart (U, φ) on M, with $X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}$ on U, each coefficient function $a^i : U \to \mathbb{R}$ is smooth.*

Proof. Suppose X was smooth on M, or $X : M \to TM$ is a smooth map. Then for any chart (U, φ) on M, the composition $\tilde{\varphi} \circ X|_U : U \to \varphi(U) \times \mathbb{R}^n$ is smooth, and so each component function $a^i : U \to \mathbb{R}$ is smooth. Conversely, suppose that for any chart (U, φ) on M, with $X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}$ on U, each coefficient function $a^i : U \to \mathbb{R}$ is smooth. Then for any chart (U, φ) on M, the composition $\tilde{\varphi} \circ X|_U : U \to \varphi(U) \times \mathbb{R}^n$ is smooth, and so $X : M \to TM$ is a smooth map. ■

Proposition 2.17. *Let X be a vector field on a smooth manifold M. Then X is smooth if and only if for any smooth function $f \in C^\infty(M)$, the function $X(f) : M \to \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth.*

Proof. Suppose X is smooth. Then for any smooth function $f \in C^\infty(M)$, the function $X(f) : M \to \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth. Conversely, suppose that for any smooth function $f \in C^\infty(M)$, the function $X(f) : M \to \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth. Let (U, φ) be a chart on M, with $\varphi = (x^1, \ldots, x^n)$. Then on U, we have $X = \sum_{i=1}^n a^i \frac{\partial}{\partial x^i}$, where each coefficient function $a^i : U \to \mathbb{R}$ is given by $a^i(p) = X_p(x^i)$. Since each coordinate function x^i is smooth, it follows that each coefficient function $a^i : U \to \mathbb{R}$ is smooth. Therefore, by the previous lemma, X is smooth on M. ■

Moreover, since X_p is a $\mathbb{R}$-linear derivation from $C^\infty(M)$ to $\mathbb{R}$, we get that X is a derivation on $C^\infty(M)$; that is, for any $f, g \in C^\infty(M)$, we have

$$X(fg)(p) = X_p(fg) = f(p)X_p(g) + g(p)X_p(f) = (fX(g) + gX(f))(p) \implies X(fg) = fX(g) + gX(f). \tag{2.11}$$

2.3 Integral Curves

The following theorem will be useful in the study of vector fields and their flows on smooth manifolds.

Theorem 2.18. *Let $p_0 \in V \subseteq \mathbb{R}^n$ be an open set, and $f : V \to \mathbb{R}^n$ be a smooth map. Then the differential equation $\frac{dy}{dt} = f(y)$ with initial condition $y(0) = p_0$ has a unique smooth solution $y : (a(p_0), b(p_0)) \to V$ where $(a(p_0), b(p_0))$ is the maximal open interval containing 0 such that y is defined on $(a(p_0), b(p_0))$.*

Definition 2.19. Let X be a smooth vector field on a smooth manifold M, and let $p \in M$. An *integral curve* of X is a smooth curve $c : (a, b) \to M$ such that $c'(t) = X_{c(t)}$ for all $t \in (a, b)$. We usually presume $0 \in (a, b)$; if $c(0) = p$, we say that c is an integral curve of X starting at p.
A *maximal integral curve* is an integral curve whose domain cannot be extended to a larger interval while still satisfying the integral curve condition.

We look at a few examples.

Example 2.20. Consider the smooth manifold $M = \mathbb{R}^2$ and the smooth vector field $X = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}$. Let $p_0 = (1, 0) \in \mathbb{R}^2$. Then, an integral curve $c : (a, b) \to \mathbb{R}^2$ of X starting at p_0 satisfies the differential equation $c'(t) = (\dot{x}(t), \dot{y}(t)) = (-y(t), x(t)) = X_{c(t)}$ with initial condition $c(0) = (1, 0)$. Solving this system of differential equations, we get

$$c(t) = (\cos t, \sin t). \tag{2.12}$$

This integral curve is a circle centered at the origin with radius 1, and it is defined for all $t \in \mathbb{R}$. If, instead, we had chosen $p_0 = (x_0, y_0) \in \mathbb{R}^2$, then the integral curve of X starting at p_0 would be given by

$$c(t) = (x_0 \cos t - y_0 \sin t, x_0 \sin t + y_0 \cos t), \tag{2.13}$$

which is a circle centered at the origin with radius $\sqrt{x_0^2 + y_0^2}$, and it is also defined for all $t \in \mathbb{R}$. Moreover, one can verify that $c_s(c_t) = c_{s+t}$ for all $s, t \in \mathbb{R}$, where $c_t(p_0)$ denotes the integral curve of X starting at p_0 evaluated at time t. c_t is a diffeomorphism of $\mathbb{R}^2$ for each $t \in \mathbb{R}$, and the map

$$\mathbb{R} \to \mathrm{Diff}(\mathbb{R}^2), \quad t \mapsto c_t \tag{2.14}$$

is a smooth group homomorphism from $(\mathbb{R}, +)$ to the group of diffeomorphisms of $\mathbb{R}^2$, where the group operation on $\mathrm{Diff}(\mathbb{R}^2)$ is composition of maps.

From the above, any group homomorphism from $(\mathbb{R}, +)$ to the group of diffeomorphisms of a smooth manifold M is called a one-parameter group of diffeomorphisms of M.

Corollary 2.21. *Let X be a smooth vector field on a smooth manifold M, with $p \in M$. Then there exists an integral curve $c : (a, b) \to M$ of X such that $0 \in (a, b)$ with $c(0) = p$, and this integral curve is unique.*

Theorem 2.22. *Let $V \subseteq \mathbb{R}^n$ be open, and $f : V \to \mathbb{R}^n$ be a smooth map. Then for each point $p_0 \in V$, there exists a neighbourhood W of p_0 in V, an $\epsilon > 0$, and a smooth function $y : (-\epsilon, \epsilon) \times W \to V$ such that*

$$\frac{\partial y}{\partial t}(t, q) = f(y(t, q)), \qquad y(0, q) = q \tag{2.15}$$

for all $(t, q) \in (-\epsilon, \epsilon) \times W$.

Corollary 2.23. *Let X be a smooth vector field on a smooth manifold M, and let U be a chart around $p \in M$. Then there exists a neighbourhood W of p in U, an $\epsilon > 0$, and a smooth map $F : (-\epsilon, \epsilon) \times W \to U$ such that for all $q \in W$, $F(t, q)$ is an integral curve of X with initial point q ($F(0, q) = q$).*

In the above, we use the notation $F(t, q) = F_t(q)$. Suppose $s, t \in (-\epsilon, \epsilon)$, and $q \in W$, and that $F_t(F_s(q))$ and $F_{t+s}(q)$ are both defined. Then we have $F_t(F_s(q)) = F_{t+s}(q)$. This is because both sides are integral curves of X starting at q at time 0, and by uniqueness of integral curves, they must coincide. F, above, is termed as a *local flow* generated by the vector field X. $F_t(q)$ is then a flow line of q of the local flow F. If F can be defined as $F : \mathbb{R} \times M \to M$, then F is called a *global flow* generated by the vector field X. X is called a *complete vector field* if its flow is global.

Definition 2.24. In general, a *local flow* about a point $p \in U \subseteq M$ is a smooth map $F : (-\epsilon, \epsilon) \times W \to U$, where W is a neighbourhood of p in U and $\epsilon > 0$, such that

1. $F_0(q) = q$ for all $q \in W$, and
2. $F_t(F_s(q)) = F_{t+s}(q)$ whenever both sides are defined.

2.4 Lie Algebras and Lie Groups

September 23rd.

Recall that a vector field X on a smooth manifold M was smooth if and only if for any smooth function $f \in C^\infty(M)$, the function $Xf : M \to \mathbb{R}$ defined by $X(f)(p) = X_p(f)$ is smooth. That is, the map Xf itself is a smooth map on M; if $f \in C^\infty_p$, then $Xf \in C^\infty_p$ as well. If we have another smooth vector field Y on M, then we can define a new map $[X,Y](\cdot)(p) : C^\infty(M) \to \mathbb{R}$ by

$$f \mapsto X_p(Yf) - Y_p(Xf). \tag{2.16}$$

This map is $\mathbb{R}$-linear, and it is also a derivation on $C^\infty(M)$; that is, for any $f, g \in C^\infty(M)$, we have

$$\begin{aligned}
[X,Y](fg)(p) &= X_p(Y(fg)) - Y_p(X(fg)) = X_p(fYg + gYf) - Y_p(fXg + gXf) \\
&= f(p)X_p(Yg) + g(p)X_p(Yf) - f(p)Y_p(Xg) - g(p)Y_p(Xf) \\
&= f(p)(X_p(Yg) - Y_p(Xg)) + g(p)(X_p(Yf) - Y_p(Xf)) \\
&= f[X,Y](g)(p) + g[X,Y](f)(p).
\end{aligned} \tag{2.17}$$

Thus, $[X,Y](\cdot)(p)$ is a derivation on $C^\infty(M)$, and so there exists a unique tangent vector $[X,Y]_p \in T_pM$ such that $[X,Y]_p(f) = [X,Y](f)(p)$ for all $f \in C^\infty(M)$. This defines a new vector field $[X,Y]$ on M, defined as

$$XY - YX = [X,Y] : M \to TM, \quad p \mapsto [X,Y]_p. \tag{2.18}$$

This is called the Lie bracket of the vector fields X and Y, and is smooth when X and Y are smooth vector fields on M. $\mathfrak{X}(U)$ then, the set of all smooth vector fields on an open set $U \subseteq M$, is termed a Lie algebra.

Definition 2.25. Let k be a field and V be a vector space over k. V is termed a *Lie algebra* over k when equipped with an operation $[\cdot,\cdot] : V \times V \to V$ called the Lie bracket, such that the following conditions hold:

1. bilinearity: $[av_1 + bv_2, w] = a[v_1, w] + b[v_2, w]$ and $[v, aw_1 + bw_2] = a[v, w_1] + b[v, w_2]$ for all $v, v_1, v_2, w, w_1, w_2 \in V$ and $a, b \in k$,
2. anti-commutativity: $[v, w] = -[w, v]$ for all $v, w \in V$, and
3. Jacobi identity: $[v, [w, u]] + [w, [u, v]] + [u, [v, w]] = 0$ for all $v, w, u \in V$.

Verify that the set of smooth vector fields $\mathfrak{X}(U)$ on an open set $U \subseteq M$ forms a Lie algebra over $\mathbb{R}$ with the Lie bracket $[\cdot,\cdot] : \mathfrak{X}(U) \times \mathfrak{X}(U) \to \mathfrak{X}(U)$ defined above as $[X,Y] = XY - YX$ for all $X, Y \in \mathfrak{X}(U)$.

Any vector V over the field k can be viewed as a Lie algebra over k with the trivial Lie bracket $[v, w] = 0$ for all $v, w \in V$. This is also the only abelian Lie algebra over k; that is, a Lie algebra V over k is abelian if and only if $[v, w] = 0$ for all $v, w \in V$. The vector space $M_n(k)$ of all $n \times n$ matrices over a field k is a Lie algebra over k with the Lie bracket defined as $[A, B] = AB - BA$ for all $A, B \in M_n(k)$.

Definition 2.26. Let $F : N \to M$ be a smooth map between smooth manifolds, and let $p \in N$ with the induced linear map $F_{*,p} : T_pN \to T_{F(p)}M$. If X is a vector field on N and Y is a vector field on M, we say that X and Y are *F-related* if $F_{*,p}(X_p) = Y_{F(p)}$ for all $p \in N$. Y is called the pushforward of X by F.

Lemma 2.27. *With notation as above, X and Y are F-related if and only if for any smooth function $g \in C^\infty(M)$, we have $X(g \circ F) = (Yg) \circ F$.*

Proof. Suppose X and Y are F-related. Then for any smooth function $g \in C^\infty(M)$, we have

$$X(g \circ F) = F_{*,p}(X_p)(g) = Y_{F(p)}(g) = (Yg) \circ F.$$

Conversely, suppose that for any smooth function $g \in C^\infty(M)$, we have $X(g \circ F) = (Yg) \circ F$. Then for any smooth function $g \in C^\infty(M)$, we have

$$F_{*,p}(X_p)(g) = X(g \circ F) = (Yg) \circ F = Y_{F(p)}(g).$$

Thus, $F_{*,p}(X_p) = Y_{F(p)}$ for all $p \in N$, and so X and Y are F-related. ■

Lemma 2.28. *With notation as above, if X_1, X_2 are vector fields on N and Y_1, Y_2 are vector fields on M such that X_i and Y_i are F-related for $i = 1, 2$, then the Lie brackets $[X_1, X_2]$ and $[Y_1, Y_2]$ are also F-related.*

Proof. Suppose X_1, X_2 are vector fields on N and Y_1, Y_2 are vector fields on M such that X_i and Y_i are F-related for $i = 1, 2$. Then for any smooth function $g \in C^\infty(M)$, we have

$$\begin{aligned}
[X_1, X_2](g \circ F) &= X_1(X_2(g \circ F)) - X_2(X_1(g \circ F)) \\
&= X_1((Y_2 g) \circ F) - X_2((Y_1 g) \circ F) \\
&= (Y_1(Y_2 g)) \circ F - (Y_2(Y_1 g)) \circ F \\
&= ([Y_1, Y_2]g) \circ F.
\end{aligned}$$

Thus, by the previous lemma, we have that $[X_1, X_2]$ and $[Y_1, Y_2]$ are F-related. ■

September 25th.

Definition 2.29. A *Lie group* G is both a group and a smooth manifold such that the maps

$$G \times G \to G, \ \ (g, h) \mapsto gh, \qquad G \to G, \ \ g \mapsto g^{-1}. \tag{2.19}$$

are smooth maps.

For example, both $GL_n(\mathbb{R})$ and $SL_n(\mathbb{R})$ are Lie groups. Similarly, $O(n)$ is also a Lie group. $(\mathbb{R}^n, +)$ is also a Lie group. If G and H are both Lie groups, then so is $G \times H$.

Definition 2.30. If G and H are Lie groups, a *map of Lie groups* $f : G \to H$ is a map that is both a homomorphism of groups and a smooth map between smooth manifolds.

The determinant $\det : GL_n(\mathbb{R}) \to \mathbb{R}^*$ is a map of Lie groups.

Definition 2.31. Let G be a Lie group. A subset $H \subseteq G$ is termed a *Lie subgroup* if H is a subgroup and an immersed manifold of G, and the product and inverse operations restricted to H remain smooth.

Example 2.32. We look at the space $S^1 \times S^1 = \mathbb{R}^2/\mathbb{Z}^2$. Let f be the natural quotient map $f : \mathbb{R}^2 \to \mathbb{R}^2/\mathbb{Z}^2$. Look at the map $g_\alpha : \mathbb{R} \to \mathbb{R}^2$ defined as $t \mapsto (t, \alpha t)$ for some $\alpha \in \mathbb{R}$. Let $L_\alpha = g_\alpha(\mathbb{R}) = \{(t, \alpha t) \mid t \in \mathbb{R}\}$ be the image line, and now set $H_\alpha = f(L_\alpha) = f \circ g_\alpha(\mathbb{R})$. We claim that $f \circ g_\alpha$ is a one-to-one immersion, and further that H_α is a closed curve if and only if α is rational. The map f may be explicitly written as $(s, t) \mapsto (e^{2\pi i s}, e^{2\pi i t})$. Note that it is enough to prove that $f^{-1}(H_\alpha) = \{(t + n, \alpha t + m) \mid t \in \mathbb{R}, n, m \in \mathbb{Z}\}$ is dense in $\mathbb{R}^2$ if and only if α is irrational. If $\alpha = \frac{p}{q}$ is rational, then $f^{-1}(H_\alpha) = \{(t + n, \frac{p}{q}t + m) \mid t \in \mathbb{R}, n, m \in \mathbb{Z}\}$. Then for any $t \in \mathbb{R}$, we have $(t+n, \frac{p}{q}t+m) = (t+n, \frac{p}{q}(t+qn)+(m-pn))$, and so $f^{-1}(H_\alpha) = \{(t+n, \frac{p}{q}t+m) \mid t \in [0, q), n, m \in \mathbb{Z}\}$. This is a union of q lines in $\mathbb{R}^2$, and so it cannot be dense in $\mathbb{R}^2$. Conversely, if α is irrational, then for any $(x, y) \in \mathbb{R}^2$ and $\epsilon > 0$, we can find integers n, m such that $|x - n| < \epsilon$ and $|y - m - \alpha n| < \epsilon$. This shows that $f^{-1}(H_\alpha)$ is dense in $\mathbb{R}^2$ if and only if α is irrational.

2.4.1 Matrix Lie Groups and Lie Algebras

September 28th.

Definition 2.33. For a matrix $A \in M_n(\mathbb{R})$, we define the exponential map $\exp : M_n(\mathbb{R}) \to GL_n(\mathbb{R})$ as

$$\exp(A) = e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots \tag{2.20}$$

To show that the exponential map is well-defined and lands in $GL_n(\mathbb{R})$, we need to show that the series converges and that the resulting matrix is invertible. For $X \in M_n(\mathbb{R})$, define the norm $\|X\| = \sqrt{\sum_{i,j=1}^n x_{ij}^2}$; this norm is also sub-multiplicative, meaning that $\|XY\| \leq \|X\| \, \|Y\|$ for all $X, Y \in M_n(\mathbb{R})$. Then, looking at the partial sums of the series, we have

$$\left\| \sum_{k=0}^{N} \frac{A^k}{k!} \right\| \leq \sum_{k=0}^{N} \frac{\|A\|^k}{k!} \leq \|I\| + \sum_{k=1}^{\infty} \frac{\|A\|^k}{k!} = e^{\|A\|} + \sqrt{n} - 1 \tag{2.21}$$

which implies that the series converges absolutely, and hence converges. We also state a few properties of the trace map $\operatorname{tr} : M_n(\mathbb{R}) \to \mathbb{R}$ defined as $\operatorname{tr}(A) = \sum_{i=1}^n a_{ii}$ for $A = (a_{ij}) \in M_n(\mathbb{R})$; in particular, the following properties hold:

1. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for all $A, B \in M_n(\mathbb{R})$.
2. If $A \in GL_n(\mathbb{R})$, then $\operatorname{tr}(AXA^{-1}) = \operatorname{tr}(X)$ for all $X \in M_n(\mathbb{R})$.
3. For $A \in M_n(\mathbb{R})$, $\operatorname{tr}(A) = \sum_{i=1}^n \lambda_i$, where $\lambda_1, \ldots, \lambda_n$ are the (complex) eigenvalues of A.
4. For all $A \in M_n(\mathbb{R})$, there exists $B \in GL_n(\mathbb{C})$ such that BAB^{-1} is upper triangular, and has the eigenvalues of A on the diagonal. In particular, $\det(A) = \prod_{i=1}^n \lambda_i$, where $\lambda_1, \ldots, \lambda_n$ are the (complex) eigenvalues of A.

Proposition 2.34. *For $A \in M_n(\mathbb{R})$, $\det e^A = e^{\operatorname{tr}(A)}$.*

Proof. Without the loss of generality, we may assume that A is upper triangular, with eigenvalues $\lambda_1, \ldots, \lambda_n$ on the diagonal (if not, we can find $B \in GL_n(\mathbb{C})$ such that BAB^{-1} is upper triangular, and $\det e^A = \det e^{BAB^{-1}}$ and $\operatorname{tr}(A) = \operatorname{tr}(BAB^{-1})$). Then e^A is also upper triangular, with eigenvalues $e^{\lambda_1}, \ldots, e^{\lambda_n}$ on the diagonal. Therefore, we have

$$\det e^A = \prod_{i=1}^{n} e^{\lambda_i} = e^{\sum_{i=1}^n \lambda_i} = e^{\operatorname{tr}(A)}. \tag{2.22}$$

■

In particular, if $A \in M_n(\mathbb{R})$, then $e^A \in GL_n(\mathbb{R})$, and so the exponential map $\exp : M_n(\mathbb{R}) \to GL_n(\mathbb{R})$ is well-defined. We use this exponential map with smooth curves. Define $c : \mathbb{R} \to GL_n(\mathbb{R}) \subseteq M_n(\mathbb{R})$ as $c(t) = e^{tA}$ for some $A \in M_n(\mathbb{R})$. Then c is a smooth curve in $GL_n(\mathbb{R})$, with $c(0) = I$ and $c'(0) = A$. In fact, the derivative is given as

$$c'(t) = \frac{d}{dt} e^{tA} = \frac{d}{dt} \left(\sum_{k=0}^{\infty} \frac{t^k A^k}{k!} \right) = \sum_{k=1}^{\infty} \frac{k t^{k-1} A^k}{k!} = \sum_{k=1}^{\infty} \frac{t^{k-1} A^k}{(k-1)!} = A \sum_{k=0}^{\infty} \frac{t^k A^k}{k!} = A e^{tA}. \tag{2.23}$$

Now, let G be a Lie group, and let $g \in G$. Define the left translation map $\ell_g : G \to G$ as $\ell_g(h) = gh$ for all $h \in G$. Then ℓ_g is a diffeomorphism of G, with inverse $\ell_{g^{-1}}$. Let $G = GL_n(\mathbb{R})$. Then the induced linear map at the identity is

$$(\ell_g)_{*,I} : T_I(GL_n(\mathbb{R})) \to T_g(GL_n(\mathbb{R})) \tag{2.24}$$

With $c : \mathbb{R} \to GL_n(\mathbb{R})$ a smooth curve with $c(0) = I$ and $c'(0) = A$, we have

$$(\ell_g)_{*,I}(A) = (\ell_g \circ c)'(0) = (gc)'(0) = gc'(0) = gA. \tag{2.25}$$

The exact same map is induced. If we instead look at the map $\det : GL_n(\mathbb{R}) \to \mathbb{R}^*$, then the induced linear map at the identity is

$$(\det)_{*,I} : T_I(GL_n(\mathbb{R})) \to T_1(\mathbb{R}^*) \tag{2.26}$$

and repeating the above, we have

$$(\det)_{*,I}(A) = (\det \circ c)'(0) = \frac{d}{dt}\det(c(t))\bigg|_{t=0} = \frac{d}{dt}\det(e^{tA})\bigg|_{t=0} = \frac{d}{dt}e^{\mathrm{tr}(tA)}\bigg|_{t=0} = \mathrm{tr}(A). \tag{2.27}$$

That is, the induced linear map $(\det)_{*,I} : T_I(GL_n(\mathbb{R})) \to T_1(\mathbb{R}^*)$ is given as $A \mapsto \mathrm{tr}(A)$. Now, pre-add the inclusion map $i : SL_n(\mathbb{R}) \to GL_n(\mathbb{R})$, and consider the composition $\det \circ i : SL_n(\mathbb{R}) \to \mathbb{R}^*$. Then the induced linear map at the identity is

$$(\det \circ i)_{*,I} : T_I(SL_n(\mathbb{R})) \to T_1(\mathbb{R}^*). \tag{2.28}$$

But with the chain rule for induced linear maps, we have

$$(\det \circ i)_{*,I} = (\det)_{*,I} \circ (i)_{*,I} : T_I(SL_n(\mathbb{R})) \to T_1(\mathbb{R}^*), \qquad A \mapsto \mathrm{tr}(A) = 0. \tag{2.29}$$

That is, the induced linear map $(\det \circ i)_{*,I} : T_I(SL_n(\mathbb{R})) \to T_1(\mathbb{R}^*)$ is the zero map. This shows that $T_I(SL_n(\mathbb{R})) = \ker((\det \circ i)_{*,I}) = \{A \in M_n(\mathbb{R}) \mid \mathrm{tr}(A) = 0\}$, which is a vector space of dimension $n^2 - 1$. From here, we define

$$\mathfrak{sl}_n(\mathbb{R}) := T_I(SL_n(\mathbb{R})) = \{A \in M_n(\mathbb{R}) \mid \mathrm{tr}(A) = 0\}, \tag{2.30}$$

the space of traceless matrices. Moving on, we look at the Lie algebra of $O(n)$. Let $G = O(n) \subseteq GL_n(\mathbb{R})$ and consider the map $f : GL_n(\mathbb{R}) \to GL_n(\mathbb{R})$ defined as $f(A) = A^T A$. Then $O(n) = f^{-1}(I)$, and so the induced linear map at the identity is

$$f_{*,I} : T_I(GL_n(\mathbb{R})) \to T_I(GL_n(\mathbb{R})). \tag{2.31}$$

Pre-composing the smooth curve $c : \mathbb{R} \to GL_n(\mathbb{R})$ defined as $c(t) = e^{tA}$, we have

$$f_{*,I}(A) = (f \circ c)'(0) = (f(e^{tA}))'(0) = (e^{tA^T}e^{tA})'(0) = (e^{t(A^T+A)})'(0). \tag{2.32}$$

Thus, we have $T_I(O(n)) = \ker(f_{*,I}) = \{A \in M_n(\mathbb{R}) \mid A^T + A = 0\}$, which is a vector space of dimension $\frac{n(n-1)}{2}$. From here, we define

$$\mathfrak{so}_n(\mathbb{R}) := T_I(O(n)) = \{A \in M_n(\mathbb{R}) \mid A^T + A = 0\}, \tag{2.33}$$

the space of skew-symmetric matrices. We also denote $\mathfrak{gl}_n(\mathbb{R}) := T_I(GL_n(\mathbb{R})) = M_n(\mathbb{R})$, the space of all $n \times n$ matrices. These structures are all Lie algebras (subalgebras of $\mathfrak{gl}_n(\mathbb{R})$) with the Lie bracket defined as $[A, B] = AB - BA$ for all $A, B \in \mathfrak{gl}_n(\mathbb{R})$. In particular, we have the inclusions

$$\mathfrak{so}_n(\mathbb{R}) \subseteq \mathfrak{sl}_n(\mathbb{R}) \subseteq \mathfrak{gl}_n(\mathbb{R}). \tag{2.34}$$

It is left as an exercise to show that the Lie bracket is closed on $\mathfrak{so}_n(\mathbb{R})$ and $\mathfrak{sl}_n(\mathbb{R})$.

One may show that if $AB = BA$, then $e^{A+B} = e^A e^B$, and $(e^A)^{-1} = e^{-A}$.

Definition 2.35. Let G be a Lie group, and X be a (not necessarily smooth) vector field on G. X, then, is called a *left-invariant vector field* on G if $(\ell_g)_* X = X$ for all $g \in G$, or

$$(\ell_g)_* X_h = X_{\ell_g(h)} = X_{gh} \tag{2.35}$$

for any $g, h \in G$. That is, X is left-invariant if it is invariant under the pushforward of left translation by any element of G.

If we pick the identity $e \in G$, the above property of the vector field X boils down to the property that $X_g = (\ell_g)_* X_e$ for all $g \in G$. That is, a left-invariant vector field is completely determined by its value at the identity.

Proposition 2.36. *Any left-invariant vector field on a Lie group G is smooth.*

Proof. Let X be a left-invariant vector field on a Lie group G. Note that it is enough to prove that if $f \in C^\infty(G)$, then $Xf \in C^\infty(G)$. Let $g \in G$, and let $c : \mathbb{R} \to G$ be a smooth curve with $c(0) = e$ and $c'(0) = X_e$. Then, we have

$$(Xf)(g) = X_g(f) = (\ell_g)_* X_e(f) = X_e(f \circ \ell_g) = \frac{d}{dt}(f \circ \ell_g \circ c(t))\bigg|_{t=0}. \tag{2.36}$$

The map $(g, t) \mapsto f \circ \ell_g \circ c(t)$ is smooth, and so the map $g \mapsto (Xf)(g)$ is smooth as well. Thus, $Xf \in C^\infty(G)$, and so X is a smooth vector field on G. ∎

Suppose X and Y are two left-invariant vector fields on a Lie group G. That means, X and X, and Y and Y are ℓ_g-related for all $g \in G$. Thus, by the result above, $[X, Y]$ and $[X, Y]$ must be ℓ_g-related, which tells us $[X, Y]$ is a left-invariant vector field on G. Thus, we have, from before,

$$\mathfrak{X}(G) = \{X : G \to TG \mid X \text{ is smooth}\}, \tag{2.37}$$

the set of all smooth vector fields on G along with the Lie bracket $[X, Y] = XY - YX$ is a Lie algebra, and

$$L(G) := \{X : G \to TG \mid X \text{ is left-invariant}\} \subseteq \mathfrak{X}(G) \tag{2.38}$$

is a Lie subalgebra of $\mathfrak{X}(G)$. That is, there is a bijection between left-invariant vector fields on G and the tangent space at the identity T_eG. In particular, we have

$$L(G) \cong T_eG, \qquad X \mapsto X_e \tag{2.39}$$

where the inverse map is given as $v \mapsto X_v$, where X_v is the unique left-invariant vector field on G such that $X_{v,e} = v$. Thus, we have that $L(G)$ is a finite-dimensional Lie algebra over $\mathbb{R}$, with dimension $\dim L(G) = \dim T_eG = \dim G$. This bijection is as $\mathbb{R}$-linear spaces, and the Lie bracket on $L(G)$ is given as $[X_v, X_w] = X_{[v,w]}$ for all $v, w \in T_eG$. The Lie algebra $L(G)$ is called the Lie algebra of the Lie group G. In particular, we have that $L(GL_n(\mathbb{R})) = \mathfrak{gl}_n(\mathbb{R})$, $L(SL_n(\mathbb{R})) = \mathfrak{sl}_n(\mathbb{R})$, and $L(O(n)) = \mathfrak{so}_n(\mathbb{R})$.

September 30th.

We state a few facts, summarising the above discussion.

1. Any left-invariant vector field X on a Lie group G is smooth.
2. If X annd Y are two left-invariant vector fields on G, then $[X, Y]$ is also a left-invariance vector field.
3. $L(G)$, the set of all (smooth) left-invariant vector fields on G, is a Lie subalgebra of $\mathfrak{X}(G)$, the Lie algebra of all smooth vector fields on G.
4. There is an $\mathbb{R}$-linear isomorphism $T_eG \cong L(G)$; specifically, the map $X \mapsto X_e$ for $X \in L(G)$ and the map $A \mapsto \tilde{A}$ for $A \in T_eG$, define as $(\tilde{A})_h = (\ell_h)_* A$ for all $h \in G$, are inverses of each other.
5. T_eG becomes a Lie algebra via the above isomorphism, with the Lie bracket defined as $[A, B] = [\tilde{A}, \tilde{B}]_e$ for all $A, B \in T_eG$.

For $F : G \to H$ is a homomorphism of Lie groups, define $F_* : L(G) \to L(H)$ as $F_*X = \widetilde{F_{*,e}(X_e)}$ for all $X \in L(G)$. Then F_* is a homomorphism of Lie algebras. F_* is termed the pushforward of a left-invariant vector field X on G by the homomorphism F.

Proposition 2.37. *Let $F : G \to H$ be a homomorphism of Lie groups. Let $X \in L(G)$ and $F_*X \in L(H)$ be the pushforward of X by F. Then X and F_*X are F-related.*

Proof. We are to show that $F_{*,g}(X_g) = (F_*X)_{F(g)}$ for all $g \in G$. We have, via $X_g = (\ell_g)_{*,e}(X_e)$,

$$\begin{aligned} F_{*,g}((\ell_g)_{*,e}(X_e)) &= (F \circ \ell_g)_{*,e}(X_e) = (\ell_{F(g)} \circ F)_{*,e}(X_e) \\ &= (\ell_{F(g)})_{*,F(e)}(F_{*,e}(X_e)) = (\ell_{F(g)})_{*,F(e)}((F_*X)_{F(e)}) = (F_*X)_{F(g)}. \end{aligned} \tag{2.40}$$

∎

Proposition 2.38. *Let $F : G \to H$ be a homormophism of Lie groups, and set $\mathfrak{g} = T_eG$ and $\mathfrak{h} = T_eH$, their associated Lie algebras. Then the induced linear map $F_{*,e} : \mathfrak{g} \to \mathfrak{h}$ is a homomorphism of Lie algebras.*

Proof. Let $A, B \in \mathfrak{g}$. Since $\tilde{A}$ and $F_*\tilde{A}$, and $\tilde{B}$ and $F_*\tilde{B}$ are F-related, we have that $[\tilde{A}, \tilde{B}]$ and $[F_*\tilde{A}, F_*\tilde{B}]$ are also F-related. Thus, we have

$$F_{*,e}([A, B]) = F_{*,e}([\tilde{A}, \tilde{B}]_e) = [F_*\tilde{A}, F_*\tilde{B}]_{F(e)} = [F_{*,e}(A), F_{*,e}(B)]. \tag{2.41}$$

■

Theorem 2.39. *Let G be a Lie group, and $\mathfrak{g} = T_eG$ be its associated Lie algebra. Let $\mathfrak{h} \subseteq \mathfrak{g}$ be a Lie subalgebra. Then there exists a unique connected Lie subgroup $H \subseteq G$ such that $T_eH = \mathfrak{h}$.*

Corollary 2.40. *There exists a bijective correspondence between connected Lie subgroups of a Lie group G and Lie subalgebras of its associated Lie algebra $\mathfrak{g} = T_eG$.*

Theorem 2.41. *Let H be a Lie subgroup of a Lie group G. Then H is a closed subgroup of G if and only if H is a regular submanifold of G.*

Theorem 2.42. *Let G and H be Lie groups with associated Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Suppose G is simply connected. Then, given a Lie algebra homomorphism $\psi : \mathfrak{g} \to \mathfrak{h}$, there exists a unique homomorphism of Lie groups $\varphi : G \to H$ such that $\varphi_{*,e} = \psi$.*

Corollary 2.43. *If G and H are simply connected Lie groups with isomorphic associated Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, then G and H are isomorphic as Lie groups.*

Corollary 2.44. *There exits a bijective correspondence between isomorphism classes of Lie algebras and isomorphism classes of simply connected Lie groups.*

Theorem 2.45. *Let G and H be Lie groups. Then any continuous Lie group homomorphism $\varphi : G \to H$ is smooth.*

Corollary 2.46. *A second countable Hausdorff locally Euclidean topological group G has at most one differentiable structure making it into a Lie group.*

Theorem 2.47. *Let G be a lie group, and H be a closed (not necessarily Lie) subgroup of G. Then H has a unique manifold structure making it into a Lie subgroup of G.*

Theorem 2.48. *Let $\psi : G \to H$ be a homomorphism of Lie groups, and set $A = \ker \psi$ and $\mathfrak{A} = \ker \psi_{*,e}$, where $\psi_{*,e} : \mathfrak{g} \to \mathfrak{h}$ is the induced linear map between the associated Lie algebras. Then A is a closed Lie subgroup of G whose associated Lie algebra is $\mathfrak{A}$.*

Chapter 3

FORMS, ORIENTATION, AND INTEGRATION

3.1 Differential Forms

October 5th.

We translate our theory of differential forms on $\mathbb{R}^n$ onto smooth manifolds. A 1-form on a manifold M is section of the cotangent bundle T^*M. That is, a 1-form ω on M is a map $\omega : M \to T^*M = \bigsqcup_{p\in M} T_p^*M$ such that $\omega(p) \in T_p^*M$ for all $p \in M$. This is analagous to a vector field X which is a map $X : M \to TM = \bigsqcup_{p\in M} T_pM$ such that $X(p) \in T_pM$ for all $p \in M$. Given a smooth function $f : M \to \mathbb{R}$, we have the differential $df : M \to T^*M$ defined as $df(p) = df_p$, where $df_p : T_pM \to \mathbb{R}$ is the linear map defined as $df_p(X_p) = X_p(f)$ for all $X_p \in T_pM$. In particular, we have that df is a 1-form on M.

Moreover, given a chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on M, the x^i's can, again, be thought of as coordinate functions. With $\{\frac{\partial}{\partial x^i}\}_{i=1}^n$ the standard basis for T_pM, p being in U, the partials $\{(dx^i)_p\}$ forms the respective dual basis for T_p^*M. Thus, any $\omega_p \in T_p^*M$ can be written as

$$\omega_p = \sum_{i=1}^{n} a_i(p)(dx^i)_p \tag{3.1}$$

where $a_i(p)$'s are real numbers. More generally, any 1-form ω on U can be written as

$$\omega = \sum_{i=1}^{n} a_i dx^i \tag{3.2}$$

where $a_i : U \to \mathbb{R}$ are real-valued functions on U.

Definition 3.1. The *cotangent bundle* of a smooth manifold M is the vector bundle $T^*M = \bigsqcup_{p\in M} T_p^*M$ over M, with projection $\pi : T^*M \to M$ defined as $\pi(\omega_p) = p$ for all $\omega_p \in T_p^*M$. A *1-form* on M is a section of the cotangent bundle, that is, a map $\omega : M \to T^*M$ such that $\omega(p) \in T_p^*M$ for all $p \in M$.

We can, again, put a topology on T^*M by using the charts on M, just as we did with the tangent bundle TM. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M. Also adopt the notation $T^*U = \pi^{-1}(U) = \bigsqcup_{p\in U} T_p^*M$. Then we have the bijective map

$$\tilde{\varphi} : T^*U \to \varphi(U) \times \mathbb{R}^n, \qquad \omega_p \mapsto (\varphi(p), a_1(p), \dots, a_n(p)). \tag{3.3}$$

With the standard topology on $\varphi(U) \times \mathbb{R}^n$, we can give T^*U the topology induced by $\tilde{\varphi}$. Then, letting $\mathcal{B}_U$ be the collection of all open sets in T^*U, we can give T^*M the topology generated by $\mathcal{B} = \bigcup_{(U,\varphi)} \mathcal{B}_U$, where the union is taken over all charts (U, φ) on M. The exact same analysis as tangent bundles shows that T^*M is a smooth manifold of dimension $2n$, and the projection $\pi : T^*M \to M$ is a smooth map.

A 1-form ω on M is said to be smooth if ω is a smooth section of π. That is, ω is smooth if $\omega : M \to T^*M$ is a smooth map between smooth manifolds.

Proposition 3.2. *Let ω be a 1-form on a smooth manifold M. Then the following are equivalent.*

1. *ω is smooth.*
2. *For any chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on M, the functions $a_i : U \to \mathbb{R}$ defined as $\omega = \sum_{i=1}^n a_i dx^i$ are smooth.*

Let X be a vector field on a smooth manifold M, and let ω be a 1-form on M. Then we can define the map $\omega(X) : M \to \mathbb{R}$ as $\omega(X)(p) = \omega_p(X_p)$ for all $p \in M$. That is, $\omega(X)$ is the function that takes a point $p \in M$ and returns the value of the linear map $\omega_p : T_pM \to \mathbb{R}$ evaluated at the vector $X_p \in T_pM$.

Proposition 3.3. *Let ω be a 1-form on a smooth manifold M. Then ω is a smooth 1-form if and only if $\omega(X)$ is a smooth function on M for all smooth vector fields X on M.*

Proof. Let ω be a smooth 1-form on M. Let X be a smooth vector field on M. Then the map $X : M \to TM$ is smooth, and so the composition $\omega(X) = \omega \circ X : M \to \mathbb{R}$ is smooth. Conversely, suppose that $\omega(X)$ is a smooth function on M for all smooth vector fields X on M. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M. Then the partials $\{\frac{\partial}{\partial x^i}\}_{i=1}^n$ form a basis for the tangent space T_pM for all $p \in U$, and so we can write

$$\omega = \sum_{i=1}^n a_i dx^i \tag{3.4}$$

for some functions $a_i : U \to \mathbb{R}$. Then we have

$$a_i = dx^i \left(\frac{\partial}{\partial x^i} \right) = \omega \left(\frac{\partial}{\partial x^i} \right) \tag{3.5}$$

for all $i = 1, \dots, n$. Since $\frac{\partial}{\partial x^i}$ is a smooth vector field on U, we have that $\omega\left(\frac{\partial}{\partial x^i}\right)$ is a smooth function on U, and so each of the functions a_i are smooth. Thus, ω is a smooth 1-form on M. ■

Recall that $\mathfrak{X}(M)$ is the real vector space of all smooth vector fields on a smooth manifold M, which is also a $C^\infty(M)$-module via $(fX)(p) = f(p)X_p$ for all $f \in C^\infty(M)$, $X \in \mathfrak{X}(M)$, and $p \in M$. Similarly, we have the real vector space $\Omega^1(M)$ of all smooth 1-forms on M, which is also a $C^\infty(M)$-module via $(f\omega)(p) = f(p)\omega_p$ for all $f \in C^\infty(M)$, $\omega \in \Omega^1(M)$, and $p \in M$. Therefore, we can define the map $\Omega^1(M) \times \mathfrak{X}(M) \to C^\infty(M)$ as $(\omega, X) \mapsto \omega(X)$, which is $\mathbb{R}$-bilinear and $C^\infty(M)$-bilinear. That is, we have

$$\begin{aligned} (\omega_1 + \omega_2)(X) = \omega_1(X) + \omega_2(X), \qquad \omega(X_1 + X_2) = \omega(X_1) + \omega(X_2), \\ (f\omega)(X) = f(\omega(X)), \qquad \omega(fX) = f(\omega(X)) \end{aligned} \tag{3.6}$$

Now, if $T : V \to W$ is a linear map between finite-dimensional vector spaces V and W, then the dual map $T^* : W^* \to V^*$ is defined as $T^*(\omega) = \omega \circ T$ for all $\omega \in W^*$. That is, T^* takes a linear functional $\omega : W \to \mathbb{R}$ and returns the linear functional $\omega \circ T : V \to \mathbb{R}$. The dual map is also linear, and satisfies $(S \circ T)^* = T^* \circ S^*$ for all linear maps $T : V \to W$ and $S : W \to U$ between finite-dimensional vector spaces V, W, and U. We can extend this notion to smooth maps between smooth manifolds.

Let $F : N \to M$ be a smooth map between smooth manifolds N and M. Then the differential $F_{*,p} : T_pN \to T_{F(p)}M$ is a linear map between the tangent spaces at $p \in N$ and $F(p) \in M$. Therefore, we can define the dual map $F_p^* : T_{F(p)}^*M \to T_p^*N$ as $F_p^*(\omega) = \omega \circ F_{*,p}$ for all $\omega \in T_{F(p)}^*M$. That is, F_p^* takes a linear functional $\omega_{F(p)} : T_{F(p)}M \to \mathbb{R}$ and returns the linear functional $\omega_{F(p)} \circ F_{*,p} : T_pN \to \mathbb{R}$. The dual map is also linear, and satisfies $(G \circ F)_p^* = F_p^* \circ G_{F(p)}^*$ for all smooth maps $F : N \to M$ and $G : M \to P$ between smooth manifolds N, M, and P. If ω is a 1-form on M, then we can define the pullback of ω by F as the 1-form $F^*\omega$ on N defined as $(F^*\omega)_p = F_p^*(\omega_{F(p)})$ for all $p \in N$.

Proposition 3.4. *Let ω be a smooth 1-form on a smooth manifold M, and let $F : N \to M$ be a smooth map between smooth manifolds N and M. Then the pullback $F^*\omega$ is a smooth 1-form on N.*

Proof. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M, and let $(V, \psi) = (V, y^1, \dots, y^m)$ be a chart on N. Then we have

$$\omega = \sum_{i=1}^n a_i dx^i, \qquad F^*\omega = \sum_{j=1}^m b_j dy^j \tag{3.7}$$

for some smooth functions $a_i : U \to \mathbb{R}$ and $b_j : V \to \mathbb{R}$. Then we have

$$b_j(p) = (F^*\omega)_p\left(\frac{\partial}{\partial y^j}\right) = F_p^*(\omega_{F(p)})\left(\frac{\partial}{\partial y^j}\right) = \omega_{F(p)}(F_{*,p}\left(\frac{\partial}{\partial y^j}\right)) = \sum_{i=1}^{n} a_i(F(p))dx^i(F_{*,p}\left(\frac{\partial}{\partial y^j}\right)). \tag{3.8}$$

Since $F_{*,p}\left(\frac{\partial}{\partial y^j}\right)$ is a smooth vector field on N, we have that $dx^i(F_{*,p}\left(\frac{\partial}{\partial y^j}\right))$ is a smooth function on N. Therefore, each of the functions b_j are smooth, and so $F^*\omega$ is a smooth 1-form on N. ■

3.1.1 k-forms

October 7th.

Moving on to k-forms, recall that $A_k(V)$ is the space of all alternating k-linear maps $\omega : V^k \to \mathbb{R}$ on a finite-dimensional vector space V. It is of dimension $\binom{n}{k}$, and if $\{\alpha_1, \ldots, \alpha_n\}$ is a basis for V^*, then the wedge products $\{\alpha_{i_1} \wedge \cdots \wedge \alpha_{i_k} \mid 1 \le i_1 < \cdots < i_k \le n\}$ form a basis for $A_k(V)$. We can extend this notion to smooth manifolds.

Let M be a smooth manifold of dimension n, and let $p \in M$. We have

$$A_k(T_pM) := \{\omega_p : T_pM^k \to \mathbb{R} \mid \omega_p \text{ is alternating and } k\text{-linear}\} = \bigwedge^k T_p^*M. \tag{3.9}$$

A k-form on M is a map

$$\omega : M \to \bigsqcup_{p\in M} A_k(T_pM) = \bigsqcup_{p\in M} \bigwedge^k T_p^*M \tag{3.10}$$

such that $\omega(p) \in A_k(T_pM)$ for all $p \in M$. If we denote $\mathcal{I}_{k,n} = \{(i_1, \ldots, i_k) \mid 1 \le i_1 < \cdots < i_k \le n\}$, then we have that $\dim \bigwedge^k T_p^*M = \binom{n}{k}$, and if $(U, \varphi) = (U, x^1, \ldots, x^n)$ is a chart on M, then the wedge products $\{(dx^I)_p = (dx^{i_1})_p \wedge \cdots \wedge (dx^{i_k})_p \mid (i_1, \ldots, i_k) \in \mathcal{I}_{k,n}\}$ form a basis for $\bigwedge^k T_p^*M$. Thus, any $\omega_p \in A_k(T_pM)$ can be written as

$$\omega_p = \sum_{I\in\mathcal{I}_{k,n}} a_I(p)(dx^I)_p \tag{3.11}$$

where $a_I(p)$'s are real numbers. More generally, any k-form ω on U can be written as

$$\omega = \sum_{I\in\mathcal{I}_{k,n}} a_I dx^I \tag{3.12}$$

where $a_I : U \to \mathbb{R}$ are real-valued functions on U. If $f_1, \ldots, f_k \in C^\infty(U)$ are smooth functions on U, then we have

$$(df_1 \wedge \cdots \wedge df_k)\left(\frac{\partial}{\partial x^{i_1}}, \ldots, \frac{\partial}{\partial x^{i_k}}\right) = \det\left(\frac{\partial f_j}{\partial x^{i_l}}\right)_{1\le j,l\le k}. \tag{3.13}$$

We also have the notion of bundles of k-forms on M. The case $k = 1$ gives the cotangent bundle T^*M. These bundles are defined as

$$\bigwedge^k T^*M = \bigsqcup_{p\in M} \bigwedge^k T_p^*M = \bigsqcup_{p\in M} A_k(T_pM) \tag{3.14}$$

with projection $\pi : \bigwedge^k T^*M \to M$ defined as $\pi(\omega_p) = p$ for all $\omega_p \in \bigwedge^k T_p^*M$. A k-form on M, then, is a section of the bundle $\bigwedge^k T^*M$. To induce a topology on $\bigwedge^k T^*M$, we can use the charts on M as we did with the tangent and cotangent bundles. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M. Also adopt the notation $\bigwedge^k T^*U = \pi^{-1}(U) = \bigsqcup_{p\in U} \bigwedge^k T_p^*M$. Then we have the bijective map

$$\tilde{\varphi} : \bigwedge^k T^*U \to \varphi(U) \times \mathbb{R}^{\binom{n}{k}}, \qquad \omega_p \mapsto (\varphi(p), a_{I_1}(p), \ldots, a_{I_{\binom{n}{k}}}(p)). \tag{3.15}$$

With the standard topology on $\varphi(U) \times \mathbb{R}^{\binom{n}{k}}$, we can give $\bigwedge^k T^*U$ the topology induced by $\tilde{\varphi}$. Then, letting $\mathcal{B}_U$ be the collection of all open sets in $\bigwedge^k T^*U$, we can give $\bigwedge^k T^*M$ the topology generated by $\mathcal{B} = \bigcup_{(U,\varphi)} \mathcal{B}_U$, where the union is taken over all charts (U, φ) on M. The exact same analysis as tangent bundles shows that $\bigwedge^k T^*M$ is a smooth manifold of dimension $n + \binom{n}{k}$, and the projection $\pi : \bigwedge^k T^*M \to M$ is a smooth map. The case for $k = 0$ is trivial, as $\bigwedge^0 T^*M = M \times \mathbb{R}$ is a smooth manifold of dimension $n + 1$, and the projection $\pi : \bigwedge^0 T^*M \to M$ is the projection onto the first factor. A smooth k-form ω on M is a smooth section of π. That is, ω is smooth if $\omega : M \to \bigwedge^k T^*M$ is a smooth map between smooth manifolds. Also, a k-form ω on M is smooth if and only if for any chart $(U, \varphi) = (U, x^1, \ldots, x^n)$ on M, the functions $a_I : U \to \mathbb{R}$ defined as $\omega = \sum_{I \in \mathcal{I}_{k,n}} a_I dx^I$ are smooth. Again, for $k = 0$, a smooth section of $\pi : M \times \mathbb{R} \to M$ is a smooth map $f : M \to M \times \mathbb{R}$ such that $\pi(f(p)) = p$ for all $p \in M$. That is, $f(p) = (p, g(p))$ for some smooth function $g : M \to \mathbb{R}$. Thus, a smooth 0-form on M is simply a smooth function on M.

October 9th.

Same as before, given a smooth map $F : N \to M$ between smooth manifolds N and M, we can define the pullback of a k-form ω on M by F as the k-form $F^*\omega$ on N defined as $(F^*\omega)_p = F_p^*(\omega_{F(p)})$ for all $p \in N$. That is, $(F^*\omega)_p : T_pN^k \to \mathbb{R}$ is the alternating k-linear map defined as

$$(F^*\omega)_p(X_1, \ldots, X_k) = \omega_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_k)) \tag{3.16}$$

for all $X_1, \ldots, X_k \in T_pN$. The pullback of a k-form by a smooth map is also smooth. That is, if ω is a smooth k-form on M, then $F^*\omega$ is a smooth k-form on N. The proof is the same as the case for 1-forms.

From here, we define

$$\Omega^k(M) = \{\omega : M \to \bigwedge^k T^*M \mid \omega \text{ is smooth}\}, \tag{3.17}$$

the vector space of all smooth k-forms on M, which is also a $C^\infty(M)$-module via $(f\omega)(p) = f(p)\omega_p$ for all $f \in C^\infty(M)$, $\omega \in \Omega^k(M)$, and $p \in M$.

October 12th.

The wedge product of forms is also defined here, as follows. Let $\omega \in \Omega^k(M)$ and $\eta \in \Omega^l(M)$ be k- and l-forms on a smooth manifold M. Then we can define the wedge product $\omega \wedge \eta$ as the $(k + l)$-form on M defined as

$$(\omega \wedge \eta)_p = \omega_p \wedge \eta_p = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \omega_p(X_{\sigma(1)}, \ldots, X_{\sigma(k)}) \eta_p(X_{\sigma(k+1)}, \ldots, X_{\sigma(k+l)}) \tag{3.18}$$

for all $p \in M$ and $X_1, \ldots, X_{k+l} \in T_pM$.

Proposition 3.5. *If $\omega \in \Omega^k(M)$ and $\eta \in \Omega^l(M)$ are smooth k- and l-forms on a smooth manifold M, then $\omega \wedge \eta$ is a smooth $(k + l)$-form on M.*

Proof. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M. Then we have

$$\omega = \sum_{I \in \mathcal{I}_{k,n}} a_I dx^I, \qquad \eta = \sum_{J \in \mathcal{I}_{l,n}} b_J dx^J \tag{3.19}$$

for some smooth functions $a_I : U \to \mathbb{R}$ and $b_J : U \to \mathbb{R}$. Then we have

$$(\omega \wedge \eta) = \sum_{I,J} a_I b_J dx^I \wedge dx^J. \tag{3.20}$$

Since the product of smooth functions is smooth, we have that each of the functions $a_I b_J$ are smooth. Thus, $\omega \wedge \eta$ is a smooth $(k + l)$-form on M. ∎

Again, the wedge product is $\mathbb{R}$-bilinear, $C^\infty(M)$-bilinear, associative, and anti-commutative. From here, for a smooth manifold M of dimension n, we define

$$\Omega^*(M) = \bigoplus_{k=0}^{n} \Omega^k(M) = \bigoplus_{k=0}^{\infty} \Omega^k(M), \tag{3.21}$$

which is a graded $\mathbb{R}$-algebra with respect to the wedge product. A smooth map $F : N \to M$ between smooth manifolds N and M, then, induces a graded $\mathbb{R}$-algebra homomorphism $F^* : \Omega^*(M) \to \Omega^*(N)$. We first show the pullback for general k-forms. For a k-form ω on M, we have

$$(F^*\omega)_p(X_1, \ldots, X_k) = \omega_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_k)) \tag{3.22}$$

for all $p \in N$ and $X_1, \ldots, X_k \in T_pN$. Then we have

$$\begin{aligned}(F^*(\omega \wedge \eta))_p(X_1, \ldots, X_{k+l}) &= (\omega \wedge \eta)_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_{k+l})) \\ &= \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \operatorname{sgn}(\sigma) \omega_{F(p)}(F_{*,p}(X_{\sigma(1)}), \ldots, F_{*,p}(X_{\sigma(k)})) \\ &\qquad \cdot \eta_{F(p)}(F_{*,p}(X_{\sigma(k+1)}), \ldots, F_{*,p}(X_{\sigma(k+l)})) \\ &= (F^*\omega \wedge F^*\eta)_p(X_1, \ldots, X_{k+l})\end{aligned} \tag{3.23}$$

and

$$\begin{aligned}(F^*(f\omega + g\eta))_p(X_1, \ldots, X_k) &= (f\omega + g\eta)_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_k)) \\ &= f(F(p))\omega_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_k)) + g(F(p))\eta_{F(p)}(F_{*,p}(X_1), \ldots, F_{*,p}(X_k)) \\ &= (F^*f)(p)(F^*\omega)_p(X_1, \ldots, X_k) + (F^*g)(p)(F^*\eta)_p(X_1, \ldots, X_k) \\ &= ((F^*f)(F^*\omega) + (F^*g)(F^*\eta))_p(X_1, \ldots, X_k)\end{aligned} \tag{3.24}$$

showing that $F^*(\omega \wedge \eta) = F^*\omega \wedge F^*\eta$ and $F^*(f\omega + g\eta) = (F^*f)(F^*\omega) + (F^*g)(F^*\eta)$ for all $\omega, \eta \in \Omega^*(M)$ and $f, g \in C^\infty(M)$. Now, we show that if $\omega \in \Omega^k(M)$, then $F^*\omega \in \Omega^k(N)$. Let $(U, \varphi) = (U, x^1, \ldots, x^n)$ be a chart on M, and let $(V, \psi) = (V, y^1, \ldots, y^m)$ be a chart on N, with $F(U) \subseteq V$. Then $F^*\omega$ is smooth since

$$\begin{aligned}(F^*\omega)_p = F_p^*(\omega_{F(p)}) &= \sum_{I \in \mathcal{I}_{k,n}} a_I(F(p)) F_p^*(dx^I) = \sum_{I \in \mathcal{I}_{k,n}} a_I(F(p)) d(x^{i_1} \circ F)_p \wedge \cdots \wedge d(x^{i_k} \circ F)_p \\ &= \sum_{I \in \mathcal{I}_{k,n}} a_I(F(p)) d(F^*x^{i_1})_p \wedge \cdots \wedge d(F^*x^{i_k})_p\end{aligned} \tag{3.25}$$

where each function $a_I \circ F$ and each function $x^i \circ F$ is smooth on $F^{-1}(U)$. Hence, $F^*\omega$ is a smooth k-form on N.

3.1.2 Exterior Differentiation

Let $A = \bigoplus_{n=0}^{\infty} A_n$ be a graded algebra over $\mathbb{R}$. An $\mathbb{R}$-linear map $D : A \to A$ is said to be an antiderivation of degree $+1$ if $D(ab) = (Da)b + (-1)^{\deg a} a(Db)$ for all $a \in A_k$ and $b \in A_l$, and $Da \in A_{k+1}$ whenever $a \in A_k$. In particular, let $A = \Omega^*(M)$ be the graded algebra of smooth differential forms on a smooth manifold M, and define the exterior derivative $d : \Omega^*(M) \to \Omega^*(M)$ as the $\mathbb{R}$-linear map defined as

$$d\left(\sum a_I dx^I\right) = \sum da_I \wedge dx^I \tag{3.26}$$

after coordinates are chosen. Then d is an antiderivation of degree $+1$, satisfies $d^2 = 0$, and for any $f \in C^\infty(M)$, $df \in \Omega^1(M)$ is the differential of f. Moreover, if D is any antiderivation of degree $+1$ on $\Omega^*(M)$ such that $D^2 = 0$ and $Df = df$ for all $f \in C^\infty(M)$, then $D = d$. That is, the exterior derivative is the unique antiderivation of degree $+1$ on $\Omega^*(M)$ that squares to zero and agrees with the differential on smooth functions.

3.2 Orientable Manifolds

October 14th.

We first provide the theory for a finite dimensional vector space V over $\mathbb{R}$, of dimension n. On the collection of all ordered bases of V, we define an equivalence relation as $(v_1, \dots, v_n) \sim (w_1, \dots, w_n)$ if and only if the change of basis matrix from $(v_1, \dots, v_n)$ to $(w_1, \dots, w_n)$ has a positive determinant. Reflexivity, symmetry, and transitivity may be easily verified. This equivalence relation on the set of ordered basis of V partitions it into 2 equivalence classes. An orientation on V is a choice between these 2 equivalence clases or, equivalently, a choice of an ordered basis and its corresponding equivalence class. We say V is oriented after making such a choice. For a finite dimensional vector space V over $\mathbb{R}$, we shall use μ and $-\mu$ to denote the two orientations.

Lifting this theory, let M be a smooth manifold of dimension n. Recall that a vector field X is a section of the tangent bundle TM such that $X : M \to TM$ satisfies $\pi \circ X = \mathrm{id}_M$, and a(n ordered) frame $(X_1, X_2, \dots, X_n)$ on M is a collection of vector fields on M such that for all $p \in M$, $(X_{1,p}, X_{2,p}, \dots, X_{n,p})$ is a(n ordered) basis for T_pM.

Definition 3.6. A *pointwise orientation* μ on a smooth manifold M is a choice of orientation μ_p on T_pM for all $p \in M$. It is said to be continuous at a point $p \in M$ if p has a neighbourhood U on which μ is represented by a continuous frame on U; that is, there exist continuous vector fields $X_1, \dots, X_n$ on U such that for all $q \in U$, $[(X_{1,p}, X_{2,p}, \dots, X_{n,p})] = \mu_p$.
An *orientation* on M is a pointwise orientation μ that is continuous for all $p \in M$. A manifold on which there exists an orientation is called an *orientable manifold*. A manifold with a choice of orientation is called a *oriented manifold*.

Proposition 3.7. *Let M be a smooth orientable connected manifold. Then M has exactly 2 orientations.*

Proof. Take two orientations μ, ν on M. Define the map $f : M \to \{\pm 1\}$ as $f(p) = 1$ if $\mu_p = \nu_p$ and $f(p) = -1$ if $\mu_p \neq \nu_p$. We claim that f is a continuous map. Let $p \in U$, where U is a connected neighbourhood of p. Then continuity at p gives $\mu = (X_1, \dots, X_n)$ and $\nu = (Y_1, \dots, Y_n)$, two collections of continuous vector fields on U which form a frame. Thus, there exists a matrix of continuous functions $A : U \to M_n(\mathbb{R})$ such that

$$(Y_{1,q}, \dots, Y_{n,q}) = (X_{1,q}, \dots, X_{n,q})A(q) \qquad \text{for all } q \in U. \tag{3.27}$$

This implies that the map $\det(A(\cdot)) : U \to \mathbb{R}^\times$ is a continuous function; so it must be positive on the entirety of U, or negative. Moreover, $f : U \to \{\pm 1\}$ may be written as $f(q) = \mathrm{sgn}(\det(A(q)))$ for all $q \in U$, showing that f is locally constant. Since M is connected, f is constant, finally implying that $\mu = \nu$ or $\mu = -\nu$. ∎

October 16th.

Proposition 3.8. *Let M be a smooth manifold of dimension n. A frame $(X_1, \dots, X_n)$ on M gives a pointwise continuous orientation on M if and only if for all $p \in M$, there exists a chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on M with $p \in U$ such that $(dx^1 \wedge \cdots \wedge dx^n)_p(X_{1,p}, \dots, X_{n,p}) > 0$.*

Proof. Let $(X_1, \dots, X_n)$ be a frame on M which gives a pointwise continuous orientation on M. Let $p \in M$. Then there exists a neighbourhood U of p and a continuous frame $(Y_1, \dots, Y_n)$ on U such that for all $q \in U$, $[(Y_{1,q}, \dots, Y_{n,q})] = [(X_{1,q}, \dots, X_{n,q})]$. $[(\frac{d}{dx^1})_q, \dots, (\frac{d}{dx^n})_q]$ is an orientation on T_qM for all $q \in U$. Since $Y_1, \dots, Y_n$ are continuous vector fields on U, the coordinate functions $a_{ij} : U \to \mathbb{R}$ defined as $Y_i = \sum_{j=1}^n a_{ij}(\frac{d}{dx^j})$ are continuous. Thus, the map $\det(a_{ij}) : U \to \mathbb{R}^\times$ is continuous, and so it must be positive on the entirety of U, or negative. If this map is negative, we can replace Y_1 with $-Y_1$ to make it positive. Then we have

$$(dx^1 \wedge \cdots \wedge dx^n)_q(Y_{1,q}, \dots, Y_{n,q}) = \det(a_{ij}(q)) > 0. \tag{3.28}$$

Finally, since $[(Y_{1,q}, \dots, Y_{n,q})] = [(X_{1,q}, \dots, X_{n,q})]$, we have that

$$(dx^1 \wedge \cdots \wedge dx^n)_q(X_{1,q}, \dots, X_{n,q}) > 0. \tag{3.29}$$

For the converse, let $(X_1, \dots, X_n)$ be a frame on M such that for all $p \in M$, there exists a chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on M with $p \in U$ such that $(dx^1 \wedge \cdots \wedge dx^n)_p(X_{1,p}, \dots, X_{n,p}) > 0$. Then for all $q \in U$, we have

$$(dx^1 \wedge \cdots \wedge dx^n)_q(X_{1,q}, \dots, X_{n,q}) > 0. \tag{3.30}$$

We show that $[(X_{1,q}, \dots, X_{n,q})] = [(\frac{d}{dx^1})_q, \dots, (\frac{d}{dx^n})_q]$ for all $q \in U$. Let $A(q)$ be the change of basis matrix from $(X_{1,q}, \dots, X_{n,q})$ to $((\frac{d}{dx^1})_q, \dots, (\frac{d}{dx^n})_q)$. Then we have

$$(dx^1 \wedge \cdots \wedge dx^n)_q(X_{1,q}, \dots, X_{n,q}) = \det(A(q)) > 0. \tag{3.31}$$

This shows that the frame $(X_{1,q}, \dots, X_{n,q})$ is positively oriented with respect to the coordinate frame $((\frac{d}{dx^1})_q, \dots, (\frac{d}{dx^n})_q)$ for all $q \in U$, and hence the frame $(X_1, \dots, X_n)$ gives a pointwise continuous orientation on M. ■

Proposition 3.9. *Let M be a smooth manifold of dimension n. Then M is orientable if and only if there exists a smooth nowhere vanishing n-form $\omega \in \Omega^n(M)$.*

Proof. Suppose M is orientable, and $X_1, \dots, X_n$ is a frame on M which gives a pointwise continuous orientation on M. Then for all $p \in M$, there exists a chart $(U, \varphi) = (U, x^1, \dots, x^n)$ on M with $p \in U$ such that $(dx^1 \wedge \cdots \wedge dx^n)_p(X_{1,p}, \dots, X_{n,p}) > 0$, or on U, $(dx^1 \wedge \cdots \wedge dx^n)(X_1, \dots, X_n) > 0$. Vary $p \in M$ to get an open cover $\{U_\alpha\}$ of M with charts $(U_\alpha, \varphi_\alpha) = (U_\alpha, x^1_\alpha, \dots, x^n_\alpha)$ on M such that $(dx^1_\alpha \wedge \cdots \wedge dx^n_\alpha)(X_1, \dots, X_n) > 0$ on U_α. Let $\{\rho_\alpha\}$ be a smooth partition of unity subordinate to the open cover $\{U_\alpha\}$. Then we can define a smooth n-form $\omega \in \Omega^n(M)$ as

$$\omega = \sum_\alpha \rho_\alpha dx^1_\alpha \wedge \cdots \wedge dx^n_\alpha. \tag{3.32}$$

Then for all $p \in M$, we have

$$\omega_p(X_{1,p}, \dots, X_{n,p}) = \sum_\alpha \rho_\alpha(p)(dx^1_\alpha \wedge \cdots \wedge dx^n_\alpha)_p(X_{1,p}, \dots, X_{n,p}) > 0. \tag{3.33}$$

This shows that ω is a smooth nowhere vanishing n-form on M. For the converse, suppose there exists a smooth nowhere vanishing n-form $\omega \in \Omega^n(M)$. Then for all $p \in M$, we have $\omega_p \neq 0$, and hence there exists a basis $(X_{1,p}, \dots, X_{n,p})$ for T_pM such that $\omega_p(X_{1,p}, \dots, X_{n,p}) > 0$. Vary $p \in M$ to get a frame $(X_1, \dots, X_n)$ on M such that $\omega(X_1, \dots, X_n) > 0$. Define an orientation on T_pM by declaring an ordered basis $(v_1, \dots, v_n)$ to be positive if and only if

$$\omega_p(v_1, \dots, v_n) > 0. \tag{3.34}$$

This is well-defined, since replacing a basis by another one multiplies the value of ω_p by the determinant of the change-of-basis matrix. It remains to verify continuity. Let $p \in M$ and choose a chart $(U, x^1, \dots, x^n)$ around p. Since ω is nowhere vanishing, we can write

$$\omega = f\, dx^1 \wedge \cdots \wedge dx^n \tag{3.35}$$

on U, where $f : U \to \mathbb{R}$ is smooth and nowhere zero. After shrinking U if necessary, f has constant sign. Consequently, the coordinate frame $(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n})$, or the same frame with one vector multiplied by -1, represents the orientation defined above throughout U. Thus the pointwise orientation is continuous, so M is orientable. ■

October 21st.

Definition 3.10. Let $\{(U_\alpha, \varphi_\alpha) = (U_\alpha, x^1_\alpha, \dots, x^n_\alpha)\}_{\alpha \in \Lambda}$ be an atlas on a smooth manifold M of dimension n. It is termed an *orientable atlas* if for all $\alpha, \beta \in \Lambda$ with $U_\alpha \cap U_\beta \neq \emptyset$, the Jacobian determinant of the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is positive on $\varphi_\alpha(U_\alpha \cap U_\beta)$; that is,

$$\det\left(\frac{\partial x^i_\beta}{\partial x^j_\alpha}\right)_{1 \le i,j \le n} > 0 \text{ on } \varphi_\alpha(U_\alpha \cap U_\beta). \tag{3.36}$$

Proposition 3.11. *Let M be a smooth manifold of dimension n. Then M is orientable if and only if there exists an orientable atlas on M.*

Proof. If M is orientable, then there exists a frame $(X_1, \dots, X_n)$ on M such that there exist charts $(U, \varphi) = (U, x^1, \dots, x^n)$ covering M satisfying $(dx^1 \wedge \cdots \wedge dx^n)(X_1, \dots, X_n) > 0$. Let $\{(U_\alpha, \varphi_\alpha) = (U_\alpha, x^1_\alpha, \dots, x^n_\alpha)\}_{\alpha \in \Lambda}$ be this atlas of charts on M. Then for all $\alpha, \beta \in \Lambda$ with $U_\alpha \cap U_\beta \neq \emptyset$, we have

$$\det\left(\frac{\partial x^i_\beta}{\partial x^j_\alpha}\right) \cdot (dx^1_\alpha \wedge \cdots \wedge dx^n_\alpha)(X_1, \dots, X_n) = (dx^1_\beta \wedge \cdots \wedge dx^n_\beta)(X_1, \dots, X_n) > 0 \tag{3.37}$$

implying that the Jacobian determinant of the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is positive on $\varphi_\alpha(U_\alpha \cap U_\beta)$. Thus, $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ is an orientable atlas on M. Conversely, if there exists an orientable atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ on M, let μ be the pointwise orientation on M defined as follows. For all $p \in M$, choose $\alpha \in \Lambda$ such that $p \in U_\alpha$, and define $\mu_p = [((\frac{\partial}{\partial x^1_\alpha})_p, \dots, (\frac{\partial}{\partial x^n_\alpha})_p)]$. This is well-defined, since if $p \in U_\alpha \cap U_\beta$, then the Jacobian determinant of the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is positive on $\varphi_\alpha(U_\alpha \cap U_\beta)$, and hence

$$\left[\left(\left(\frac{\partial}{\partial x^1_\alpha}\right)_p, \dots, \left(\frac{\partial}{\partial x^n_\alpha}\right)_p\right)\right] = \left[\left(\left(\frac{\partial}{\partial x^1_\beta}\right)_p, \dots, \left(\frac{\partial}{\partial x^n_\beta}\right)_p\right)\right]. \tag{3.38}$$

■

Let M be a connected smooth manifold of dimension n. Then M is orientable if and only if there exists a smooth nowhere vanishing n-form $\omega \in \Omega^n(M)$; on the set of all smooth nowhere vanishing n-forms on M, we define an equivalence relation as $\omega \sim \eta$ if and only if $\omega = f\eta$ for some smooth function $f : M \to \mathbb{R}$ such that $f(p) > 0$ for all $p \in M$. This equivalence relation partitions the set of all smooth nowhere vanishing n-forms on M into 2 equivalence classes. Thus, an orientation determined by a smooth nowhere vanishing n-form ω on M is exactly the orientation given by a frame $(X_1, \dots, X_n)$ on M such that $\omega(X_1, \dots, X_n) > 0$. In particular, if ω and η are smooth nowhere vanishing n-forms on M, then they determine the same orientation on M if and only if $\omega = f\eta$ for some smooth function $f > 0$.

3.2.1 Manifolds with Boundary

October 24th.

Hereforth, we use $\mathbb{H}^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n \mid x^n \geq 0\}$ to denote the closed upper half-space in $\mathbb{R}^n$.

Definition 3.12. A topological space X is *locally* $\mathbb{H}^n$ if for all $p \in X$, there exists a neighbourhood U of p and a homeomorphism $\varphi : U \to V$ onto an open subset V of $\mathbb{H}^n$ such that $\varphi(p) = 0$. M is termed a *topological manifold with boundary* of dimension n if it is Hausdorff, second countable, and locally $\mathbb{H}^n$.

A smooth atlas on M is a collection of charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ such that for all $\alpha, \beta \in \Lambda$ with $U_\alpha \cap U_\beta \neq \emptyset$, the transition map $\varphi_\beta \circ \varphi_\alpha^{-1}$ is smooth on $\varphi_\alpha(U_\alpha \cap U_\beta)$. A smooth manifold with boundary is a topological manifold with boundary equipped with a smooth atlas.

For the closed half space $\mathbb{H}^n$, we define

$$\partial\mathbb{H}^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n \mid x^n = 0\}, \qquad (\mathbb{H}^n)^\circ = \{(x^1, \dots, x^n) \in \mathbb{R}^n \mid x^n > 0\} \tag{3.39}$$

so that $\mathbb{H}^n = (\mathbb{H}^n)^\circ \sqcup \partial\mathbb{H}^n$. Note that if U and V are open subsets of $\mathbb{H}^n$ and $f : U \to V$ is a diffeomorphism, then $f(\partial\mathbb{H}^n \cap U) \subseteq \partial\mathbb{H}^n \cap V$ and $f((\mathbb{H}^n)^\circ \cap U) \subseteq (\mathbb{H}^n)^\circ \cap V$. Here, $\partial\mathbb{H}^n$ denotes the boundary of $\mathbb{H}^n$, and $(\mathbb{H}^n)^\circ$ denotes the interior of $\mathbb{H}^n$. One can show that $\partial\mathbb{H}^n$ is a smooth manifold of dimension $n-1$, and $(\mathbb{H}^n)^\circ$ is a smooth manifold of dimension n. This can be extended to any smooth manifold with boundary M. We denote

$$\partial M = \{p \in M \mid \text{there exists a chart } (U, \varphi) \text{ on } M \text{ with } p \in U \text{ and } \varphi(p) \in \partial\mathbb{H}^n\}, \qquad M^\circ = M \setminus \partial M. \tag{3.40}$$

One can show that ∂M is a smooth manifold of dimension $n-1$, and M° is a smooth manifold of dimension n. We call ∂M the *boundary* of M, and M° the *interior* of M. Note that if M is a smooth manifold with boundary, then ∂M is a smooth manifold without boundary.

Smooth functions on a smooth manifold with boundary M are defined as follows. A function $f : M \to \mathbb{R}$ is smooth if for all $p \in M$, there exists a chart $(U, \varphi) = (U, x^1, \ldots, x^n)$ on M with $p \in U$ such that $f \circ \varphi^{-1}$ is smooth on $\varphi(U)$. Smooth maps between smooth manifolds with boundary are defined similarly. Germs of smooth functions on a smooth manifold with boundary M are defined just as for smooth manifolds without boundary. The tangent space T_pM at a point $p \in M$ is defined as the set of derivations on the germs of smooth functions at p. We may then, analogously, define

$$T_p^*M = \mathrm{Hom}_{\mathbb{R}}(T_pM, \mathbb{R}), \qquad \bigwedge^k T_p^*M = A_k(T_pM) \tag{3.41}$$

and

$$TM = \bigsqcup_{p \in M} T_pM, \qquad T^*M = \bigsqcup_{p \in M} T_p^*M. \tag{3.42}$$

A smooth k-form on a smooth manifold with boundary M is defined as a smooth section of the bundle $\bigwedge^k T^*M$. An orientation on a smooth manifold with boundary M, then, is one of three equivalent definitions: a smooth nowhere vanishing n-form $\omega \in \Omega^n(M)$, a pointwise continuous frame $(X_1, \ldots, X_n)$ on M, or an orientable atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in \Lambda}$ on M.

Let $\beta \in \bigwedge^k V^* = A_k(V)$ be a k-covector on a finite dimensional vector space V over $\mathbb{R}$, where $\dim V \geq k \geq 2$. If a vector $v \in V$ is fixed, then we can define a $(k-1)$-covector $i_v\beta \in \bigwedge^{k-1} V^*$ as

$$(i_v\beta)(v_1, \ldots, v_{k-1}) = \beta(v, v_1, \ldots, v_{k-1}). \tag{3.43}$$

If $k = 1$, then we define $i_v\beta \in \bigwedge^0 V^* = \mathbb{R}$ as $(i_v\beta) = \beta(v)$. $k = 0$ admits $i_v\beta = 0$. Moreover, if $\beta^1, \ldots, \beta^k \in V^*$ are 1-covectors on V, then $\beta^1 \wedge \cdots \wedge \beta^k \in \bigwedge^k V^*$ is a k-covector on V, and we have

$$i_v(\beta^1 \wedge \cdots \wedge \beta^k) = \sum_{i=1}^{k} (-1)^{i-1} \beta^1 \wedge \cdots \wedge \beta^{i-1} \wedge \beta^{i+1} \wedge \cdots \wedge \beta^k. \tag{3.44}$$

The same construction can be lifted to a smooth manifold, M. Let $\omega \in \Omega^k(M)$ be a smooth k-form on M, and let $X \in \mathfrak{X}(M)$ be a vector field on M. Then we can define a smooth $(k-1)$-form $i_X\omega \in \Omega^{k-1}(M)$ as

$$(i_X\omega)_p = i_{X_p}(\omega_p) \tag{3.45}$$

for all $p \in M$.

Note that if M is an oriented smooth manifold with boundary, then ∂M is an oriented smooth manifold without boundary.

Definition 3.13. Let M be a smooth manifold with boundary. Suppose $p \in \partial M$. A vector $X_p \in T_pM$ is said to be *inward pointing* if

1. $X_p \notin T_p(\partial M)$, and
2. there exists a smooth map $c : [0, \epsilon) \to M$ for some $\epsilon > 0$ such that $c(0) = p$, $c'(0) = X_p$, and $c((0, \epsilon)) \subseteq M^\circ$.

It is said to be *outward pointing* if $-X_p$ is inward pointing.

Definition 3.14. For a smooth manifold with boundary M, a vector field X along the boundary ∂M is a function $X : \partial M \to \bigsqcup_{p \in \partial M} T_pM$ such that $X(p) \in T_pM$ for all $p \in \partial M$. Such a vector field along the boundary is said to be smooth at $p \in \partial M$ if for any chart $(U, \varphi) = (U, x^1, \ldots, x^n)$ on M with $p \in U$, X when written as $X(q) = \sum_{i=1}^n a^i(q)(\frac{\partial}{\partial x^i})_q$ for $q \in U \cap \partial M$ has smooth coefficients $a^i : U \cap \partial M \to \mathbb{R}$. A vector field along the boundary is said to be smooth if it is smooth at all points $p \in \partial M$. It is outward pointing at $p \in \partial M$ if and only if $a^n(p) > 0$ in any chart $(U, \varphi) = (U, x^1, \ldots, x^n)$ on M with $p \in U$.

Proposition 3.15. *For any smooth manifold with boundary, there exists a smooth outward pointing vector field along the boundary.*

Proof. Let $p \in \partial M$, and choose a chart $(U_\alpha, x^1_\alpha, \dots, x^n_\alpha)$ about p. Define $X_\alpha = -\frac{\partial}{\partial x^n_\alpha}$ on $U_\alpha \cap \partial M$. Then X_α is a smooth outward pointing vector field along the boundary on $U_\alpha \cap \partial M$. Let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ be an open cover of ∂M by such charts, and let $\{\rho_\alpha\}_{\alpha\in\Lambda}$ be a smooth partition of unity subordinate to this open cover. Then we can define a smooth outward pointing vector field along the boundary as

$$X = \sum_{\alpha\in\Lambda} \rho_\alpha X_\alpha. \tag{3.46}$$

This is well-defined and smooth, and it is outward pointing at each point of ∂M. ∎

Now, let M be an oriented smooth manifold with boundary of dimension n. Then ∂M is an oriented smooth manifold without boundary of dimension $n-1$: let $\omega \in \Omega^n(M)$ be a smooth nowhere vanishing n-form on M which determines the orientation on M. Let X be a smooth outward pointing vector field along the boundary ∂M, via the above proposition. Then we can define a smooth nowhere vanishing $(n-1)$-form $\eta \in \Omega^{n-1}(\partial M)$ on ∂M as

$$\eta = i_X \omega|_{\partial M}. \tag{3.47}$$

This η cannot be zero; if it were at some point $p \in \partial M$, then $\omega_p(X_p, v_1, \dots, v_{n-1}) = 0$ for all $v_1, \dots, v_{n-1} \in T_p(\partial M)$. But since X_p is outward pointing, it is not in $T_p(\partial M)$, and hence $(X_p, v_1, \dots, v_{n-1})$ forms a basis for T_pM. This contradicts the fact that ω is nowhere vanishing. Thus, η is a smooth nowhere vanishing $(n-1)$-form on ∂M, and it determines an orientation on ∂M. Moreover, an ordered basis $(v_1, v_2, \dots, v_{n-1})$ for $T_p(\partial M)$ is positively oriented with respect to the orientation on ∂M if and only if $(X_p, v_1, v_2, \dots, v_{n-1})$ is positively oriented with respect to the orientation on M.

3.3 Integration

October 26th.

Recall the theory of integration for a a bounded function $f : [a,b] \to \mathbb{R}$ on a closed interval $[a,b] \subseteq \mathbb{R}$. Let $\mathcal{P}$ be the set of all partitions of $[a,b]$, and for each partition $P = \{x_0, \dots, x_n\} \in \mathcal{P}$, define the lower sum and upper sum of f with respect to P as

$$L(f,P) = \sum_{i=1}^n m_i(x_i - x_{i-1}), \qquad U(f,P) = \sum_{i=1}^n M_i(x_i - x_{i-1}), \tag{3.48}$$

where $m_i = \inf\{f(x) \mid x \in [x_{i-1}, x_i]\}$ and $M_i = \sup\{f(x) \mid x \in [x_{i-1}, x_i]\}$. One can show that $L(f,P) \le U(f,Q)$ for any partitions $P, Q \in \mathcal{P}$. The lower integral and upper integral of f on $[a,b]$ are defined as

$$\underline{\int_a^b} f(x)dx = \sup\{L(f,P) \mid P \in \mathcal{P}\}, \qquad \overline{\int_a^b} f(x)dx = \inf\{U(f,P) \mid P \in \mathcal{P}\}. \tag{3.49}$$

f is said to be Riemann integrable on $[a,b]$ if $\underline{\int_a^b} f(x)dx = \overline{\int_a^b} f(x)dx$, in which case we denote the common value as $\int_a^b f(x)dx$. In general in $\mathbb{R}^n$, we have seen that if $f : A \to \mathbb{R}$ is a bounded function on a set $A \subseteq \mathbb{R}^n$, then we can define $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ as $\tilde{f}(x) = f(x)$ if $x \in A$, and $\tilde{f}(x) = 0$ if $x \notin A$. If A happens to be a bounded subset, then we can consider any rectangle $R \subseteq \mathbb{R}^n$ containing A, and define the lower and upper integrals of f on A as

$$\underline{\int_A} f(x)dx = \underline{\int_R} \tilde{f}(x)dx, \qquad \overline{\int_A} f(x)dx = \overline{\int_R} \tilde{f}(x)dx. \tag{3.50}$$

f is said to be Riemann integrable on A if $\underline{\int_A} f(x)dx = \overline{\int_A} f(x)dx$, in which case we denote the common value as $\int_A f(x)dx$.

A bounded subset $A \subseteq \mathbb{R}^n$ is calle a *domain of integration* if its boundary ∂A has measure zero.

Lemma 3.16 (*Lebesgue's lemma*)**.** *Let $A \subseteq \mathbb{R}^n$ be a bounded subset, and let $f : A \to \mathbb{R}$ be a bounded function. Then f is Riemann integrable on A if and only if $\tilde{f}$ is continuous almost everywhere on a rectangle $R \subseteq \mathbb{R}^n$ containing A.*

Corollary 3.17. *Let $U \subseteq \mathbb{R}^n$ be an open subset, and let $f : U \to \mathbb{R}$ be a continuous function with compact support. Then f is Riemann integrable on U.*

Proof. Since f has compact support, there exists a compact subset $K \subseteq U$ such that $f(x) = 0$ for all $x \in U \setminus K$. Since K is compact, it is bounded, and hence there exists a rectangle $R \subseteq \mathbb{R}^n$ containing K. Then $\tilde{f}$ is continuous on R, and hence continuous almost everywhere on R. By Lebesgue's lemma, $\tilde{f}$ is Riemann integrable on R, and hence f is Riemann integrable on U. ■

Corollary 3.18. *Let $A \subseteq \mathbb{R}^n$ be a domain of integration, and let $f : A \to \mathbb{R}$ be a bounded continuous function. Then f is Riemann integrable on A.*

Proof. Since A is a domain of integration, its boundary ∂A has measure zero. Let $R \subseteq \mathbb{R}^n$ be a rectangle containing A. Then $\tilde{f}$ is continuous on A, and hence continuous almost everywhere on R. By Lebesgue's lemma, $\tilde{f}$ is Riemann integrable on R, and hence f is Riemann integrable on A. ■

Now, note that there is a bijection between the set of all n-forms on $\mathbb{R}^n$ and the set of all functions $f : \mathbb{R}^n \to \mathbb{R}$ given by

$$f \mapsto f\,dx^1 \wedge \cdots \wedge dx^n, \qquad f\,dx^1 \wedge \cdots \wedge dx^n \mapsto f. \tag{3.51}$$

The Riemann integral of an n-form $\omega = f\,dx^1 \wedge \cdots \wedge dx^n$ on a domain of integration $A \subseteq \mathbb{R}^n$ is defined as

$$\int_A \omega = \int_A f(x^1, \ldots, x^n) dx^1 \cdots dx^n \tag{3.52}$$

assuming the integral exists. For $U, V \subseteq \mathbb{R}^n$ open subsets, and a diffeomorphism $T : V \to U$, we have the change of variables formula, where with the pullback

$$T^*\omega = T^*(f(x^1, \ldots, x^n)dx^1 \wedge \cdots \wedge dx^n) = (f \circ T)(y^1, \ldots, y^n) \det\left(\frac{\partial T^i}{\partial x^j}\right) dy^1 \wedge \cdots \wedge dy^n, \tag{3.53}$$

the change of variables formula states that

$$\int_U \omega = \int_U f(x^1, \ldots, x^n)dx^1 \cdots dx^n = \int_V (f \circ T)(y^1, \ldots, y^n) \left|\det\left(\frac{\partial T^i}{\partial x^j}\right)\right| dy^1 \cdots dy^n. \tag{3.54}$$

3.3.1 Over Manifolds

We extend the theory of integration to smooth orientable manifolds. Let M be a smooth oriented manifold of dimension n with oriented atlas $\{(U_\alpha, \varphi_\alpha) = (U_\alpha, x^1_\alpha, \ldots, x^n_\alpha)\}_{\alpha \in \Lambda}$. Here, we define the set of all smooth n-forms on M with compact support as $\Omega^n_c(M) = \{\omega \in \Omega^n(M) \mid \operatorname{supp}(\omega) \text{ is compact}\}$. Now, if (U, φ) is a chart on M with $\varphi(U) \subseteq \mathbb{R}^n$, then for any $\omega \in \Omega^n_c(U)$, we define

$$\int_U \omega := \int_{\varphi(U)} (\varphi^{-1})^*\omega. \tag{3.55}$$

For well-definedness, let (V, ψ) be another chart on M with $\psi(V) \subseteq \mathbb{R}^n$, and suppose $\omega \in \Omega^n_c(U \cap V)$. Then with $T = \psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ the transition map, we have

$$\int_{\varphi(U\cap V)} (\varphi^{-1})^*\omega = \int_{\varphi(U\cap V)} T^*((\psi^{-1})^*\omega) = \int_{\psi(U\cap V)} (\psi^{-1})^*\omega. \tag{3.56}$$

For $\omega \in \Omega^n_c(M)$, we use a partition of unity $\{\rho_\alpha\}$ subordinate to the open cover $\{U_\alpha\}$ of M to define

$$\int_M \omega = \sum_\alpha \int_{U_\alpha} \rho_\alpha \omega. \tag{3.57}$$

Note that this sum is finite, since $\mathrm{supp}(\omega)$ is compact, and hence $\rho_\alpha\omega$ is nonzero for only finitely many $\alpha \in \Lambda$. Again, for well-definedness, let $\{(V_\beta, \psi_\beta)\}_{\beta\in\Gamma}$ be another oriented atlas on M which gives the same orientation on M, and let $\{\chi_\beta\}$ be a smooth partition of unity subordinate to the open cover $\{V_\beta\}$ of M. Then we have

$$\sum_\alpha \int_{U_\alpha} \rho_\alpha\omega = \sum_\alpha \sum_\beta \int_{U_\alpha\cap V_\beta} \rho_\alpha\chi_\beta\omega = \sum_\beta \sum_\alpha \int_{U_\alpha\cap V_\beta} \rho_\alpha\chi_\beta\omega = \sum_\beta \int_{V_\beta} \chi_\beta\omega. \tag{3.58}$$

Proposition 3.19. *Let M be a= connected oriented smooth manifold. Let $-M$ denote the same smooth manifold with the opposite orientation. Then for all $\omega \in \Omega^n_c(M)$, we have*

$$\int_{-M} \omega = -\int_M \omega. \tag{3.59}$$

Proof. Note that it suffices to prove the result for $\omega \in \Omega^n_c(U)$ for some chart (U, φ) on M. Let $(U, \varphi) = (U, x^1, \dots, x^n)$ be a chart on M, and let $(U, \psi) = (U, -x^1, x^2, \dots, x^n)$ be the same chart with the opposite orientation. Then we have

$$(\psi^{-1})^*\omega = f \circ \psi d(x^1 \circ \psi) \wedge d(x^2 \circ \psi) \wedge \cdots \wedge d(x^n \circ \psi) = -f \circ \psi dr^1 \wedge dr^2 \wedge \cdots \wedge dr^n = -(\varphi^{-1})^*\omega. \tag{3.60}$$

Using this, we have

$$\int_U \omega = \int_{\psi(U)} (\psi^{-1})^*\omega = -\int_{\varphi(U)} (\varphi^{-1})^*\omega = -\int_M \omega. \tag{3.61}$$

■

October 28th.

Theorem 3.20 (*Stokes' theorem*)**.** *Let M be an oriented smooth manifold with boundary of dimension n, and let $\omega \in \Omega^{n-1}_c(M)$. Then $d\omega \in \Omega^n_c(M)$, and we have*

$$\int_M d\omega = \int_{\partial M} \omega, \tag{3.62}$$

where the orientation on ∂M is induced by the orientation on M, and ω on the right-hand side is really the pullback via the inclusion map $i : \partial M \hookrightarrow M$.

We first give some easy corollaries. A smooth k-form $\omega \in \Omega^k(M)$ is said to be a *closed form* if $d\omega = 0$, and it is said to be an *exact form* if there exists a smooth $(k-1)$-form $\eta \in \Omega^{k-1}(M)$ such that $\omega = d\eta$. Note that every exact form is closed, since $d^2 = 0$. If $\partial M = \emptyset$, we define $\int_{\partial M} \omega = 0$.

Corollary 3.21. *Let M be as above.*

1. *If M, itself, happens to be compact, then $\omega \in \Omega^{n-1}(M)$ implies $\int_{\partial M} \omega = \int_M d\omega$.*
2. *If M is compact without boundary, then $\omega \in \Omega^{n-1}(M)$ implies $\int_M d\omega = 0$.*
3. *If M is compact and $\omega \in \Omega^{n-1}(M)$ is closed, then $\int_{\partial M} \omega = 0$.*

For a smooth manifold M of dimension n, a *regular domain* $D \subseteq M$ is an embedded submanifold with boundary of dimension n.

Corollary 3.22 (*Green's theorem*)**.** *Let $D \subseteq \mathbb{R}^2$ be a compact regular domain. If $P, Q : D \to \mathbb{R}$ are smooth functions, then*

$$\int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy = \int_{\partial D} Pdx + Qdy. \tag{3.63}$$

Proof. Simply apply Stokes' theorem to the 1-form $\omega = Pdx + Qdy \in \Omega^1(D)$, where $d\omega = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \wedge dy$. ■

Corollary 3.23. *Let $F = (P, Q, R) : \mathbb{R}^3 \to \mathbb{R}^3$ be a smooth vector field, and define $Fdr = Pdx + Qdy + Rdz \in \Omega^1(\mathbb{R}^3)$. Let $C = (x(t), y(t), z(t)) : [a, b] \to \mathbb{R}^3$ be a smooth curve. Suppose there exists a smooth function $f : \mathbb{R}^3 \to \mathbb{R}$ such that $F = \nabla f$. Then*

$$\int_C Fdr = f(C(b)) - f(C(a)). \tag{3.64}$$

Proof. The 1-form Fdr is exact, since $Fdr = df$, and the LHS term is $\int_C df$ and the RHS term is $\int_{\partial C} f$. Since C is a smooth curve, it is a regular domain in $\mathbb{R}^3$, and hence we can apply Stokes' theorem to get the desired result. ■

Proof of Stokes' theorem. Let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha\in\Lambda}$ be an oriented atlas on M such that each U_α is diffeomorphic to either $\mathbb{R}^n$ or $\mathbb{H}^n$, whilst preserving the orientation. Let $\{\rho_\alpha\}_{\alpha\in\Lambda}$ be a smooth partition of unity subordinate to the open cover $\{U_\alpha\}_{\alpha\in\Lambda}$ of M. Note that for each α,

$$\operatorname{supp}(\rho_\alpha\omega) \subseteq \operatorname{supp}(\rho_\alpha) \cap \operatorname{supp}(\omega) \subseteq U_\alpha \tag{3.65}$$

shows that $\rho_\alpha\omega$ is compactly supported since the support is a closed subset of the compact set $\operatorname{supp}(\omega)$. We may write $\omega = \sum_\alpha \rho_\alpha\omega$. Moreover, $\partial M \cap U_\alpha = \partial U_\alpha$ for all α. For now, assume that Stokes' theorem holds for $\mathbb{H}^n$ and $\mathbb{R}^n$. Then

$$\int_{\partial M} \omega = \sum_\alpha \int_{\partial M} \rho_\alpha\omega = \sum_\alpha \int_{\partial U_\alpha} \rho_\alpha\omega = \sum_\alpha \int_{U_\alpha} d(\rho_\alpha\omega) = \int_M d\omega. \tag{3.66}$$

We show Stokes' for $\mathbb{H}^n$ with the orientation $dx^1 \wedge \cdots \wedge dx^n$. The boundary orientation for $\partial\mathbb{H}^n$ is then $(-1)^n dx^1 \wedge \cdots \wedge dx^{n-1}$. Now let $\omega \in \Omega_c^{n-1}(\mathbb{H}^n)$. Then

$$\omega = \sum_{i=1}^n f_i dx^1 \wedge \cdots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \cdots \wedge dx^n \tag{3.67}$$

where each f_i is a smooth function with compact support on $\mathbb{H}^n$. Choose $a > 0$ large enough such that $\operatorname{supp}(f_i) \subseteq \operatorname{int}([-a, a]^{n-1} \times [0, a])$ for all i. Then we have

$$\begin{aligned} d\omega &= \sum_{i=1}^n df_i \wedge dx^1 \wedge \cdots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \cdots \wedge dx^n \\ &= \sum_{i=1}^n \frac{\partial f_i}{\partial x^i} dx^i \wedge dx^1 \wedge \cdots \wedge dx^{i-1} \wedge dx^{i+1} \wedge \cdots \wedge dx^n \\ &= \left(\sum_{i=1}^n (-1)^{i-1} \frac{\partial f_i}{\partial x^i}\right) dx^1 \wedge \cdots \wedge dx^n. \end{aligned} \tag{3.68}$$

Now,

$$\int_{\mathbb{H}^n} d\omega = \int_0^a \underbrace{\int_{-a}^a \cdots \int_{-a}^a}_{n-1 \text{ times}} \left(\sum_{i=1}^n (-1)^{i-1} \frac{\partial f_i}{\partial x^i}\right) dx^1 \cdots dx^n. \tag{3.69}$$

Note that for $i = 1, \ldots, n-1$, we have $f_i(x_1, \ldots, x_n)|_{x_i=-a}^{x_i=a} = 0$ since $\operatorname{supp}(f_i) \subseteq \operatorname{int}([-a, a]^{n-1} \times [0, a])$. Hence, by the fundamental theorem of calculus, we have

$$\begin{aligned} \int_{\mathbb{H}^n} d\omega &= \underbrace{\int_{-a}^a \cdots \int_{-a}^a}_{n-1 \text{ times}} \left(\int_0^a (-1)^{n-1} \frac{\partial f_n}{\partial x^n} dx^n\right) dx^1 \cdots dx^{n-1} \\ &= (-1)^n \int_{-a}^a \cdots \int_{-a}^a f_n(x^1, \ldots, x^{n-1}, 0) dx^1 \cdots dx^{n-1}. \end{aligned} \tag{3.70}$$

On the other hand, we have

$$\int_{\partial\mathbb{H}^n} \omega = (-1)^n \int_{\partial\mathbb{H}^n} f_n dx^1 \cdots dx^{n-1} = (-1)^n \int_{-a}^{a} \cdots \int_{-a}^{a} f_n(x^1, \ldots, x^{n-1}, 0) dx^1 \cdots dx^{n-1} \tag{3.71}$$

which is exactly the same as the previous expression. Hence, we have $\int_{\mathbb{H}^n} d\omega = \int_{\partial\mathbb{H}^n} \omega$. The proof for $\mathbb{R}^n$ is similar, there being $\int_{-a}^{a}$ instead of $\int_0^a$, and the proof is complete. ■

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